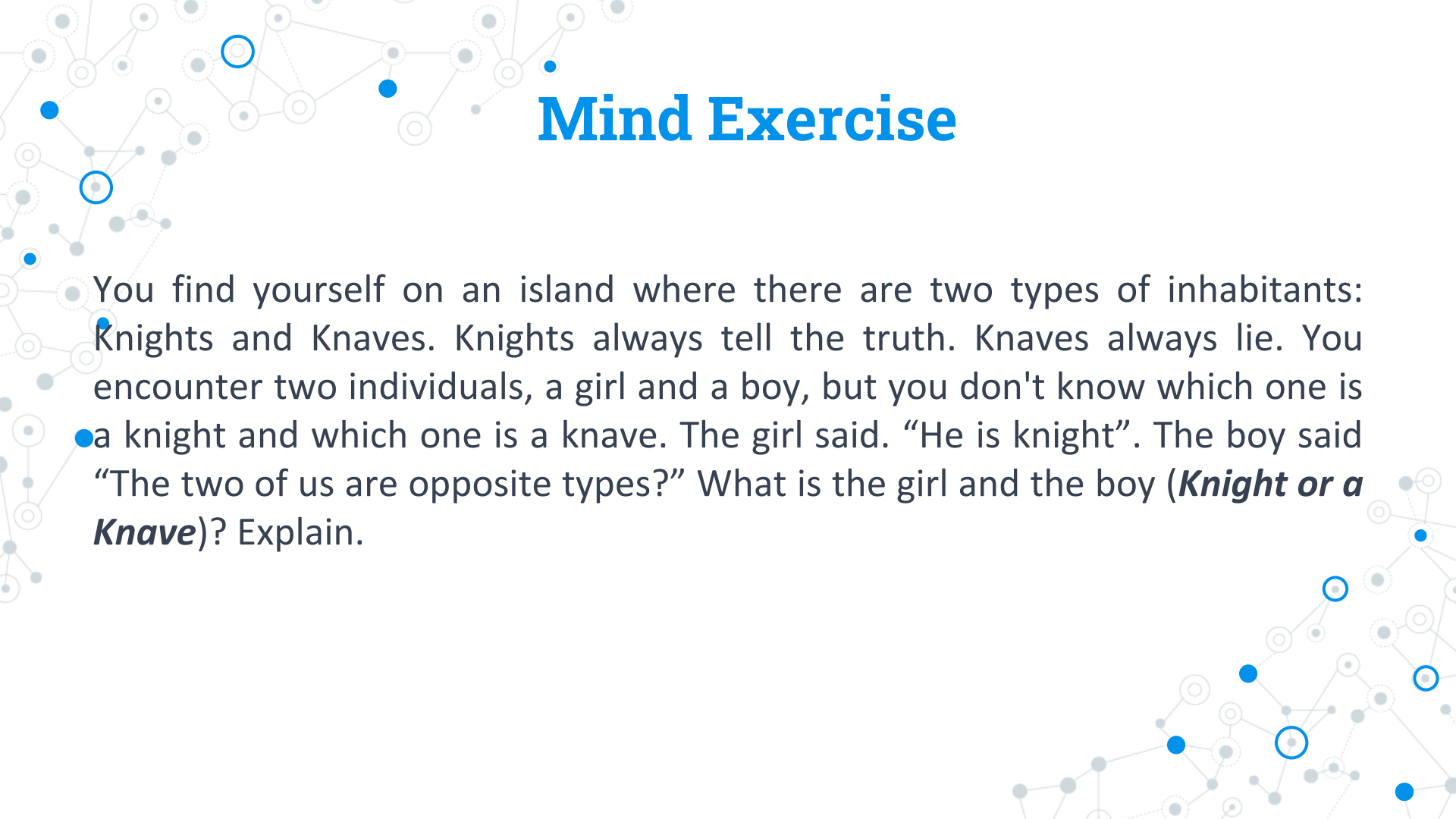




CSC 102– Discrete Structures I

# Logic





# Mind Exercise

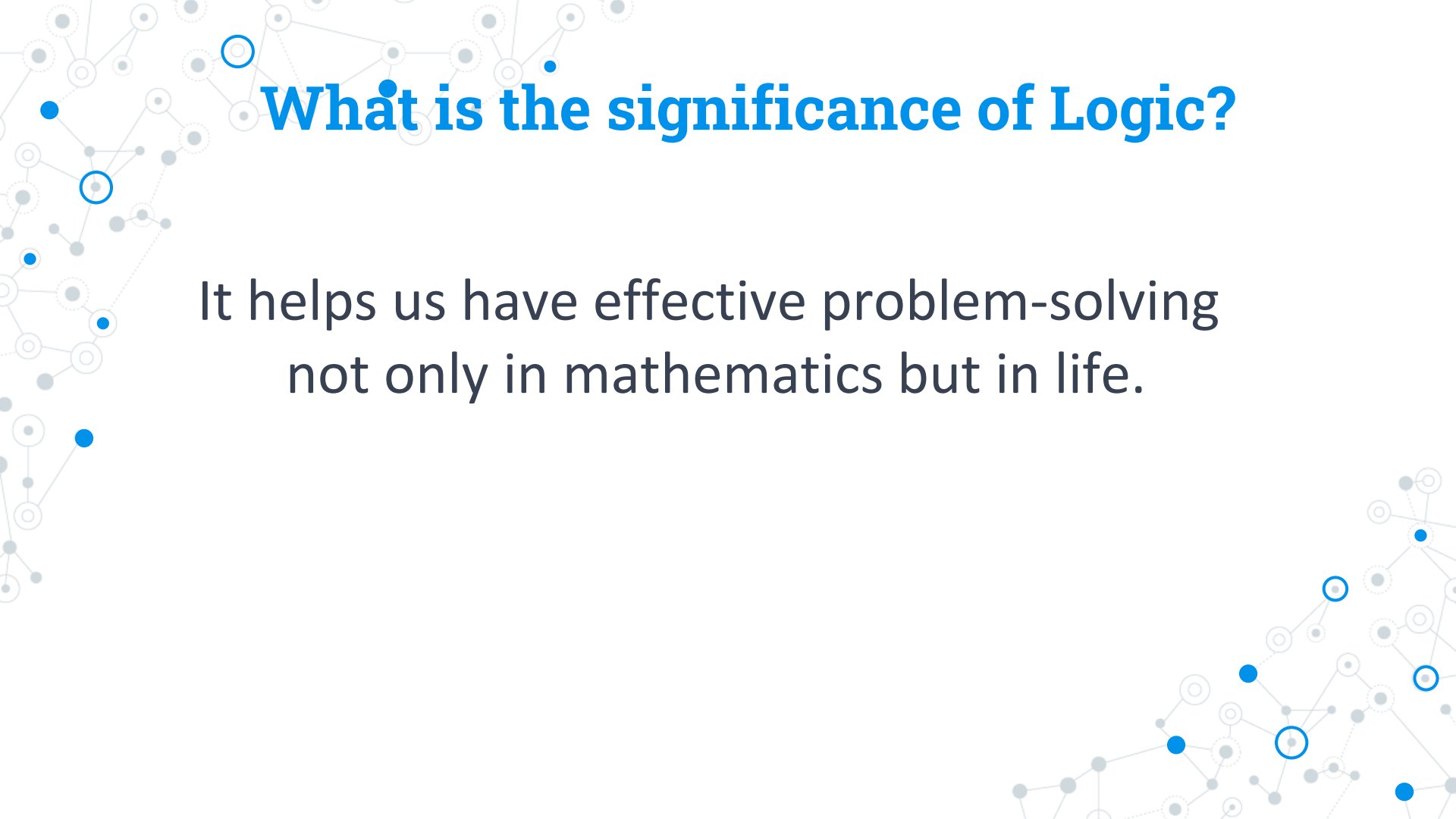
You find yourself on an island where there are two types of inhabitants: Knights and Knaves. Knights always tell the truth. Knaves always lie. You encounter two individuals, a girl and a boy, but you don't know which one is a knight and which one is a knave. The girl said. "He is knight". The boy said "The two of us are opposite types?" What is the girl and the boy (***Knight or a Knave***)? Explain.

A decorative network diagram in the top-left corner, featuring a complex web of interconnected nodes and lines. Some nodes are highlighted with blue circles, and others with blue dots.

# Logic

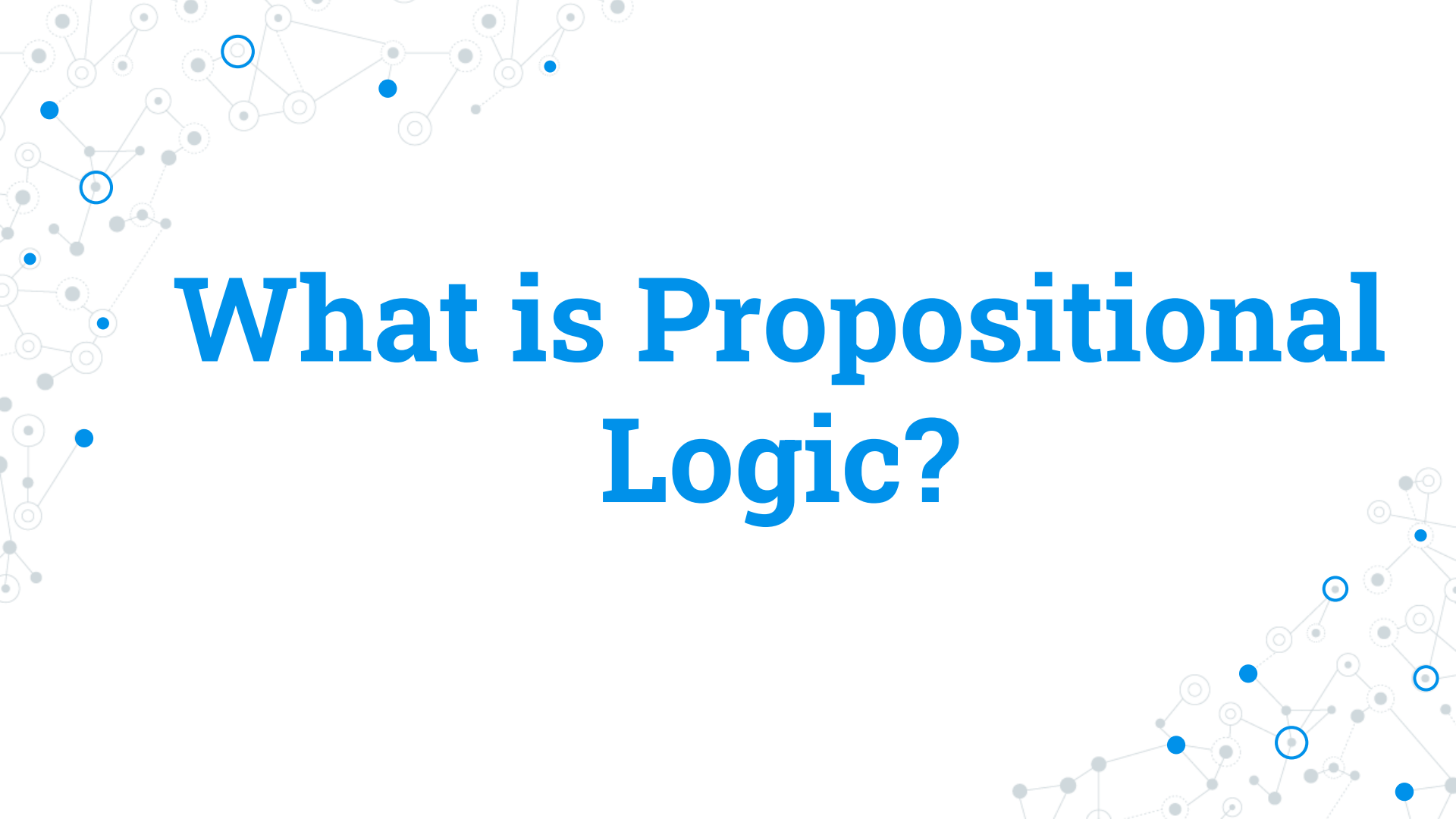
Study of reasoning

A decorative network diagram in the bottom-right corner, similar to the one in the top-left, with interconnected nodes and lines, some highlighted with blue circles and dots.

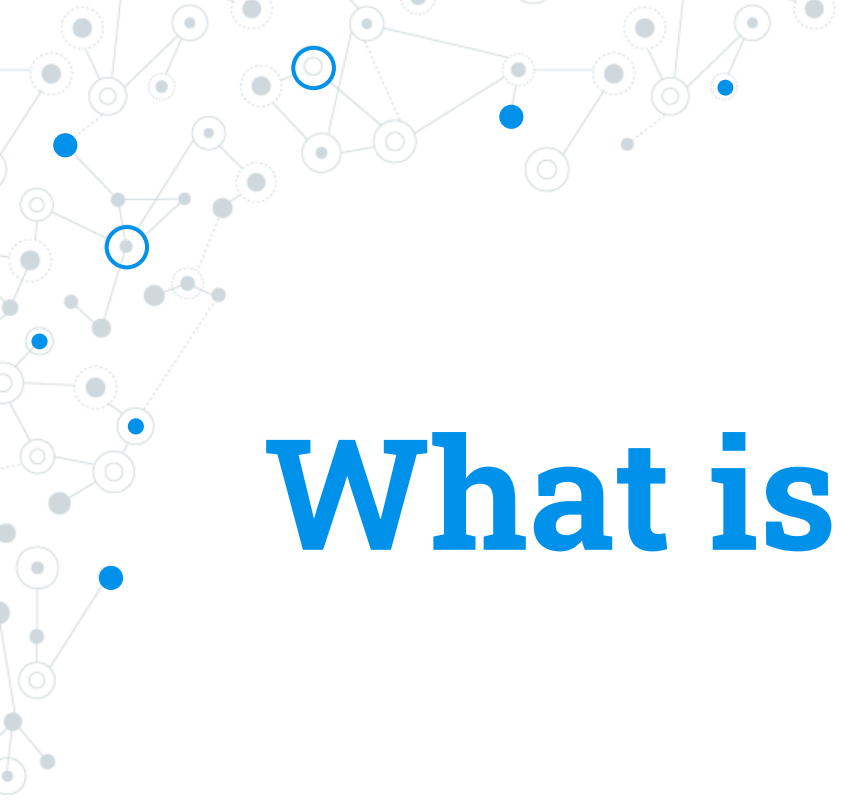
A decorative background featuring a network diagram with nodes and connecting lines. The nodes are represented by circles of varying sizes and colors (blue, grey, white), and the lines are thin and grey. The network is more dense on the left side and tapers off towards the right.

# What is the significance of Logic?

It helps us have effective problem-solving  
not only in mathematics but in life.

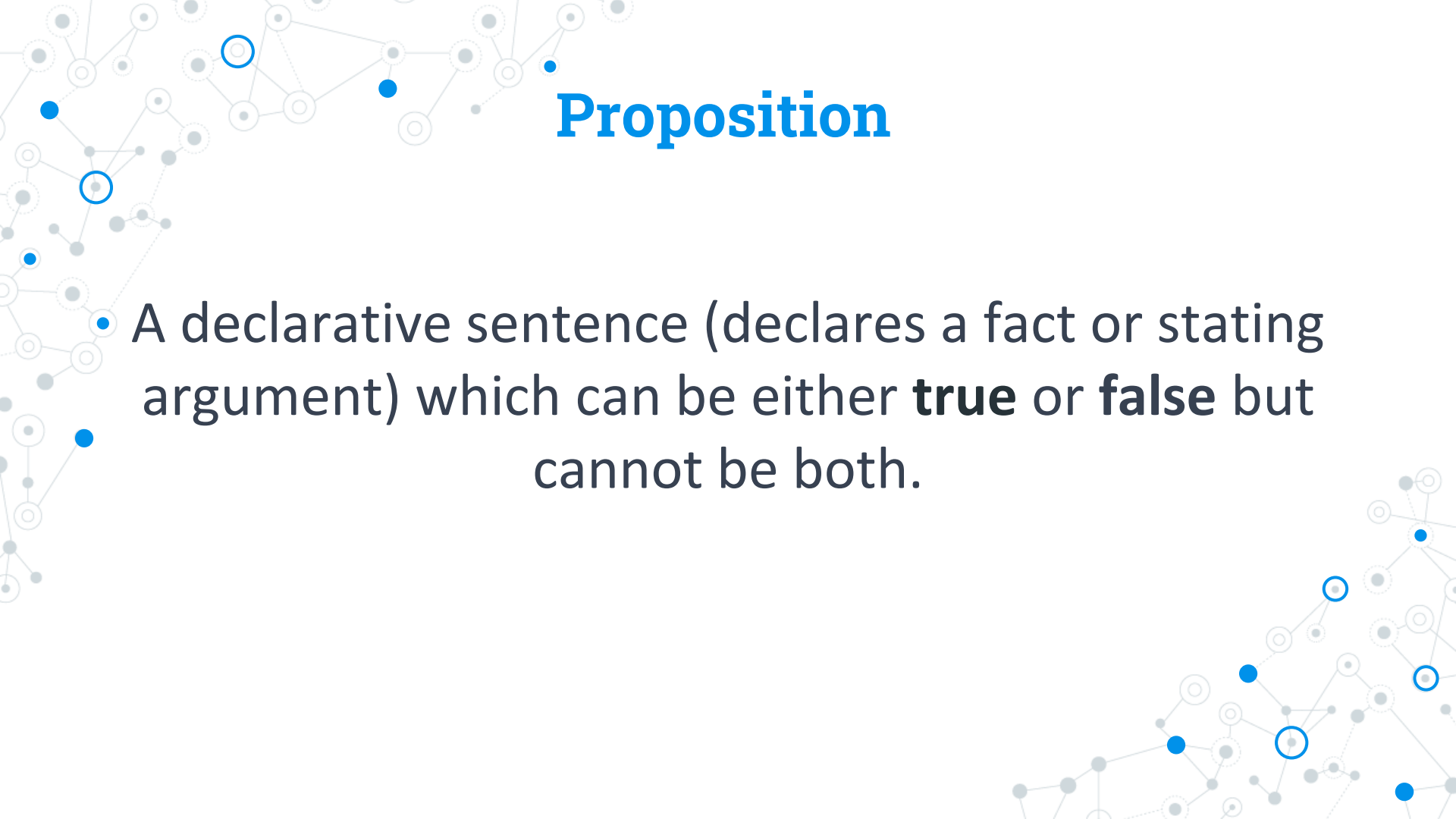


# What is Propositional Logic?

A decorative network diagram in the top-left corner, featuring a complex web of interconnected nodes. Some nodes are highlighted with blue circles, and others with blue dots. The nodes are connected by thin grey lines, forming a dense, branching structure.

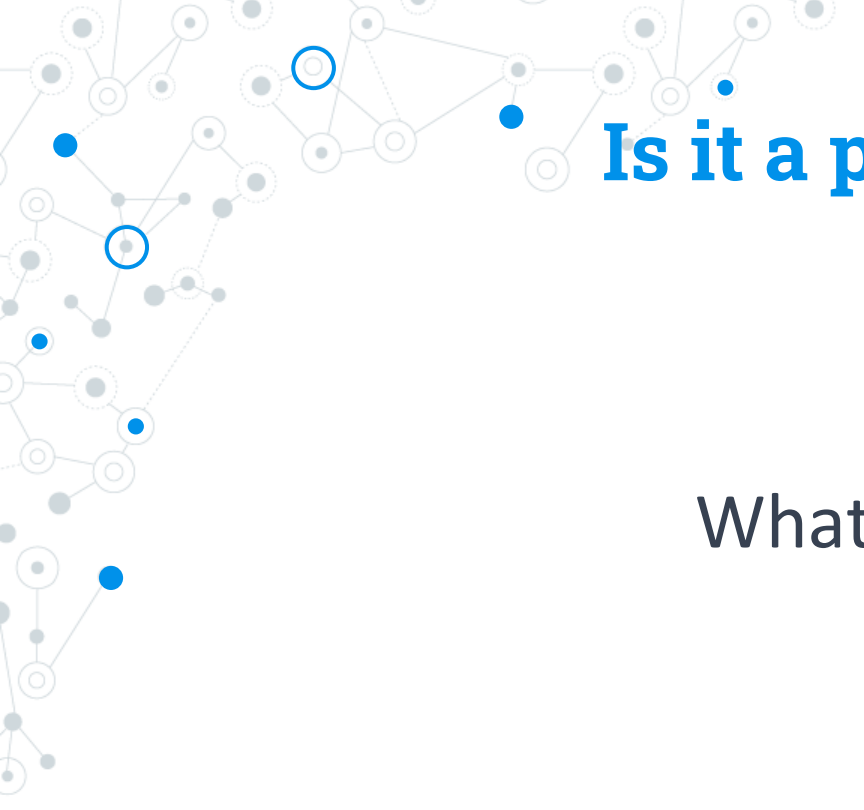
# What is Proposition?

A decorative network diagram in the bottom-right corner, similar to the one in the top-left. It shows a web of interconnected nodes, with some nodes highlighted by blue circles and others by blue dots. The connections are thin grey lines, creating a complex, branching pattern.

A decorative background graphic consisting of a network of nodes and edges. The nodes are represented by small circles, some of which are solid blue, some are solid grey, and some are hollow blue. The edges are thin grey lines connecting the nodes. The network is more dense on the left side of the slide and becomes sparser towards the right.

# Proposition

A declarative sentence (declares a fact or stating argument) which can be either **true** or **false** but cannot be both.

A decorative network diagram in the top-left corner of the slide. It features a complex web of interconnected nodes and edges. The nodes are represented by small circles, some of which are solid blue, some are solid grey, and some are hollow with a blue outline. The edges are thin grey lines connecting the nodes. The overall shape of the network is irregular and extends from the top-left towards the center of the slide.

**Is it a proposition?**

What time is it ?

A decorative network diagram in the bottom-right corner of the slide. It features a complex web of interconnected nodes and edges. The nodes are represented by small circles, some of which are solid blue, some are solid grey, and some are hollow with a blue outline. The edges are thin grey lines connecting the nodes. The overall shape of the network is irregular and extends from the bottom-right towards the center of the slide.



A decorative network graph in the top-left corner, featuring a complex web of interconnected nodes. Some nodes are solid blue circles, while others are white circles with blue outlines. The nodes are connected by thin gray lines, creating a dense, branching structure.

**Is it a proposition?**

$$x + 5 = 2$$

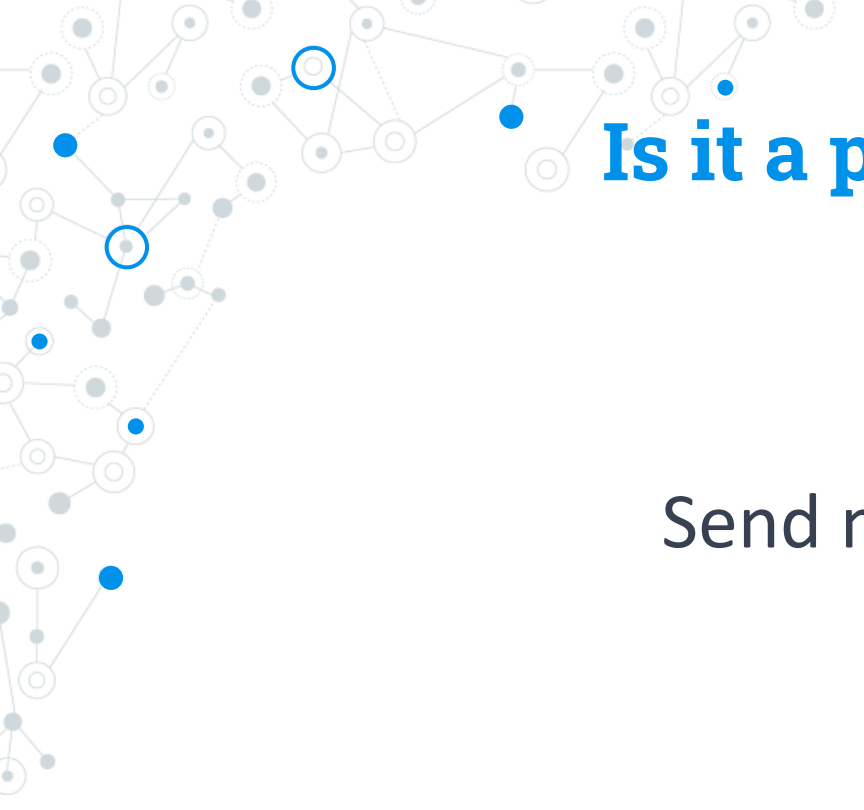


A decorative network graph in the top-left corner, featuring a complex web of interconnected nodes. Some nodes are solid blue circles, while others are white circles with blue outlines. The nodes are connected by thin grey lines, creating a dense, branching structure.

**Is it a proposition?**

$$1 + 5 = 2$$



A decorative network diagram in the top-left corner of the slide. It features a complex web of interconnected nodes and edges. The nodes are represented by small circles, some of which are solid blue, some are solid grey, and some are hollow with a blue outline. The edges are thin grey lines connecting the nodes. The overall shape of the network is irregular and extends from the top-left towards the center of the slide.

**Is it a proposition?**

Send me an email.

A decorative network diagram in the bottom-right corner of the slide. It features a complex web of interconnected nodes and edges. The nodes are represented by small circles, some of which are solid blue, some are solid grey, and some are hollow with a blue outline. The edges are thin grey lines connecting the nodes. The overall shape of the network is irregular and extends from the bottom-right towards the center of the slide.

A decorative network diagram in the top-left corner of the slide. It features a complex web of interconnected nodes and edges. The nodes are represented by small circles, some of which are solid blue, some are solid grey, and some are hollow with a blue outline. The edges are thin grey lines connecting these nodes. The overall shape of the network is irregular and extends from the top-left towards the center of the slide.

**Is it a proposition?**

CSU is located in Mindanao.

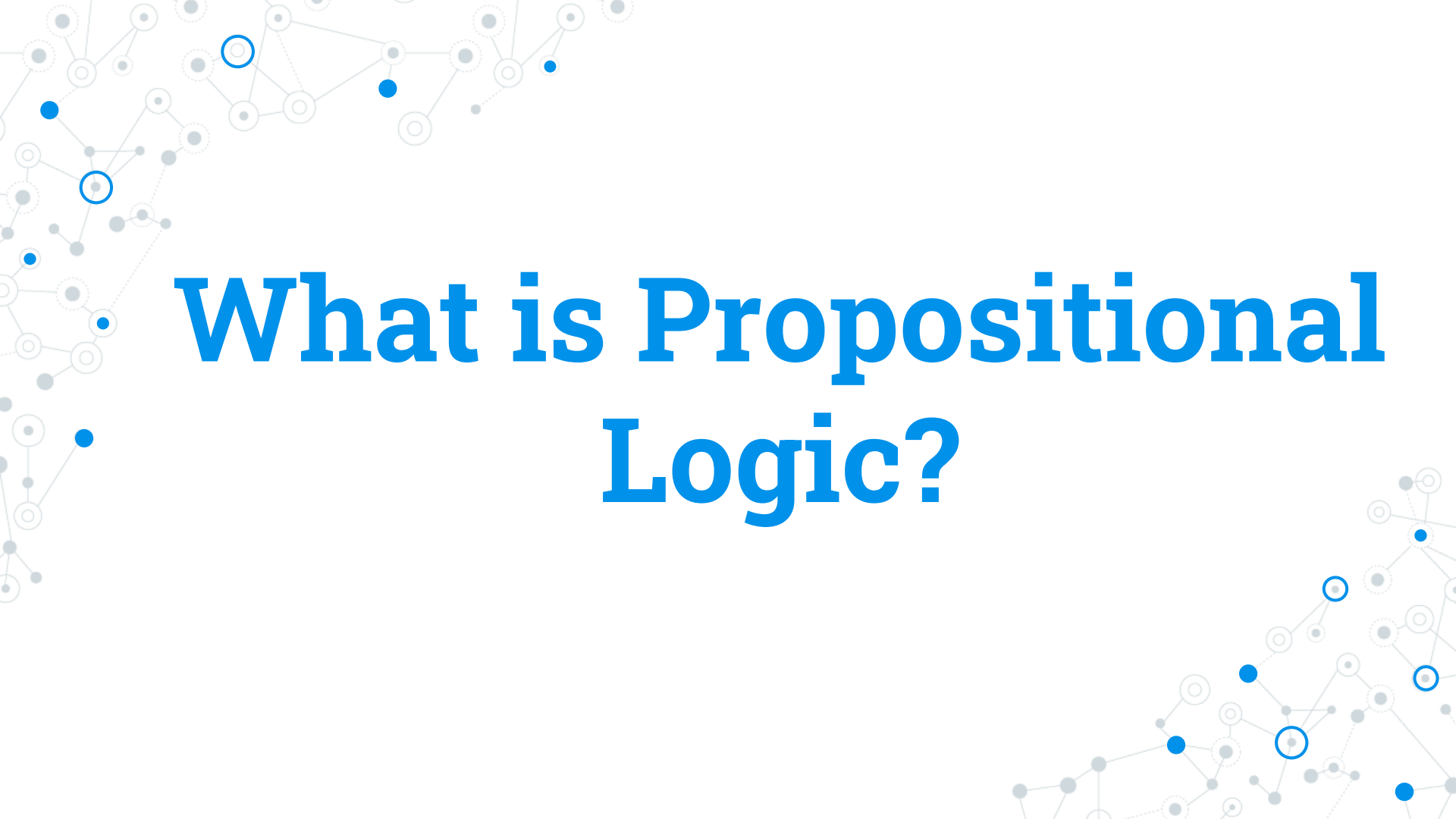
A decorative network diagram in the bottom-right corner of the slide. It is a smaller version of the network diagram in the top-left, featuring a web of interconnected nodes and edges. The nodes are represented by small circles, some solid blue, some solid grey, and some hollow with a blue outline. The edges are thin grey lines connecting the nodes. The network shape is irregular and extends from the bottom-right towards the center of the slide.

A decorative network diagram in the top-left corner of the slide. It features a complex web of interconnected nodes and edges. The nodes are represented by small circles, some of which are solid blue, some are solid grey, and some are hollow with a blue outline. The edges are thin grey lines connecting the nodes. The overall structure is dense and organic, resembling a molecular or biological network.

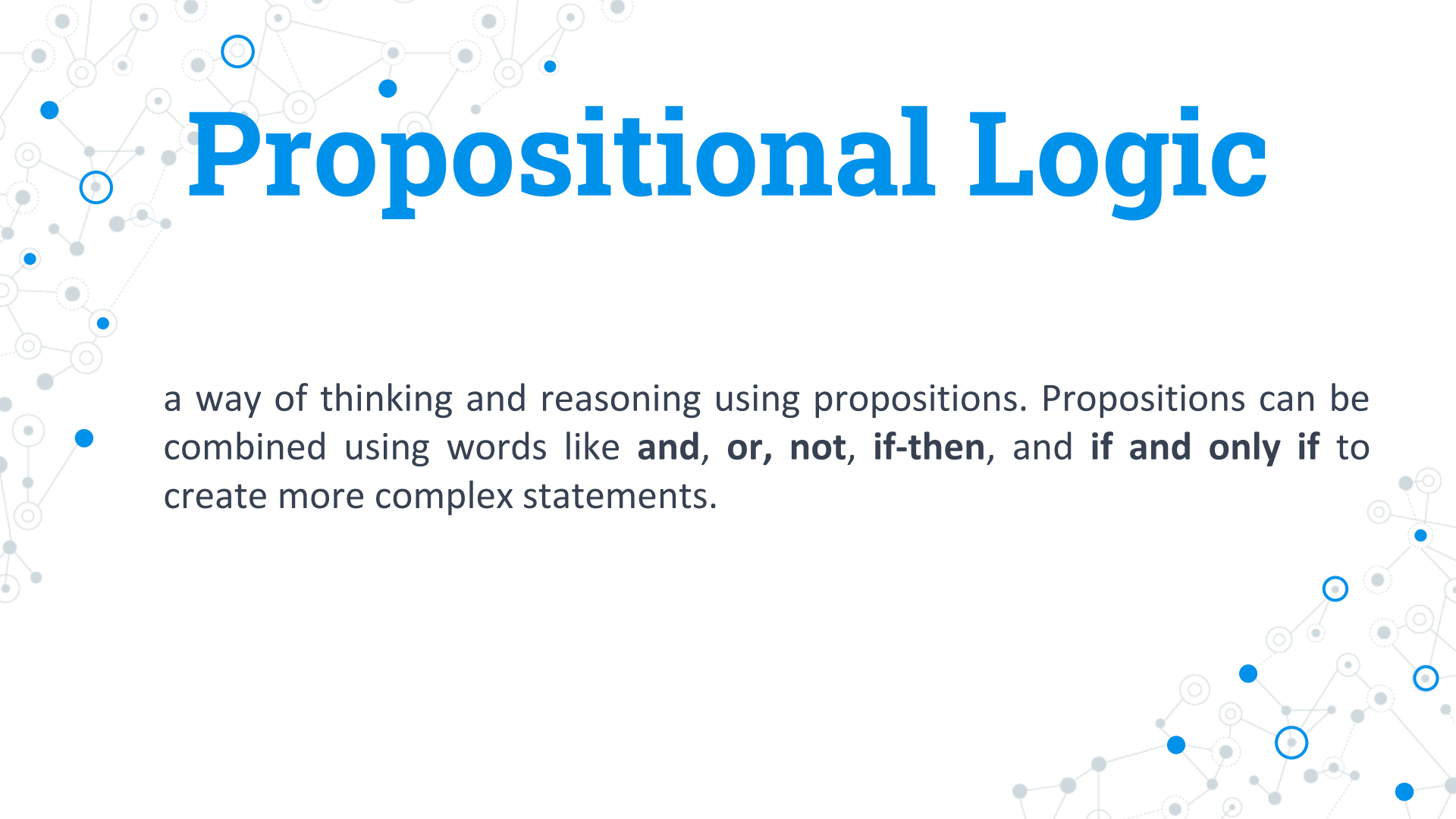
## Is it a proposition?

Eating vegetables is not good for our health.

A decorative network diagram in the bottom-right corner of the slide. It features a complex web of interconnected nodes and edges. The nodes are represented by small circles, some of which are solid blue, some are solid grey, and some are hollow with a blue outline. The edges are thin grey lines connecting the nodes. The overall structure is dense and organic, resembling a molecular or biological network.

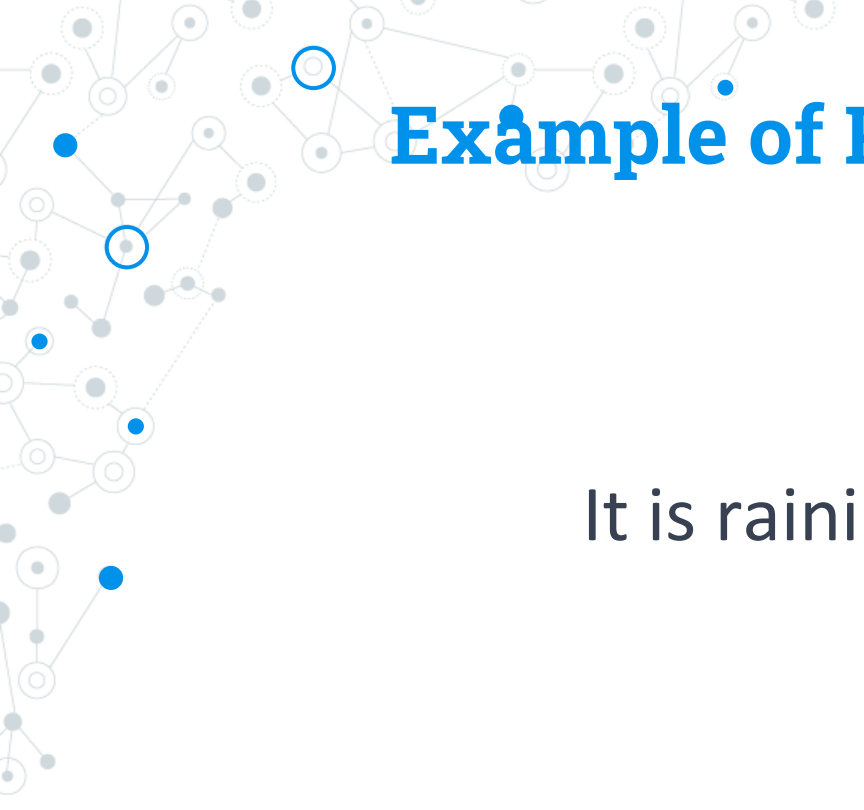


# What is Propositional Logic?

A decorative background featuring a network diagram with nodes and connecting lines. The nodes are represented by circles of varying sizes and colors (blue, grey, white), and the lines are thin and grey. The network is more dense on the left side and tapers off towards the right.

# Propositional Logic

a way of thinking and reasoning using propositions. Propositions can be combined using words like **and**, **or**, **not**, **if-then**, and **if and only if** to create more complex statements.

A decorative network diagram in the top-left corner, featuring a complex web of interconnected nodes and edges. Some nodes are highlighted with blue circles, and others with blue dots. The edges are thin grey lines, some solid and some dashed.

# Example of Propositional Logic

It is raining **and** it is cold.

A decorative network diagram in the bottom-right corner, similar to the one in the top-left, with a web of nodes and edges. Some nodes are highlighted with blue circles, and others with blue dots.





# • Compound Propositions

It is raining **and** it is cold.



# Compound Propositions

It is raining **and** it is cold.

P

Q

# Compound Propositions

It is raining **and** it is cold.

P

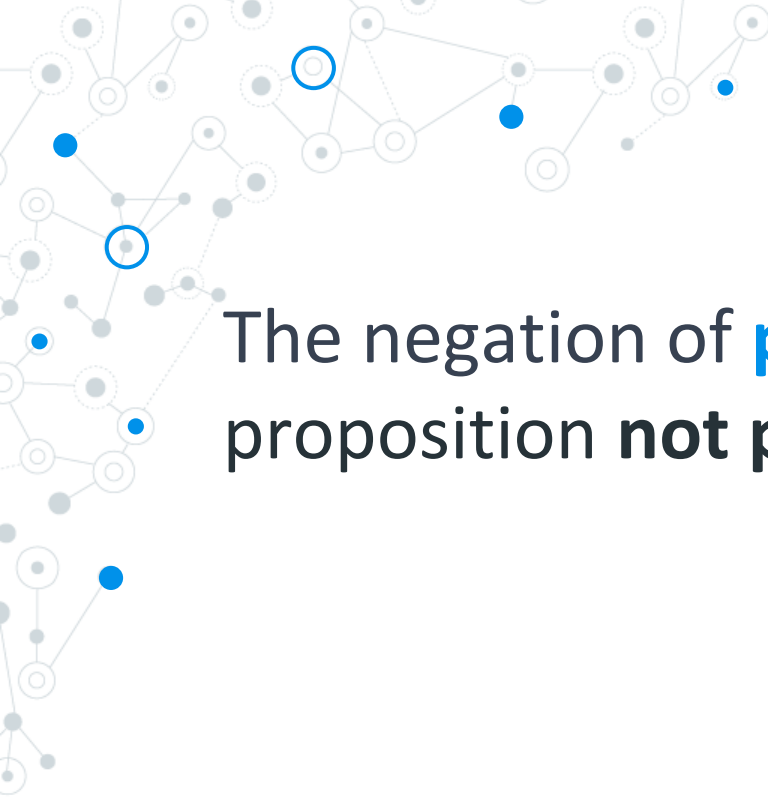
Q

P and Q



# Logical Operators

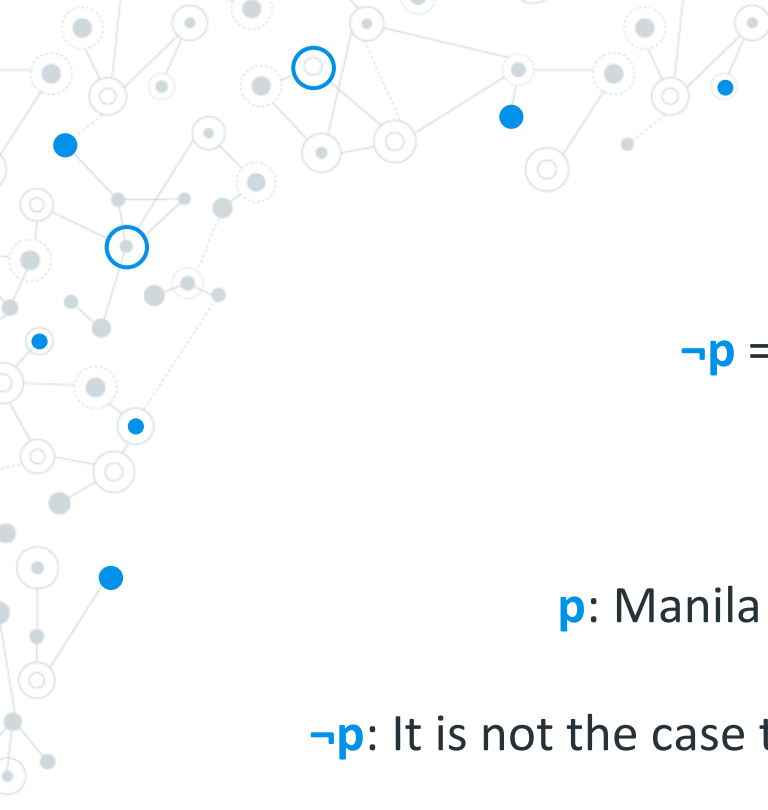
- 1) Negation
  - 2) Conjunction
  - 3) Disjunction
  - 4) Exclusive OR
  - 5) Implication
  - 6) Biconditional
- 

A decorative network diagram in the top-left corner of the slide. It consists of a complex web of interconnected nodes and edges. The nodes are represented by small circles, some of which are solid blue, some are solid grey, and some are hollow with a blue outline. The edges are thin grey lines connecting these nodes. The overall shape of the network is irregular and spreads across the top-left portion of the slide.

# Negation

The negation of **p** denoted by  $\neg p$ , is the proposition **not p**.

A decorative network diagram in the bottom-right corner of the slide. It is a smaller version of the network diagram in the top-left, featuring a cluster of interconnected nodes and edges. The nodes are a mix of solid blue, solid grey, and hollow blue circles, connected by thin grey lines.



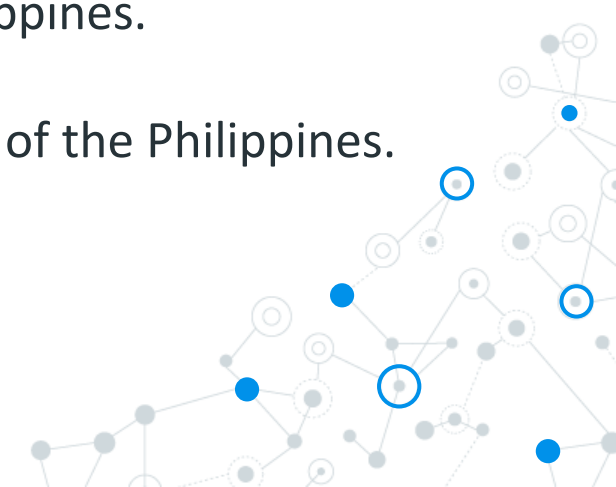
# Negation

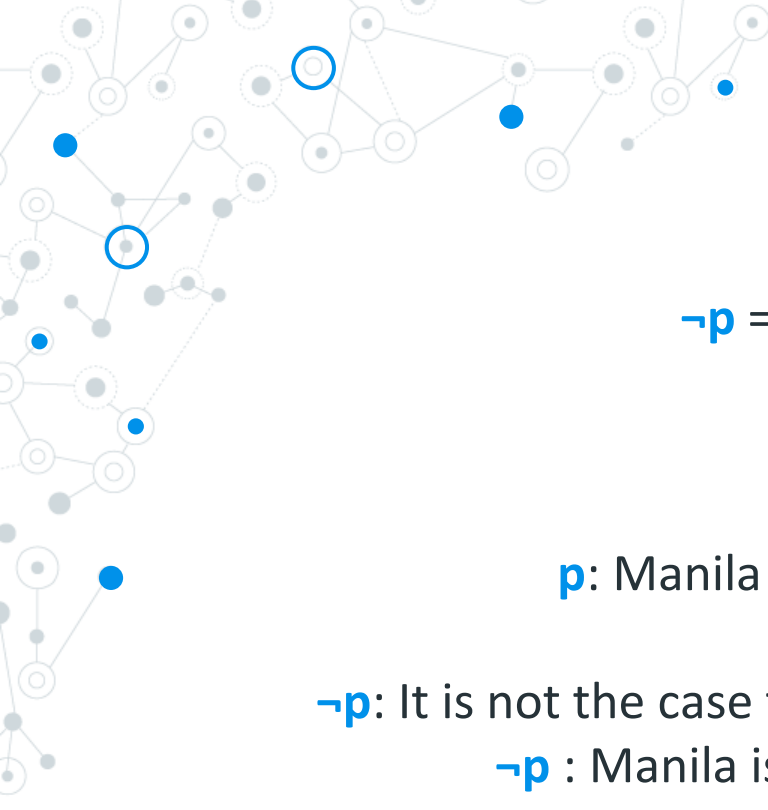
$\neg p$  = “It is not the case that  $p$ ”

Example:

$p$ : Manila is the capital of the Philippines.

$\neg p$ : It is not the case that Manila is the capital of the Philippines.





# Negation

$\neg p$  = “It is not the case that  $p$ ”

Example:

$p$ : Manila is the capital of the Philippines.

$\neg p$ : It is not the case that Manila is the capital of the Philippines

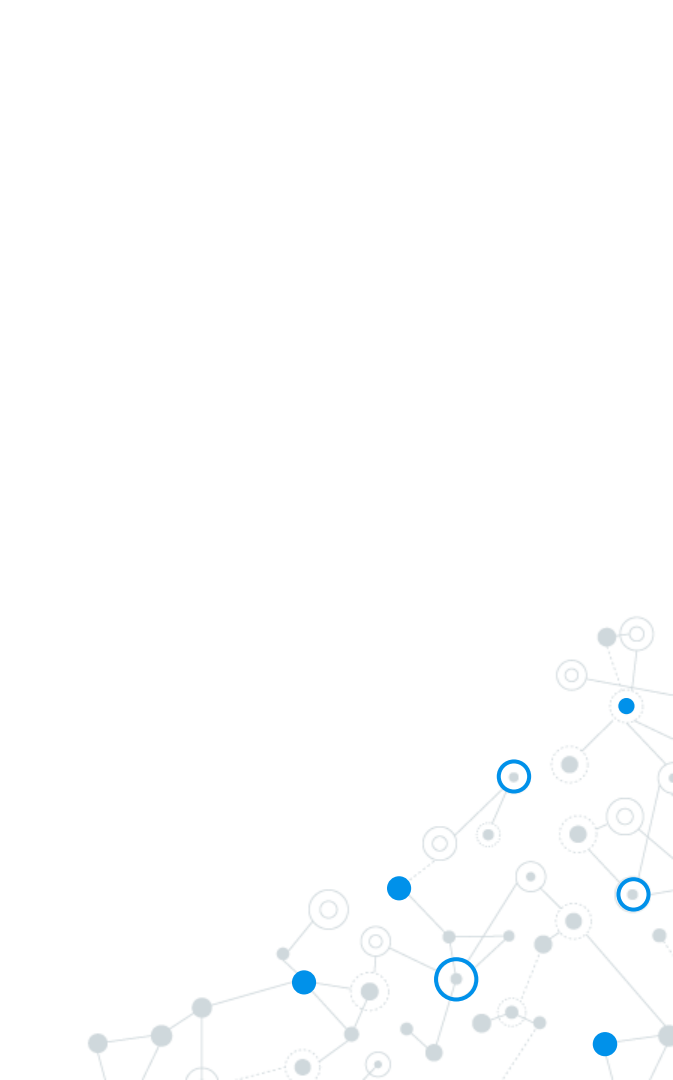
$\neg p$  : Manila is not the capital of the Philippines.



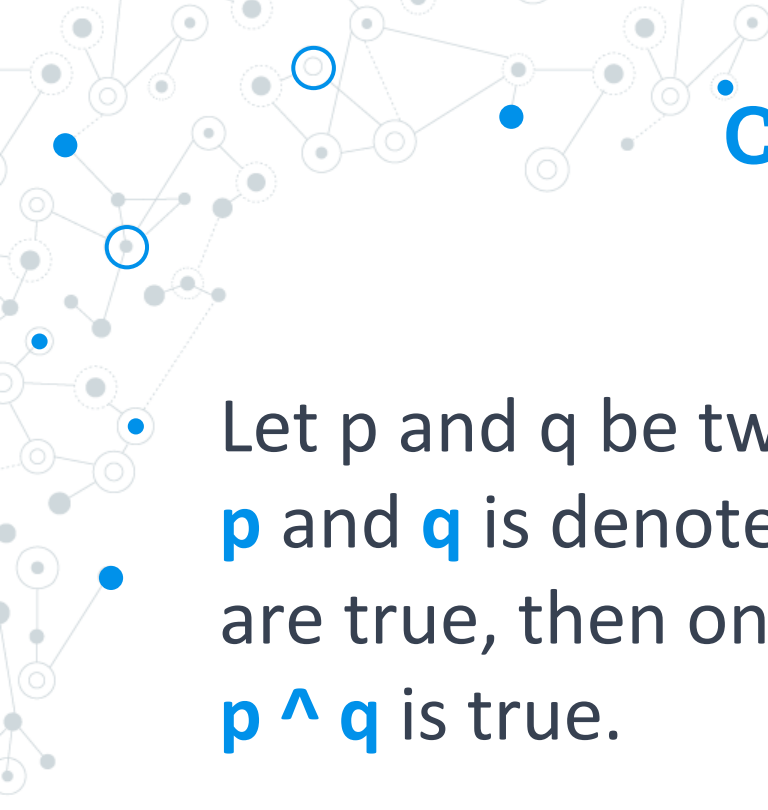


# Negation

| $p$ | $\neg p$ |
|-----|----------|
| T   | F        |
| F   | T        |







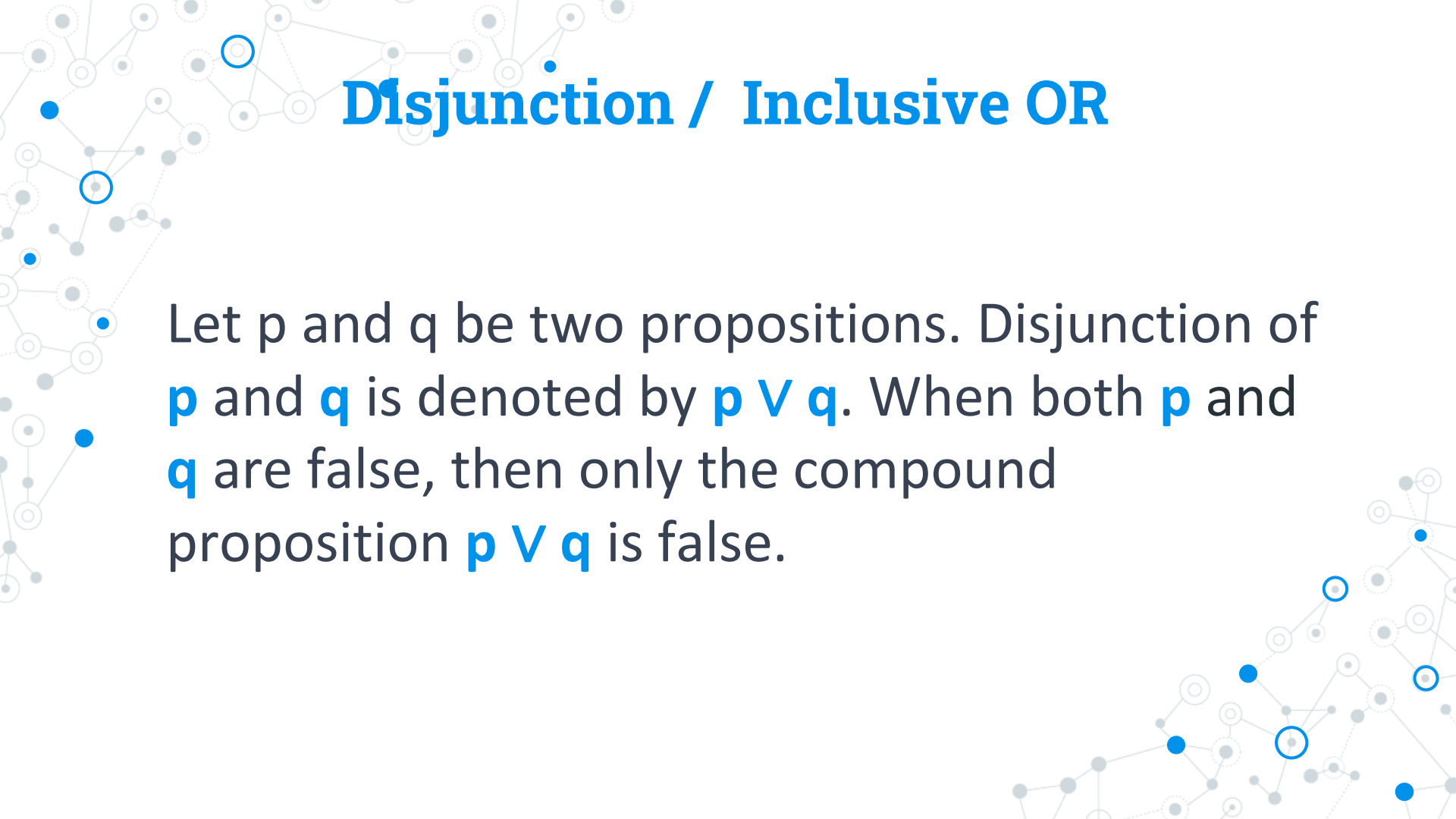
# Conjunction

Let  $p$  and  $q$  be two propositions. Conjunction of  $p$  and  $q$  is denoted by  $p \wedge q$ . When both  $p$  and  $q$  are true, then only the compound proposition  $p \wedge q$  is true.



# Conjunction: Truth Value

| $p$ | $q$ | $p \wedge q$ |
|-----|-----|--------------|
| T   | T   | T            |
| T   | F   | F            |
| F   | T   | F            |
| F   | F   | F            |

A decorative background graphic consisting of a network of nodes and edges. The nodes are represented by small circles, some of which are blue and some are grey. The edges are thin lines connecting the nodes, forming a complex, interconnected web. The overall style is modern and technical, typical of a presentation on logic or computer science.

# Disjunction / Inclusive OR

Let  $p$  and  $q$  be two propositions. Disjunction of  $p$  and  $q$  is denoted by  $p \vee q$ . When both  $p$  and  $q$  are false, then only the compound proposition  $p \vee q$  is false.

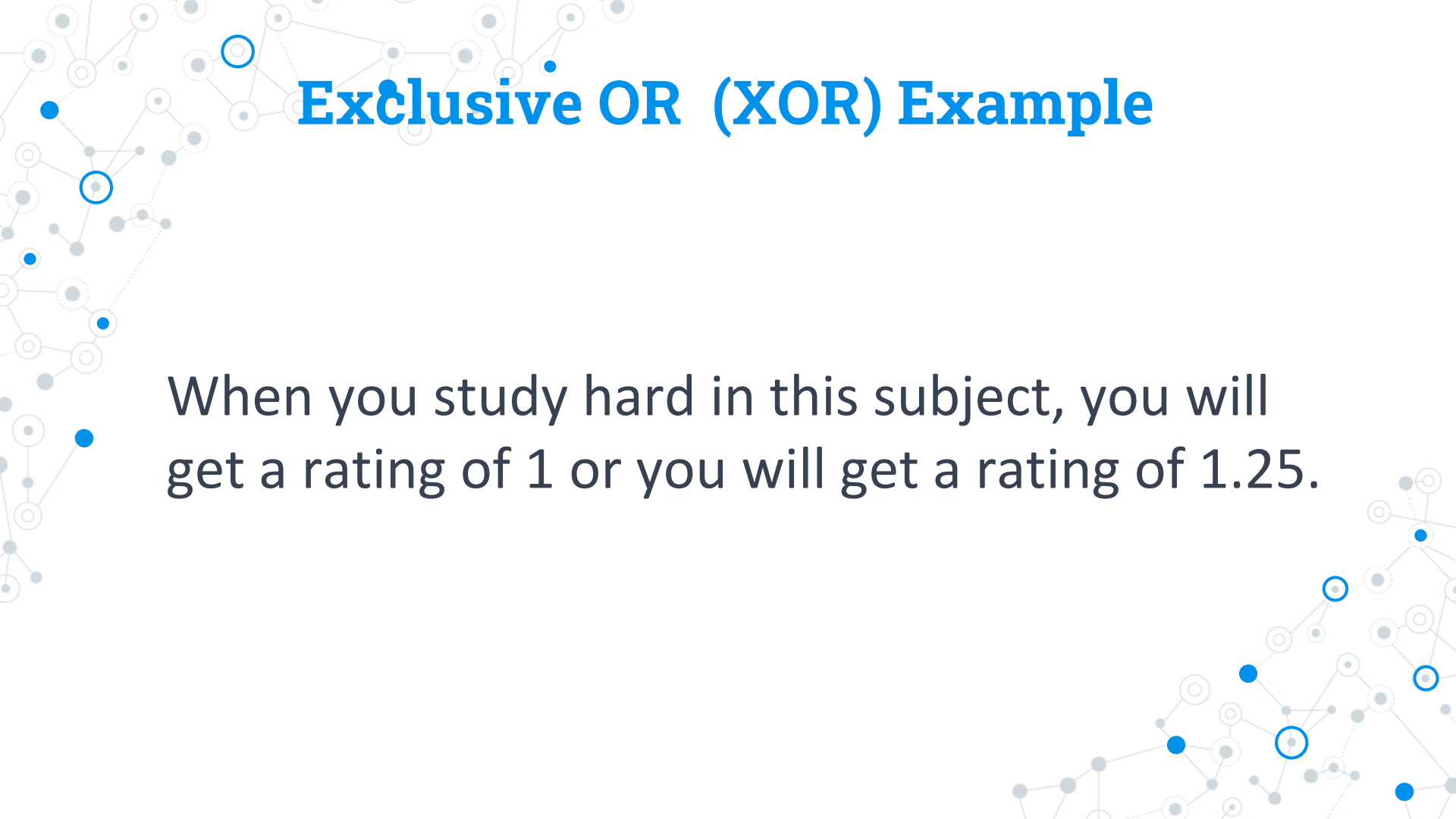
# Disjunction: Truth Value

| $p$ | $q$ | $p \vee q$ |
|-----|-----|------------|
| T   | T   | T          |
| T   | F   | T          |
| F   | T   | T          |
| F   | F   | F          |



- **Exclusive OR (XOR)**

Let  $p$  and  $q$  be two propositions. Exclusive OR of  $p$  and  $q$  is denoted by  $p \oplus q$ . This means exactly one of  $p$  and  $q$  will be true, but both cannot be true.

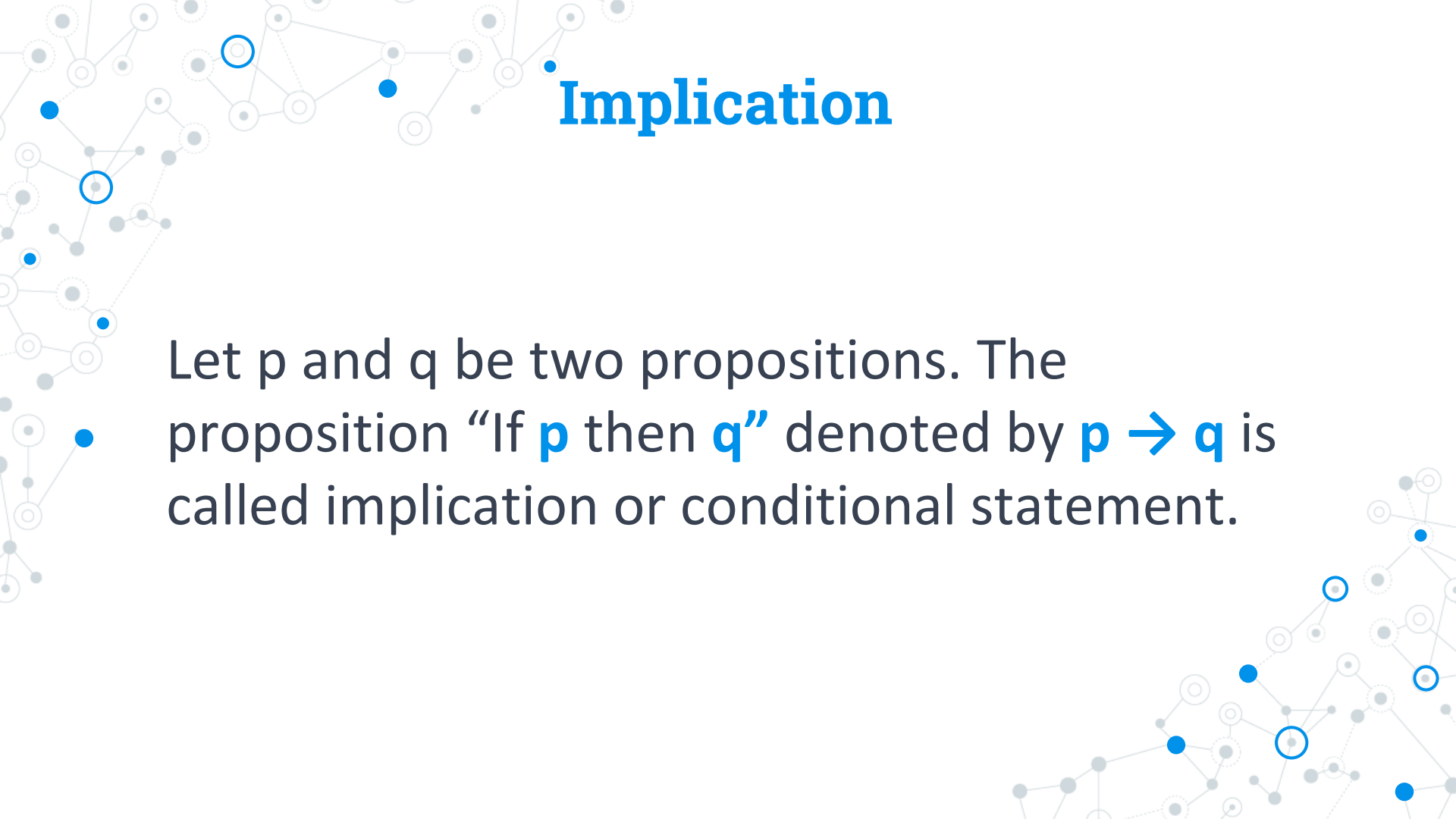
A decorative background featuring a network of nodes and connections. The nodes are represented by circles of varying shades of gray and blue, some with concentric circles. They are interconnected by thin, light gray lines, forming a complex web-like structure that spans the entire slide.

## Exclusive OR (XOR) Example

When you study hard in this subject, you will get a rating of 1 or you will get a rating of 1.25.

# Exclusive OR: Truth Value

| $p$ | $q$ | $p \oplus q$ |
|-----|-----|--------------|
| T   | T   | F            |
| T   | F   | T            |
| F   | T   | T            |
| F   | F   | F            |



# Implication

Let  $p$  and  $q$  be two propositions. The proposition “If  $p$  then  $q$ ” denoted by  $p \rightarrow q$  is called implication or conditional statement.





# Implication

“If **p** then **q**”



Hypothesis/conditional statement





# Implication

“If **p** then **q**”



Conclusion / Consequent



# Implication Truth Value

| $p$ | $q$ | $p \rightarrow q$ |
|-----|-----|-------------------|
| T   | T   | T                 |
| T   | F   | F                 |
| F   | T   | T                 |
| F   | F   | T                 |

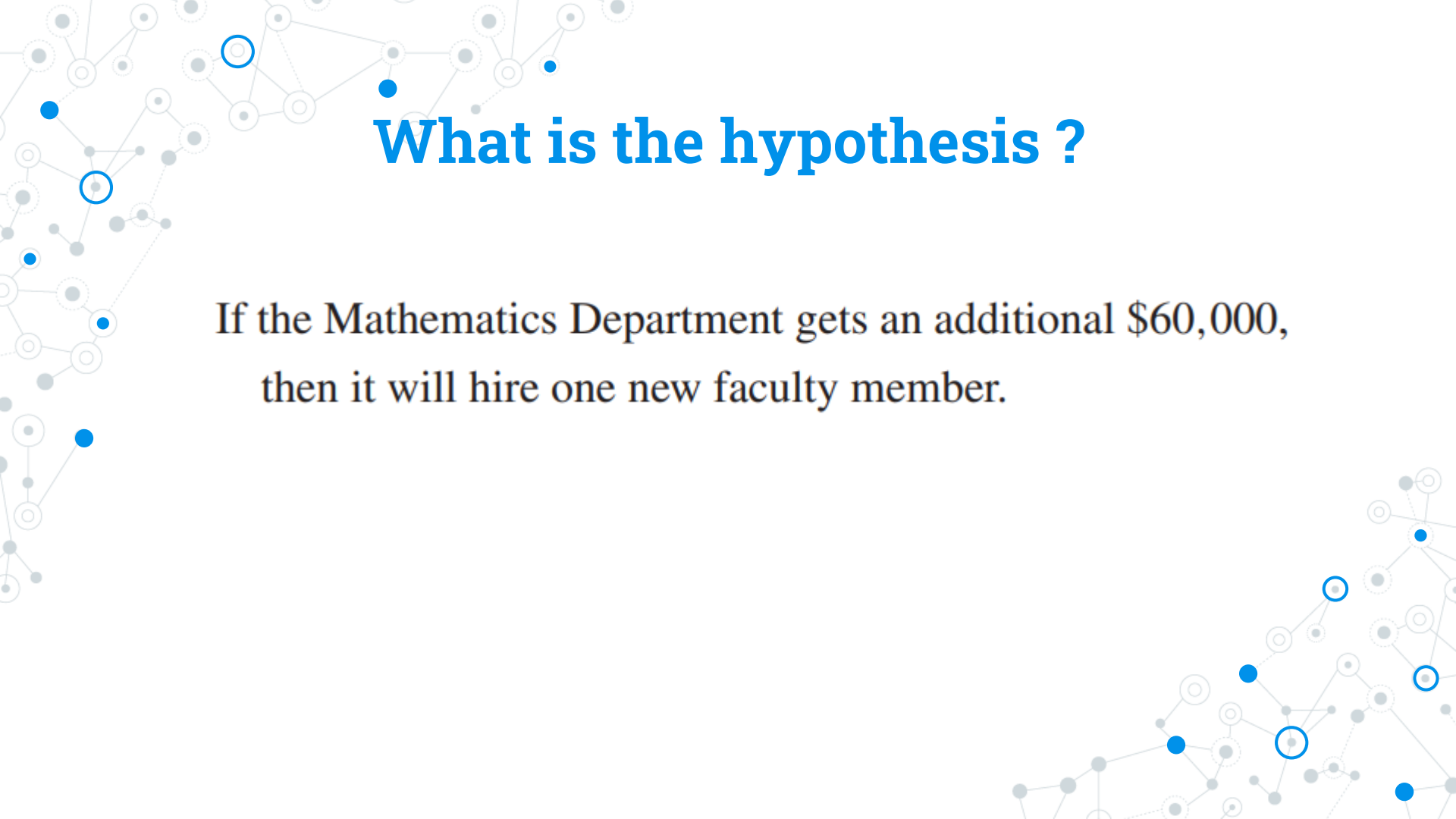


- **Implication Example**

“ If the Mathematics Department gets an additional \$60,000,  
then it will hire one new faculty member.

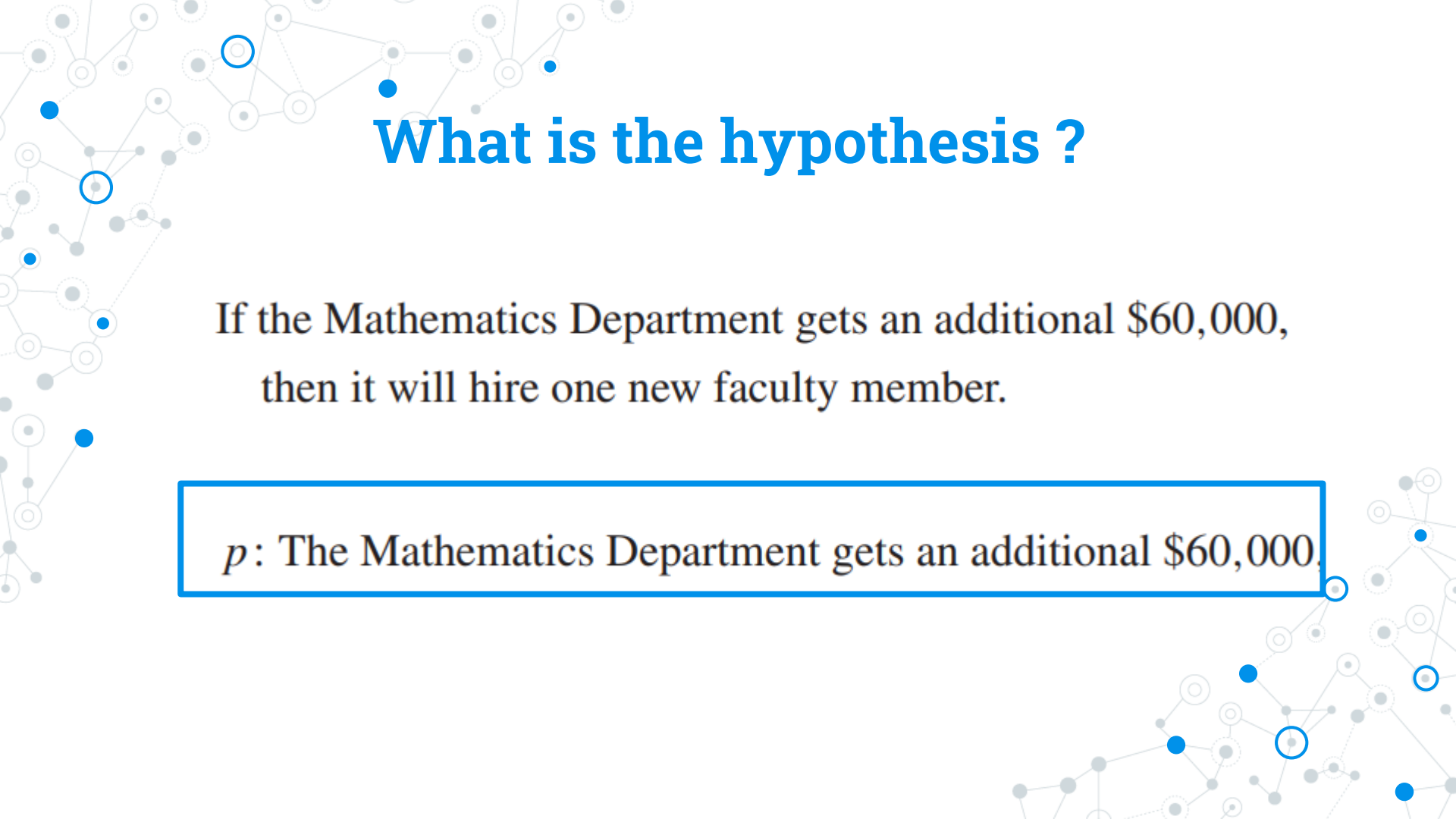
**Dean's Statement**





# What is the hypothesis ?

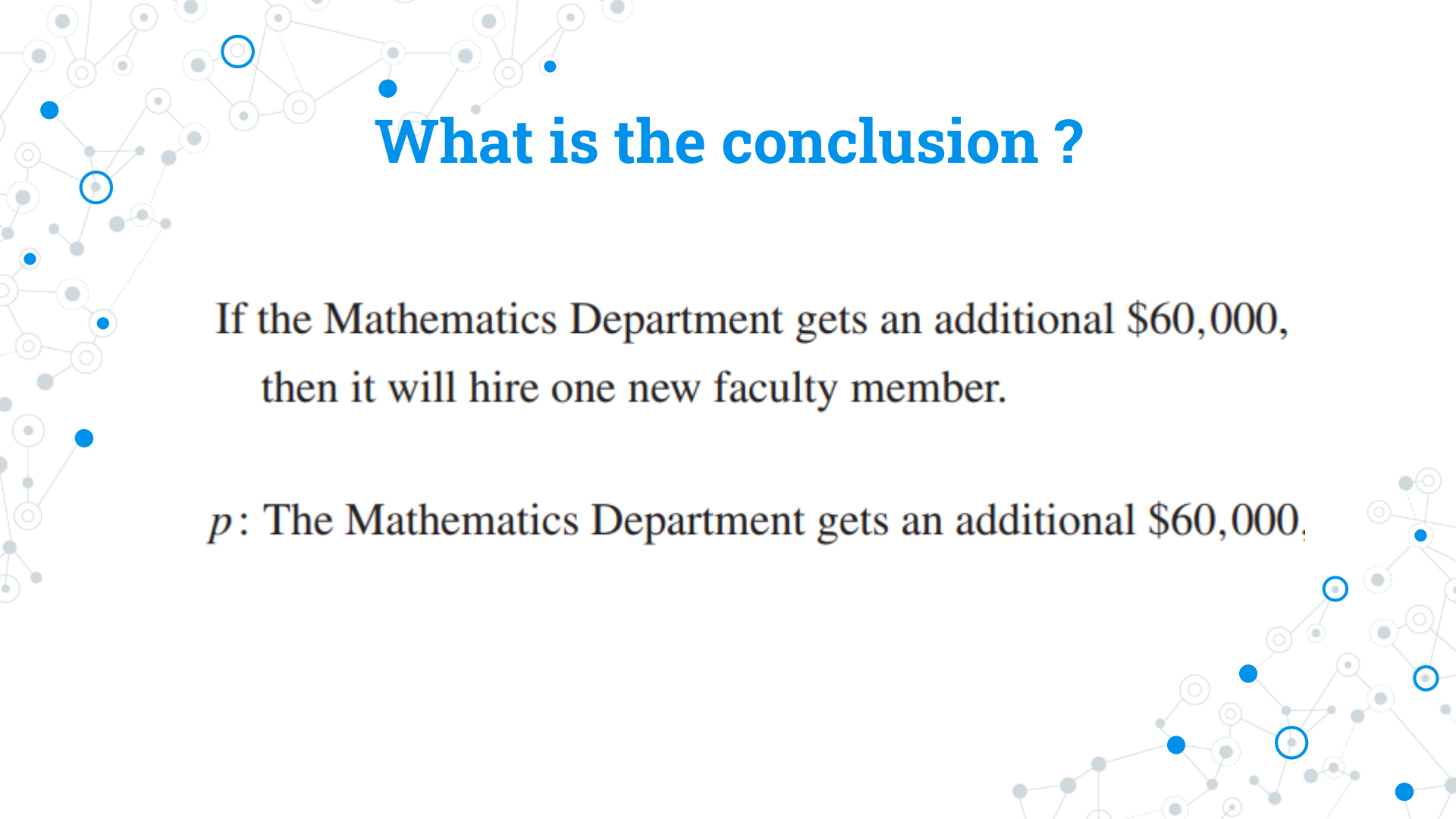
If the Mathematics Department gets an additional \$60,000,  
then it will hire one new faculty member.



## What is the hypothesis ?

If the Mathematics Department gets an additional \$60,000,  
then it will hire one new faculty member.

$p$ : The Mathematics Department gets an additional \$60,000.



## What is the conclusion ?

If the Mathematics Department gets an additional \$60,000,  
then it will hire one new faculty member.

$p$ : The Mathematics Department gets an additional \$60,000,

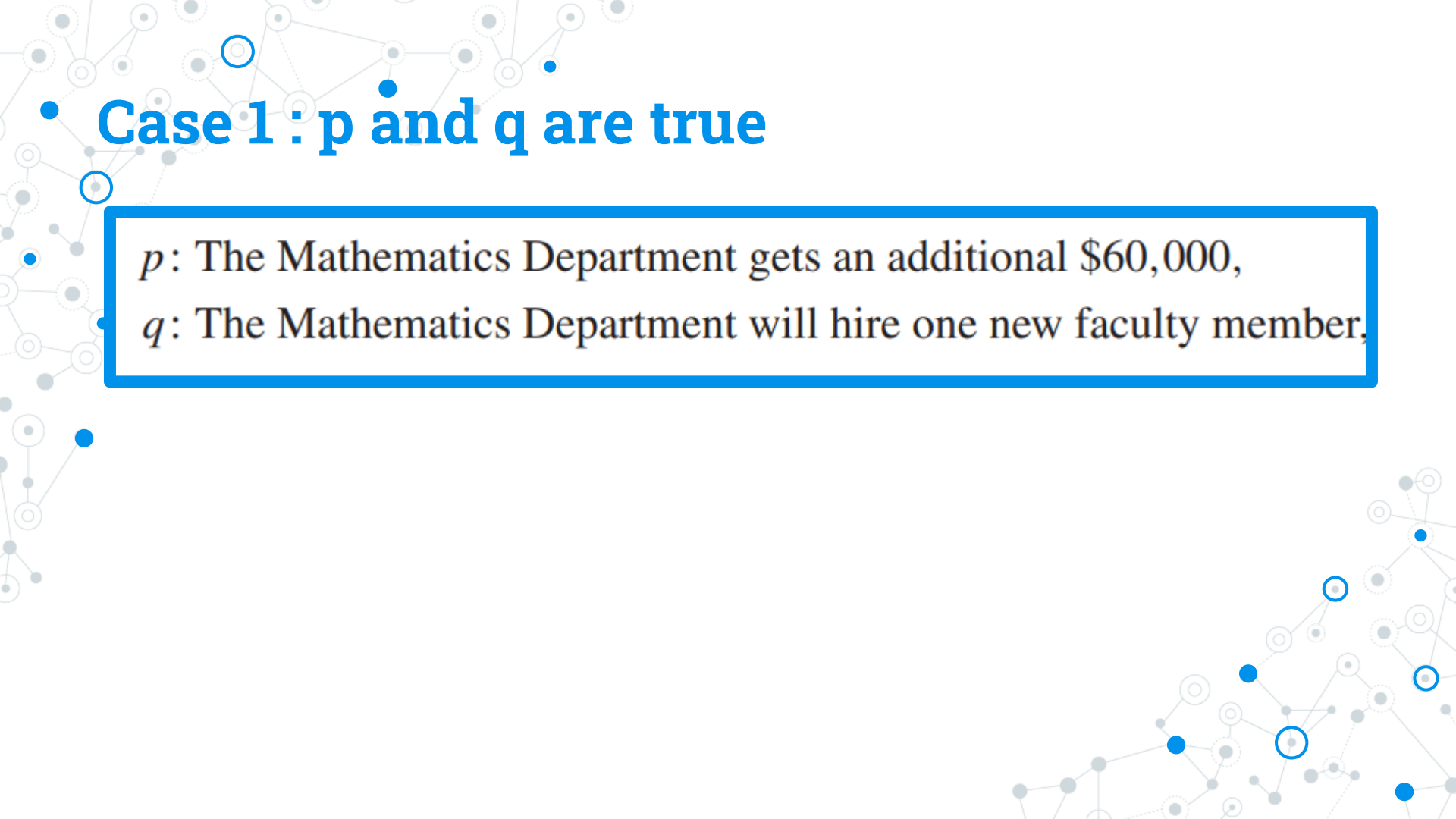
# What is the conclusion ?

If the Mathematics Department gets an additional \$60,000,  
then it will hire one new faculty member.

$p$ : The Mathematics Department gets an additional \$60,000,

$q$ : The Mathematics Department will hire one new faculty member.



A decorative background featuring a network graph with nodes and edges. Some nodes are highlighted with blue circles, and some edges are solid blue lines, while others are gray dashed lines.

## • Case 1 : $p$ and $q$ are true

$p$ : The Mathematics Department gets an additional \$60,000,

$q$ : The Mathematics Department will hire one new faculty member,

## • Case 1 : $p$ and $q$ are true

$p$ : The Mathematics Department gets an additional \$60,000,

$q$ : The Mathematics Department will hire one new faculty member,

If the Mathematics Department gets an additional \$60,000,  
then it will hire one new faculty member.

Dean's Statement is true



- **Case 2 :  $p$  is true and  $q$  is false**

$p$ : The Mathematics Department gets an additional \$60,000,

×  $q$ : The Mathematics Department will hire one new faculty member.

## • Case 2 : $p$ is true and $q$ is false

$p$ : The Mathematics Department gets an additional \$60,000,

×  $q$ : The Mathematics Department will hire one new faculty member.

If the Mathematics Department gets an additional \$60,000,  
then it will hire one new faculty member.

Dean's Statement is false



- **Case 3 :  $p$  is false and  $q$  is true**

×  $p$ : The Mathematics Department gets an additional \$60,000,  
 $q$ : The Mathematics Department will hire one new faculty member,

### • Case 3 : $p$ is false and $q$ is true

×  $p$ : The Mathematics Department gets an additional \$60,000,  
 $q$ : The Mathematics Department will hire one new faculty member,

If the Mathematics Department gets an additional \$60,000,  
then it will hire one new faculty member.

Dean's statement will still be true



- **Case 4 :  $p$  is false and  $q$  is false**

- ×  $p$ : The Mathematics Department gets an additional \$60,000,
- ×  $q$ : The Mathematics Department will hire one new faculty member,

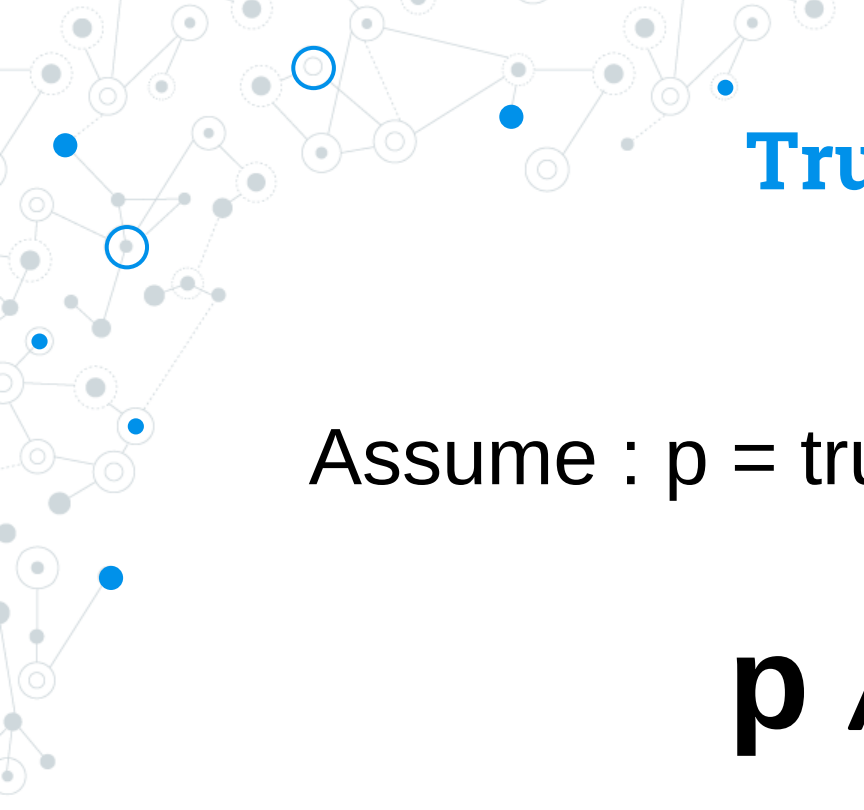
## • Case 4 : $p$ is false and $q$ is false

- ×  $p$ : The Mathematics Department gets an additional \$60,000,
- ×  $q$ : The Mathematics Department will hire one new faculty member,

If the Mathematics Department gets an additional \$60,000,  
then it will hire one new faculty member.

Dean's statement will still be true

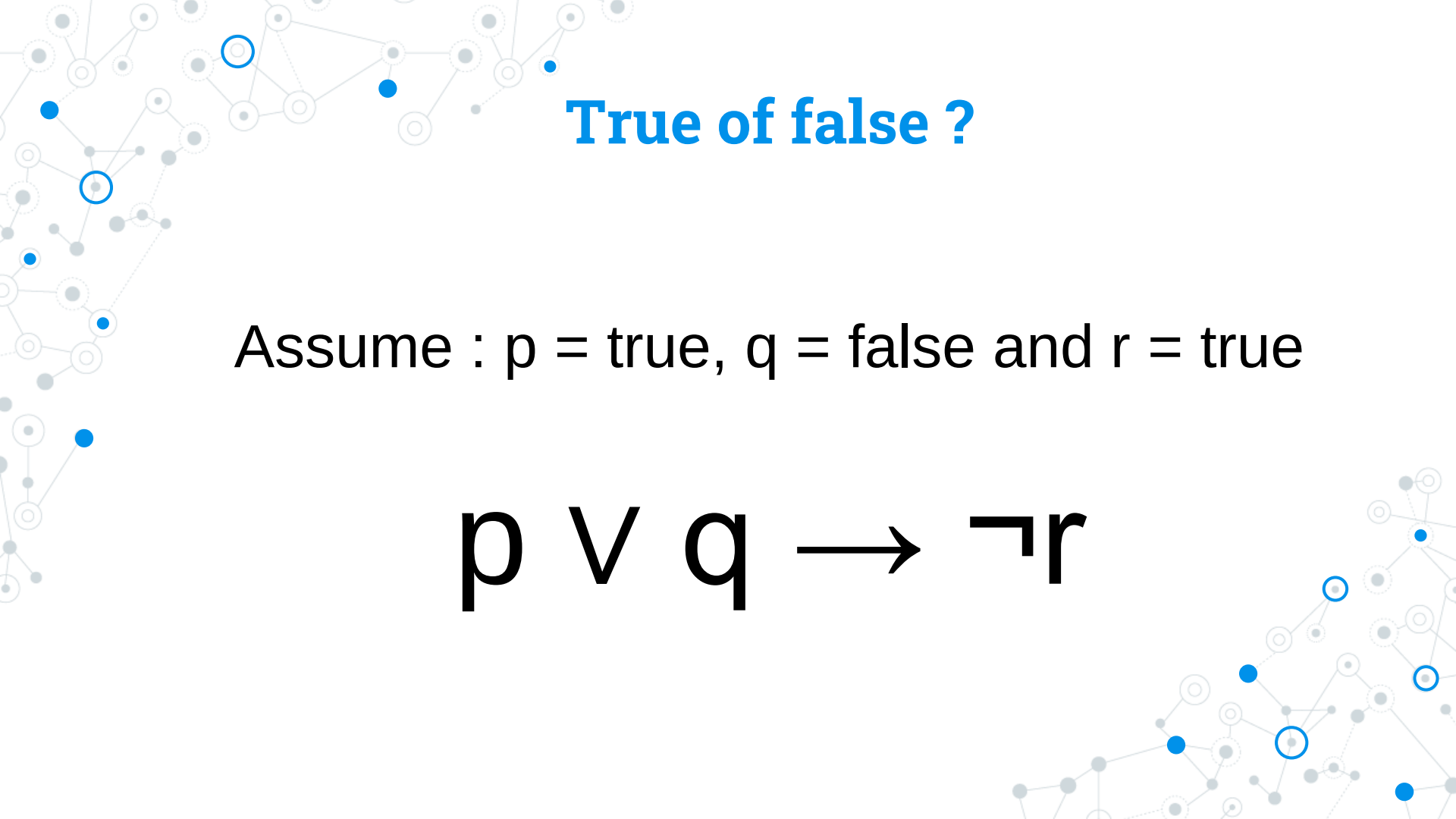


A decorative network graph in the top-left corner, featuring a complex web of interconnected nodes. Some nodes are highlighted with blue circles, and others with blue dots. The lines connecting the nodes are thin and grey.

**True or false ?**

Assume :  $p = \text{true}$ ,  $q = \text{false}$  and  $r = \text{true}$

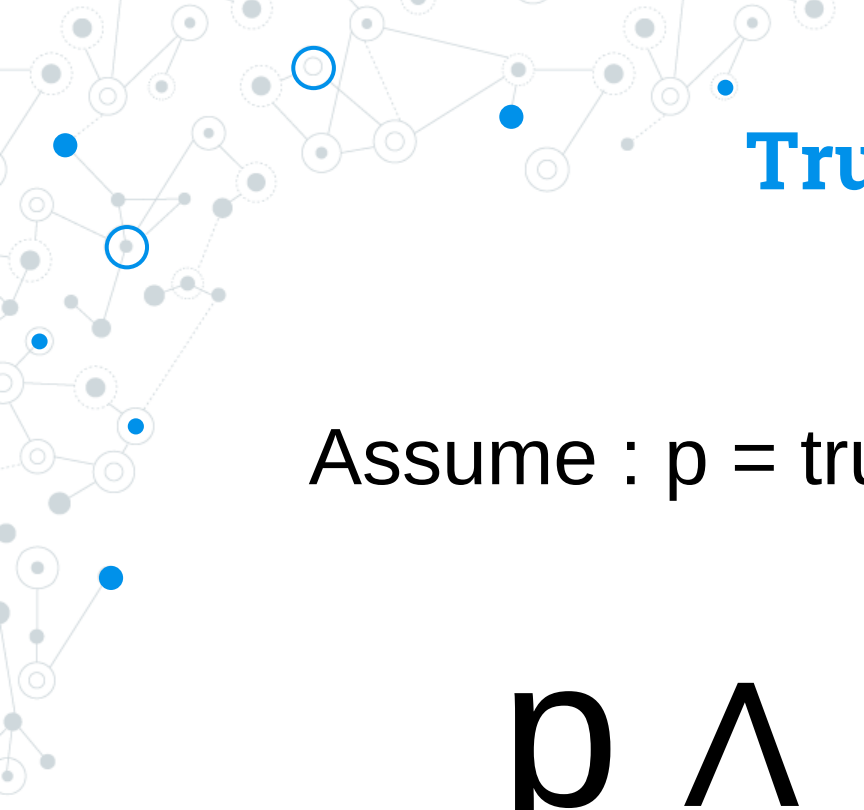
$$p \wedge q \rightarrow r$$
A decorative network graph in the bottom-right corner, similar to the one in the top-left. It shows a cluster of nodes with some highlighted in blue and others in grey, connected by thin grey lines.



**True or false ?**

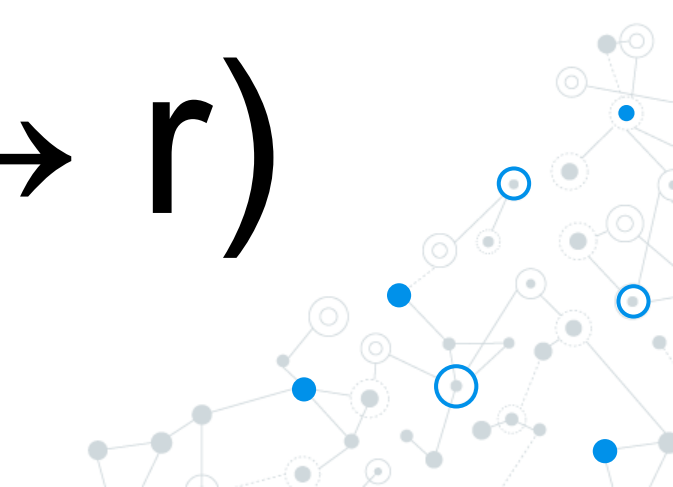
Assume :  $p = \text{true}$ ,  $q = \text{false}$  and  $r = \text{true}$

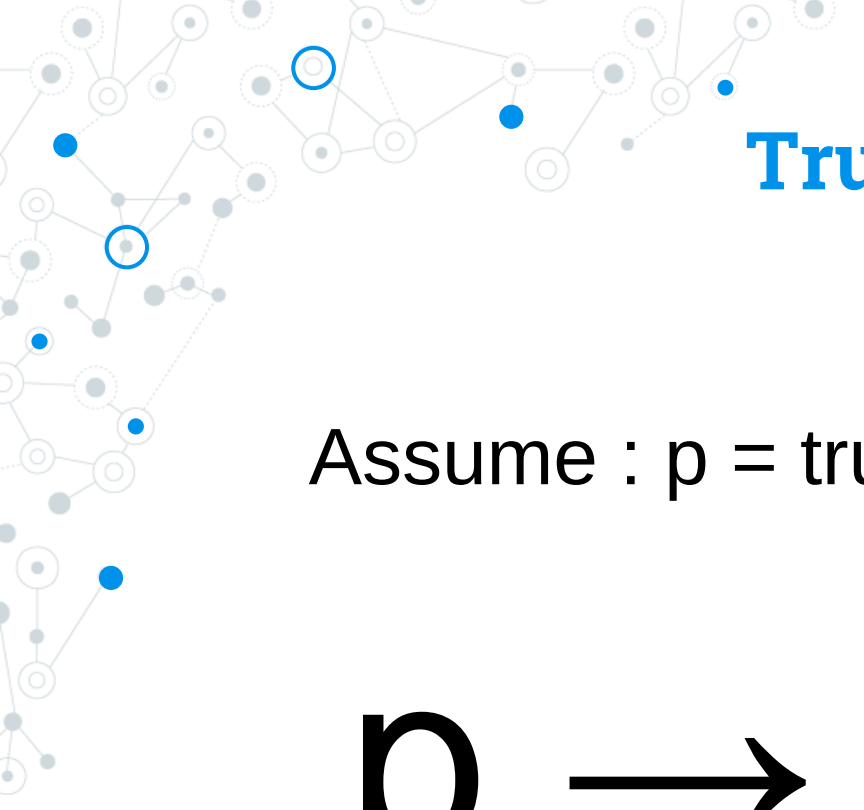
$$p \vee q \rightarrow \neg r$$



**True or false ?**

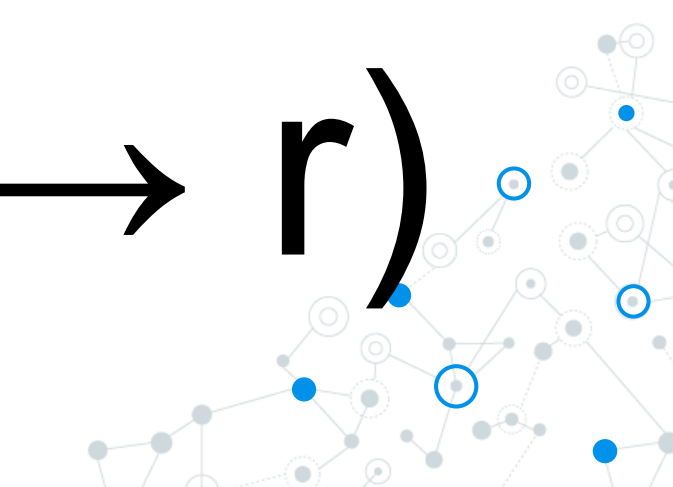
Assume :  $p = \text{true}$ ,  $q = \text{false}$  and  $r = \text{true}$

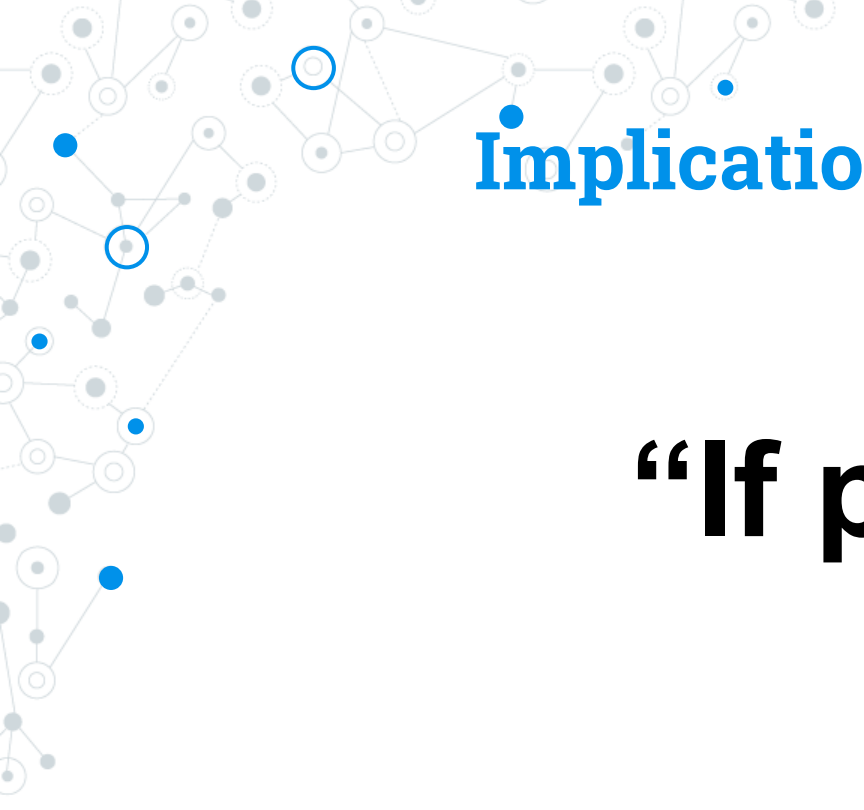
$$p \wedge (q \rightarrow r)$$




**True or false ?**

Assume :  $p = \text{true}$ ,  $q = \text{false}$  and  $r = \text{true}$

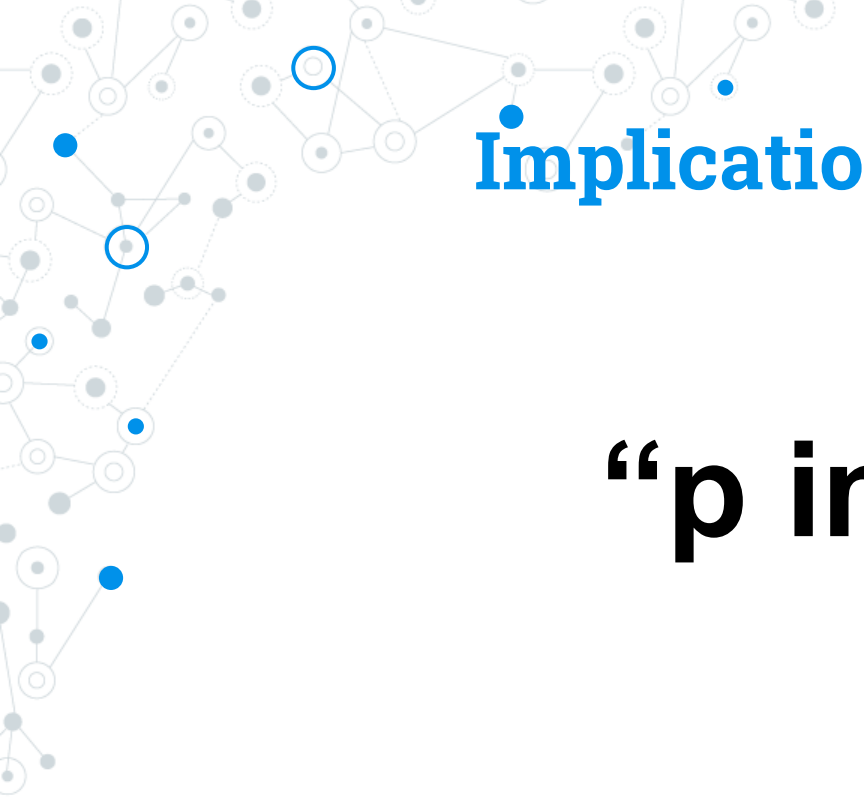
$$p \rightarrow (q \rightarrow r)$$


A decorative network graph in the top-left corner, featuring a complex web of interconnected nodes. Some nodes are highlighted with blue circles, and others with blue dots. The lines connecting the nodes are thin and grey.

# Implication Representations

**“If  $p$  then  $q$ ”**

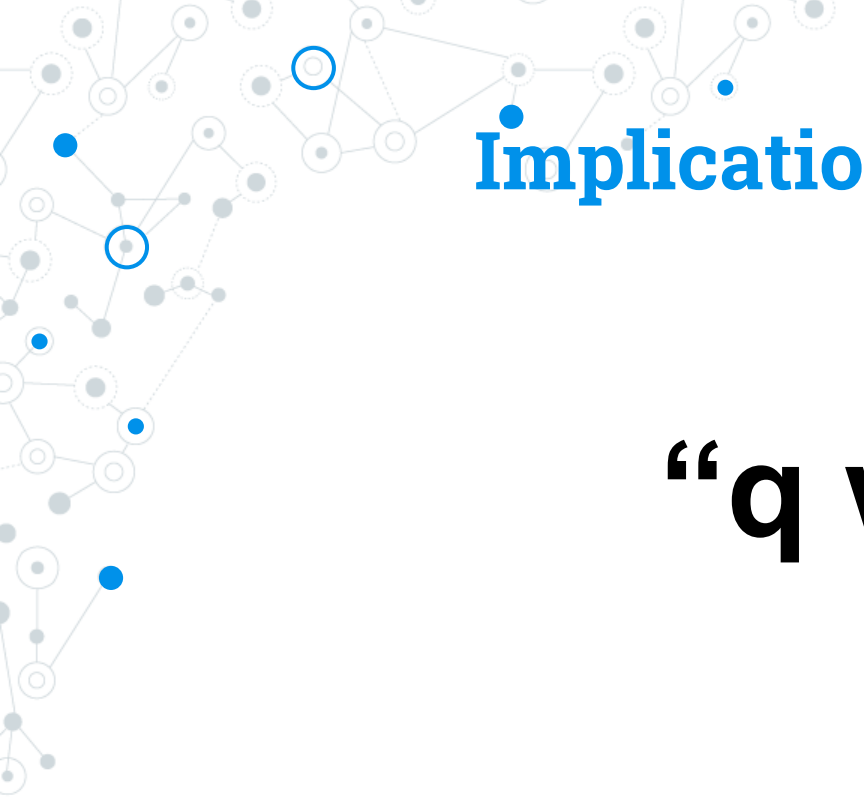
A decorative network graph in the bottom-right corner, similar to the one in the top-left, with a web of interconnected nodes and some highlighted with blue circles and dots.

A decorative network graph in the top-left corner, featuring a complex web of interconnected nodes and edges. Some nodes are highlighted with blue circles, and others with blue dots. The graph is composed of light gray lines and dots.

# Implication Representations

**“p implies q”**

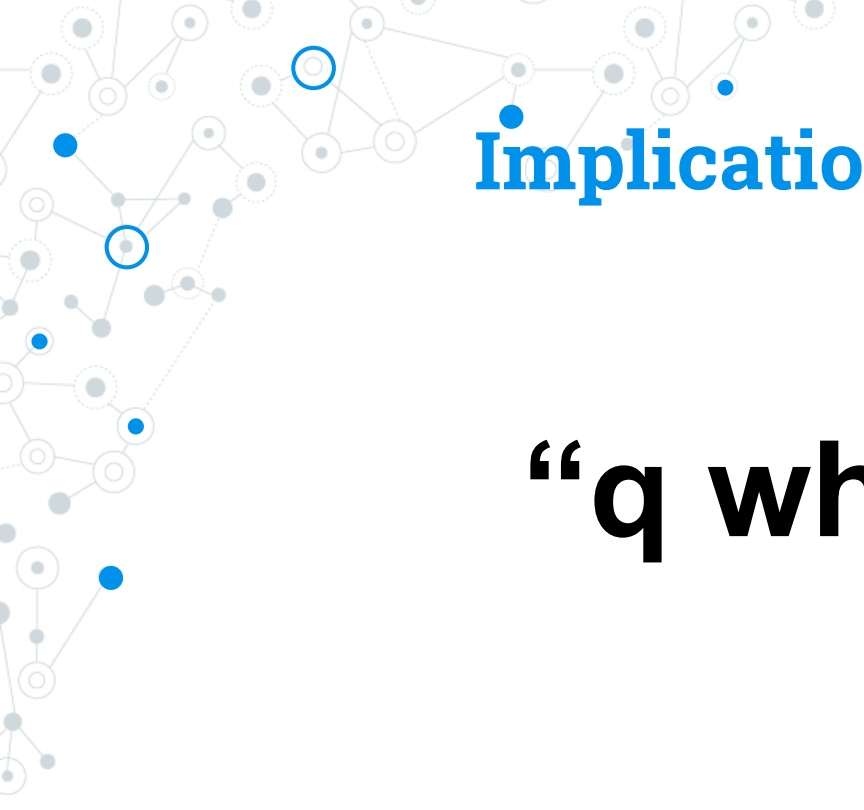
A decorative network graph in the bottom-right corner, similar to the one in the top-left, with a complex web of interconnected nodes and edges. Some nodes are highlighted with blue circles, and others with blue dots. The graph is composed of light gray lines and dots.

A decorative network graph in the top-left corner, featuring a complex web of interconnected nodes. Some nodes are solid blue circles, while others are white circles with blue outlines. The nodes are connected by thin, light gray lines, creating a dense, branching structure.

# Implication Representations

**“q when p”**

A decorative network graph in the bottom-right corner, similar to the one in the top-left. It consists of a network of nodes connected by lines. Some nodes are solid blue circles, and others are white circles with blue outlines, all set against a light gray background.

A decorative network graph pattern in the top-left corner, featuring a complex web of interconnected nodes and edges. Some nodes are highlighted with blue circles, and others with blue dots. The edges are thin gray lines, some solid and some dashed.

# Implication Representations

**“q whenever p”**

A decorative network graph pattern in the bottom-right corner, similar to the one in the top-left, with interconnected nodes and edges, some highlighted with blue circles and dots.

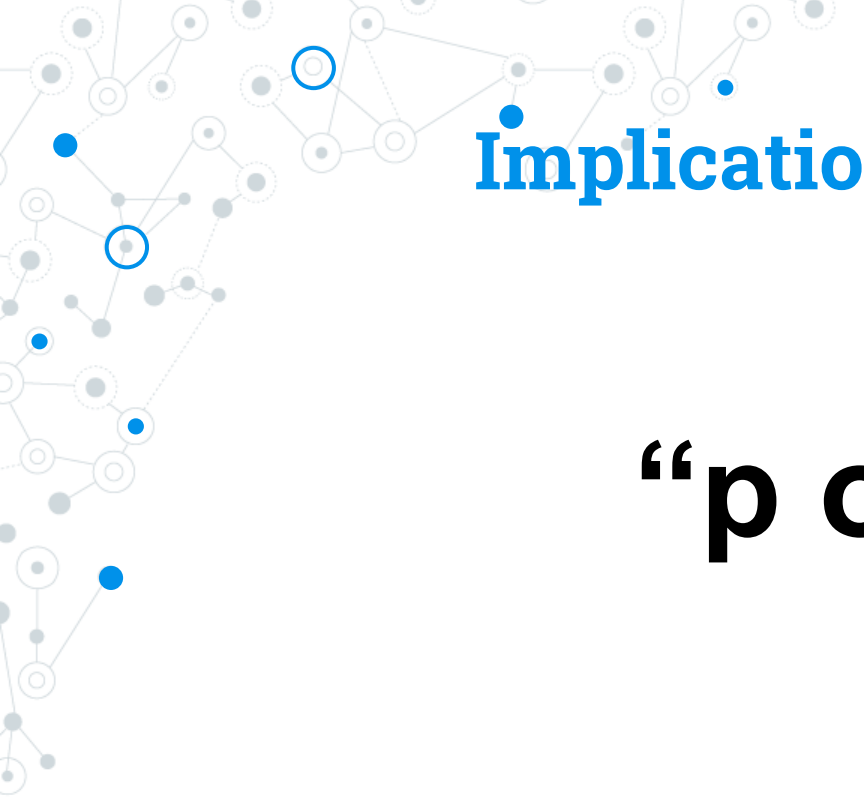


A decorative network graph in the top-left corner, featuring a complex web of interconnected nodes. Some nodes are highlighted with blue circles, and others with blue dots. The lines connecting the nodes are thin and grey.

# Implication Representations

**“q follows from p”**

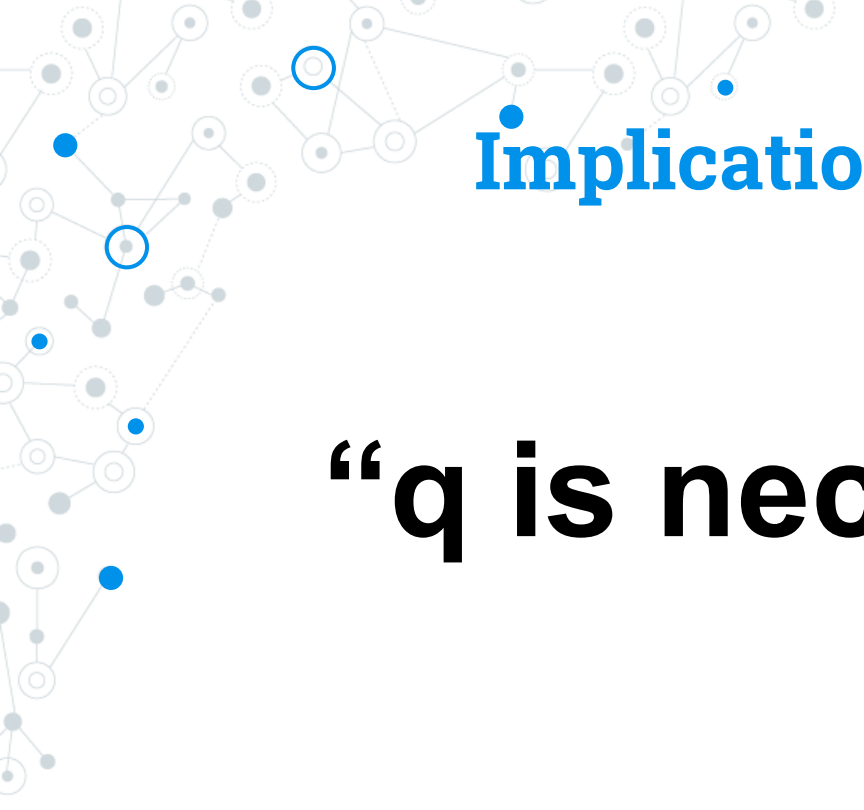
A decorative network graph in the bottom-right corner, similar to the one in the top-left, with interconnected nodes and some highlighted with blue circles and dots.

A decorative network graph in the top-left corner, featuring a complex web of interconnected nodes and edges. Some nodes are highlighted with blue circles, and others with blue dots. The graph is composed of solid and dashed lines.

# Implication Representations

**“p only if q”**

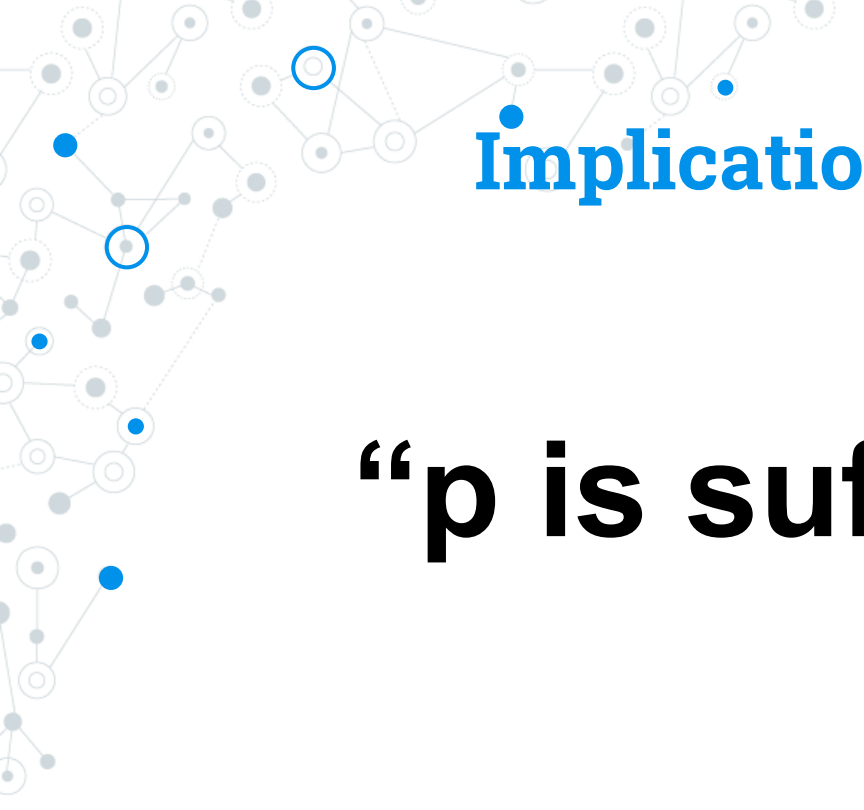
A decorative network graph in the bottom-right corner, similar to the one in the top-left, with interconnected nodes and edges. Some nodes are highlighted with blue circles, and others with blue dots. The graph is composed of solid and dashed lines.

A decorative network graph in the top-left corner, featuring a complex web of interconnected nodes. Some nodes are highlighted with blue circles, and others with blue dots. The lines connecting the nodes are thin and grey.

## Implication Representations

**“q is necessary for p”**

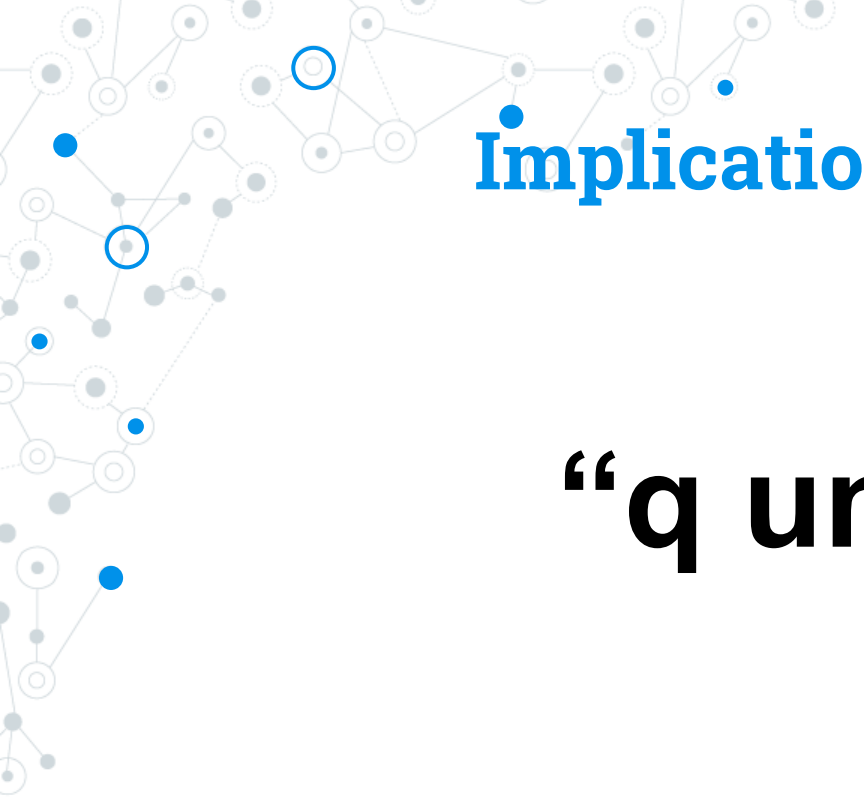
A decorative network graph in the bottom-right corner, similar to the one in the top-left, with interconnected nodes and some highlighted with blue circles and dots.

A decorative network graph in the top-left corner, featuring a complex web of interconnected nodes. Some nodes are highlighted with blue circles, and others with blue dots. The lines connecting the nodes are thin and grey.

# Implication Representations

**“p is sufficient for q”**

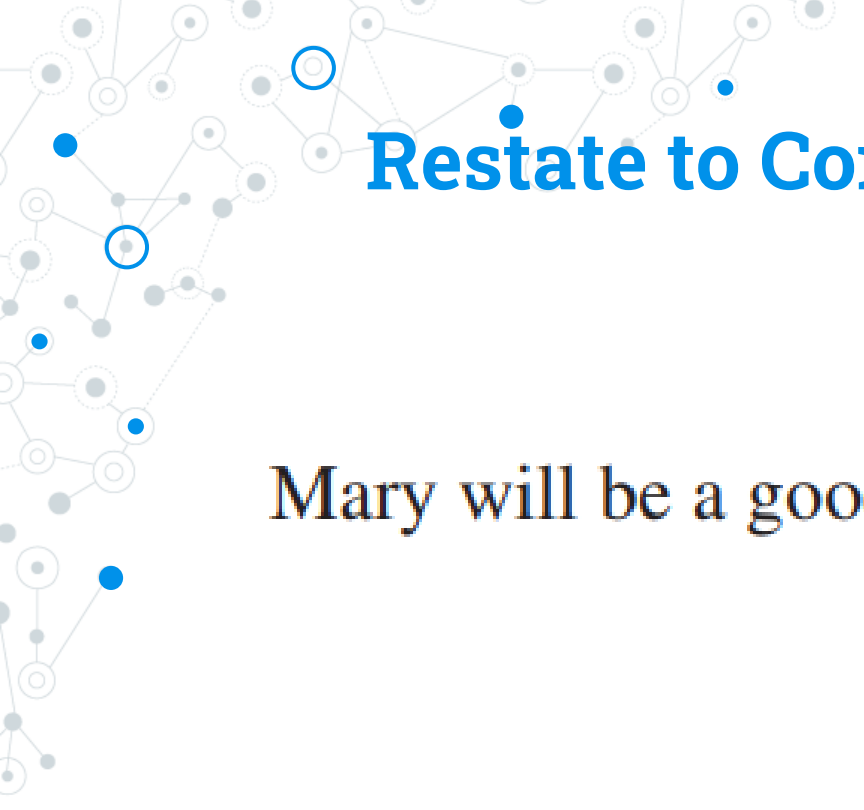
A decorative network graph in the bottom-right corner, similar to the one in the top-left, with interconnected nodes and some highlighted with blue circles and dots.

A decorative network graph in the top-left corner, featuring a complex web of interconnected nodes. Some nodes are solid blue circles, while others are white circles with blue outlines. The nodes are connected by thin, light gray lines, creating a dense, organic structure.

# Implication Representations

**“q unless  $\neg$  p”**

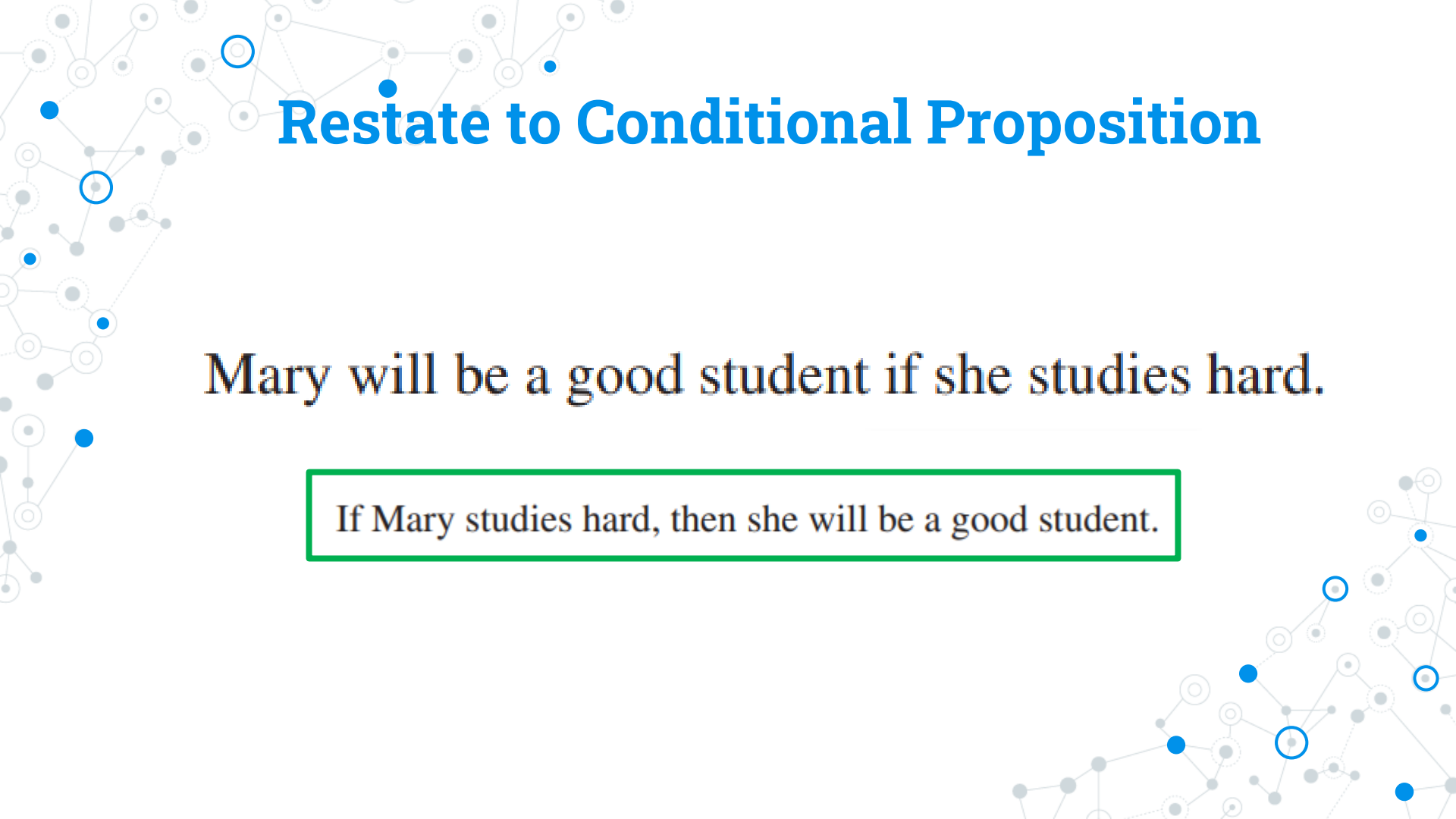
A decorative network graph in the bottom-right corner, similar to the one in the top-left. It consists of a cluster of nodes connected by lines. Some nodes are solid blue, and others are white with blue outlines. The overall shape is more triangular and less spread out than the top-left graph.

A decorative network graph pattern in the top-left corner, consisting of interconnected nodes and lines. Some nodes are solid blue circles, while others are hollow blue circles. The lines are thin and gray.

# Restate to Conditional Proposition

Mary will be a good student if she studies hard.

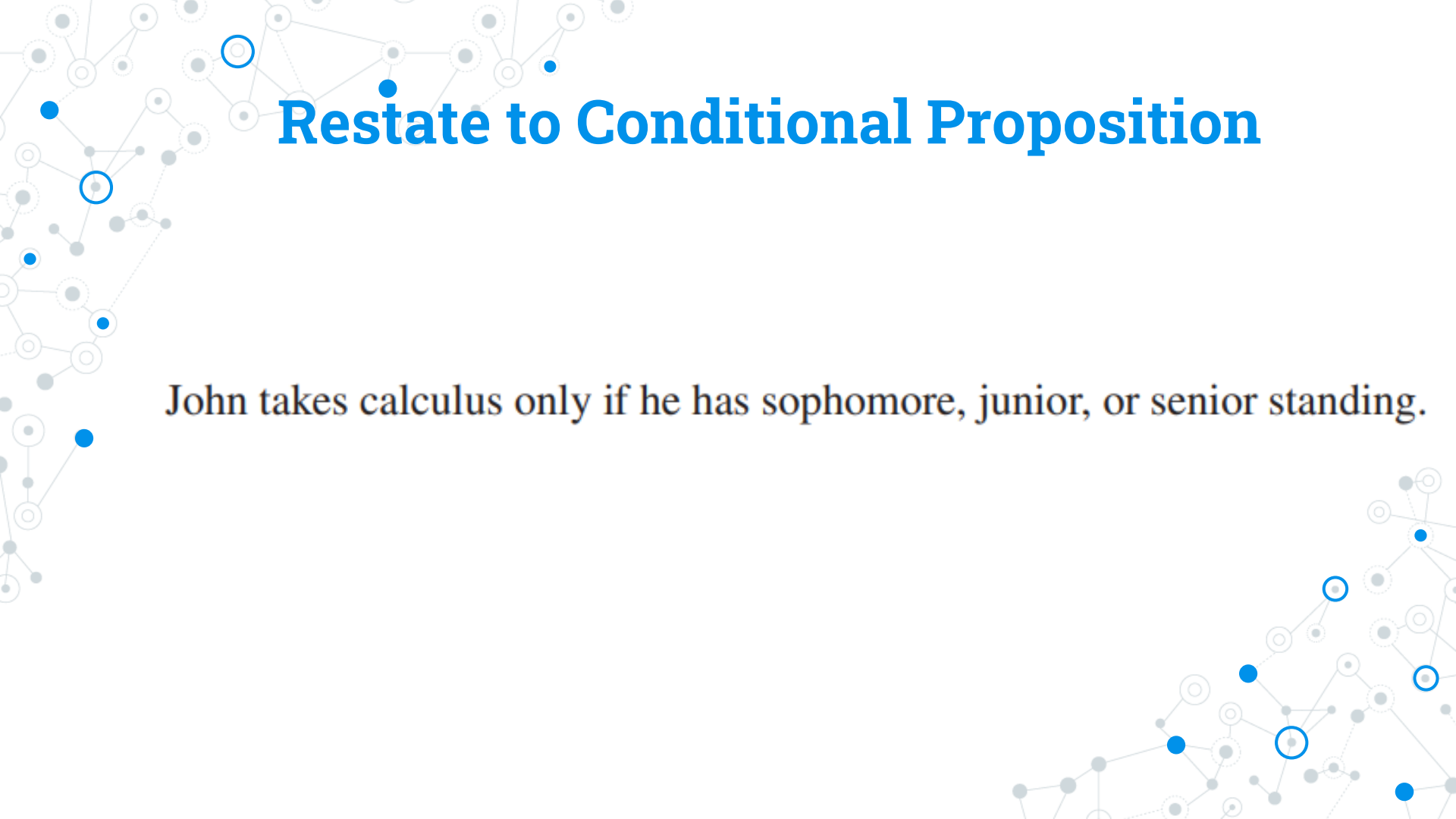
A decorative network graph pattern in the bottom-right corner, consisting of interconnected nodes and lines. Some nodes are solid blue circles, while others are hollow blue circles. The lines are thin and gray.



# Restate to Conditional Proposition

Mary will be a good student if she studies hard.

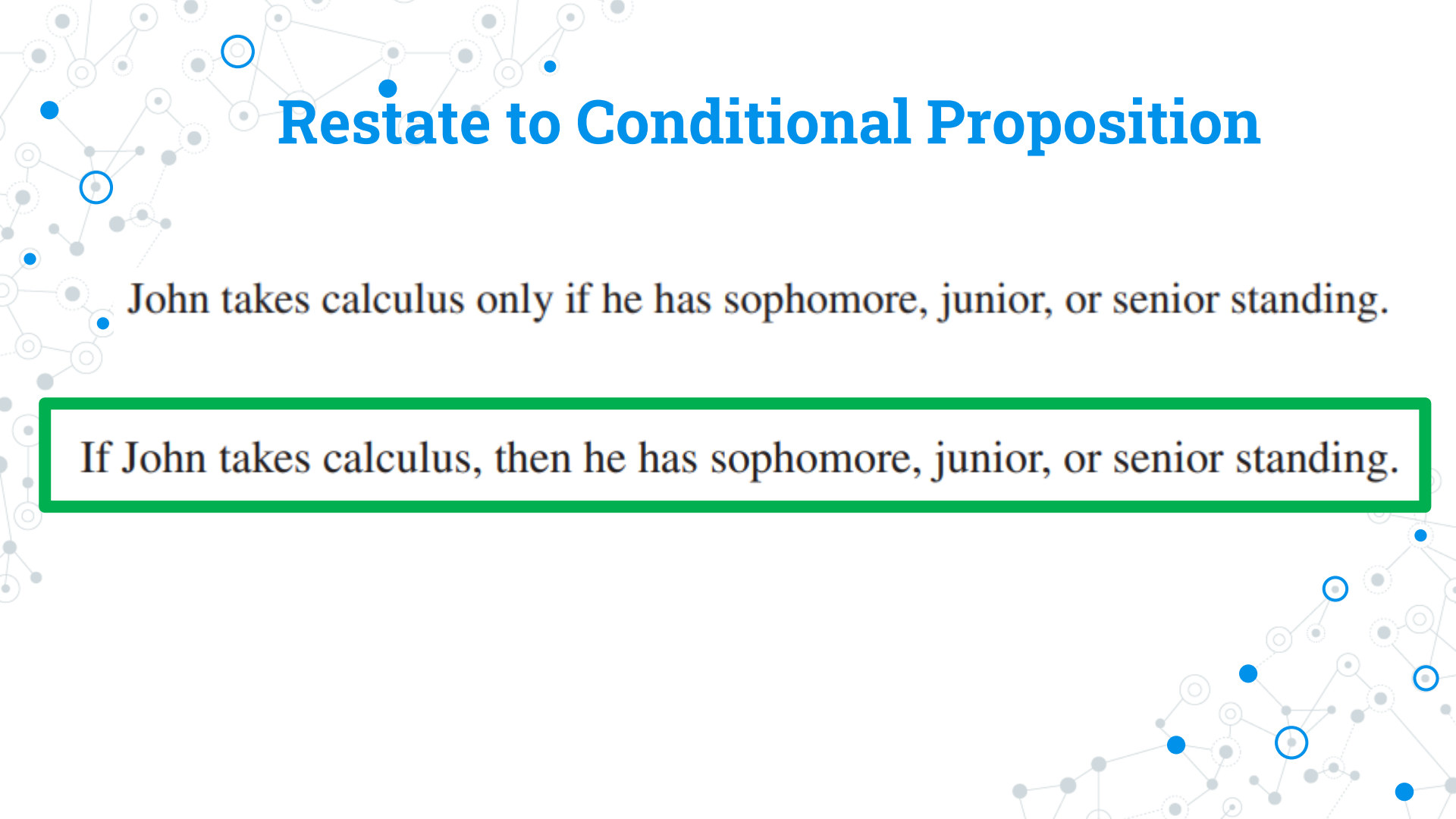
If Mary studies hard, then she will be a good student.

A decorative background graphic consisting of a network of nodes and connections. The nodes are represented by small circles, some of which are solid blue, some are hollow blue, and others are solid grey. They are interconnected by thin grey lines, forming a complex web-like structure that is more dense on the left and right sides of the slide.

# Restate to Conditional Proposition

John takes calculus only if he has sophomore, junior, or senior standing.

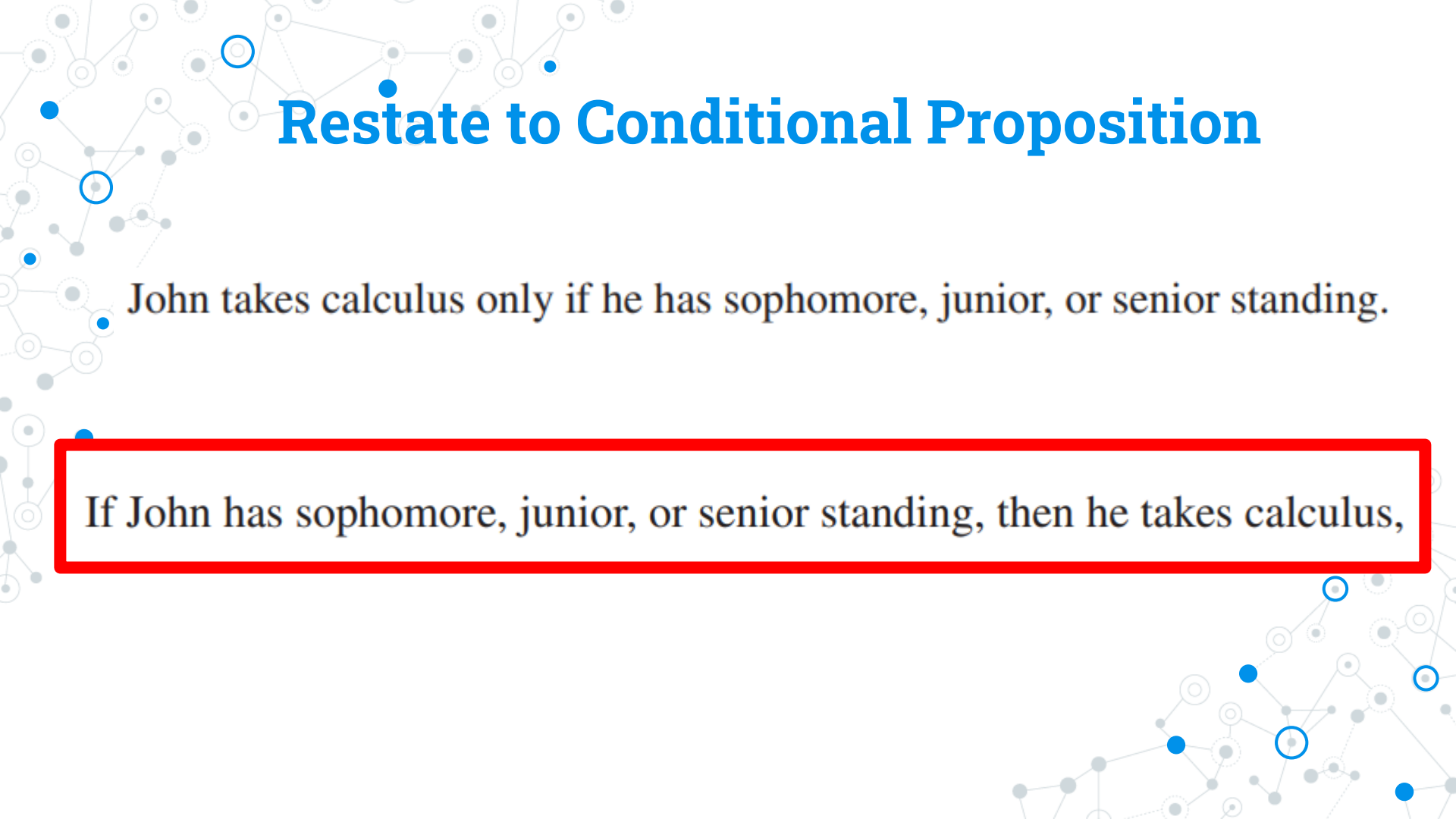




# Restate to Conditional Proposition

John takes calculus only if he has sophomore, junior, or senior standing.

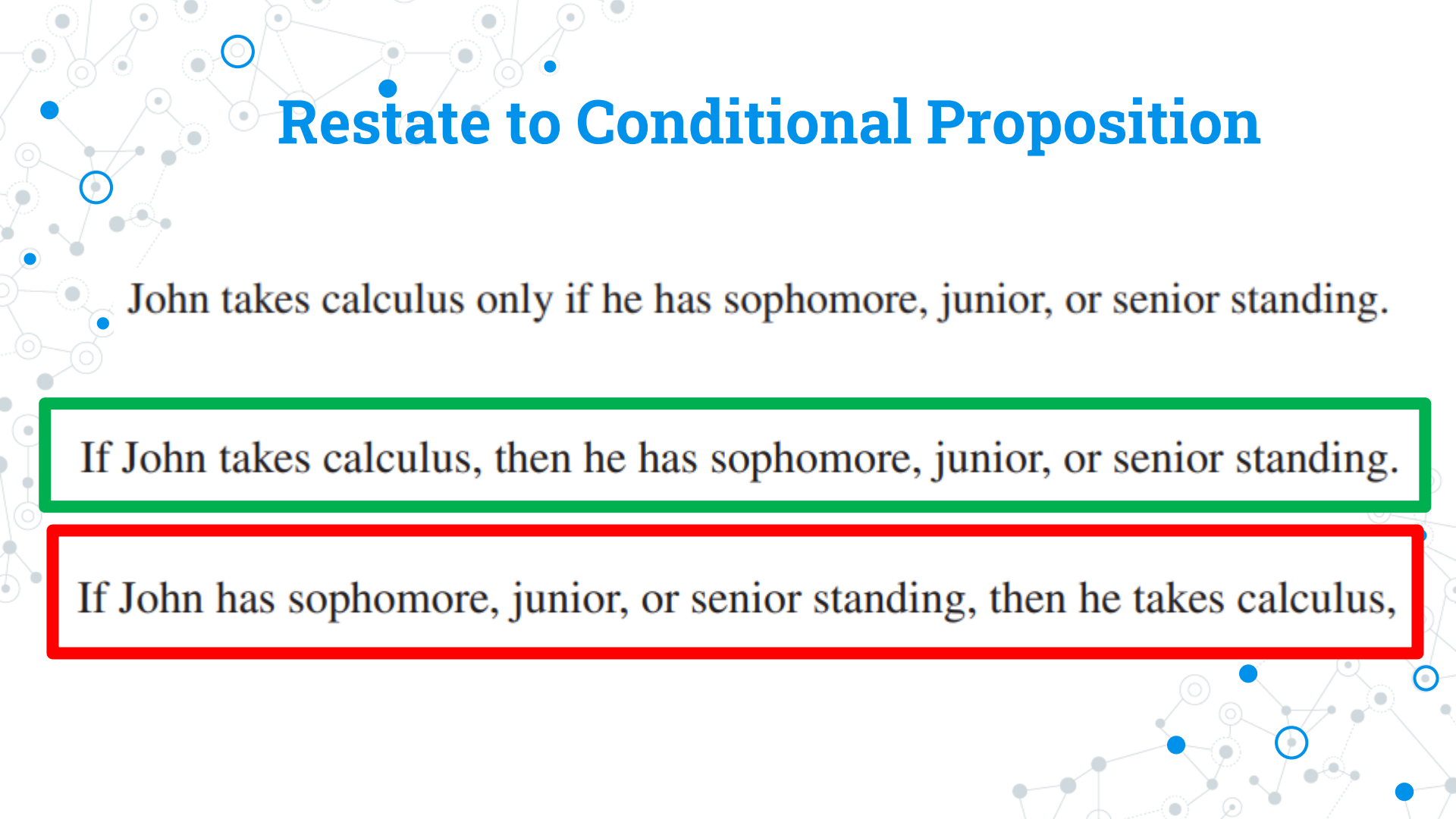
If John takes calculus, then he has sophomore, junior, or senior standing.



# Restate to Conditional Proposition

John takes calculus only if he has sophomore, junior, or senior standing.

If John has sophomore, junior, or senior standing, then he takes calculus,

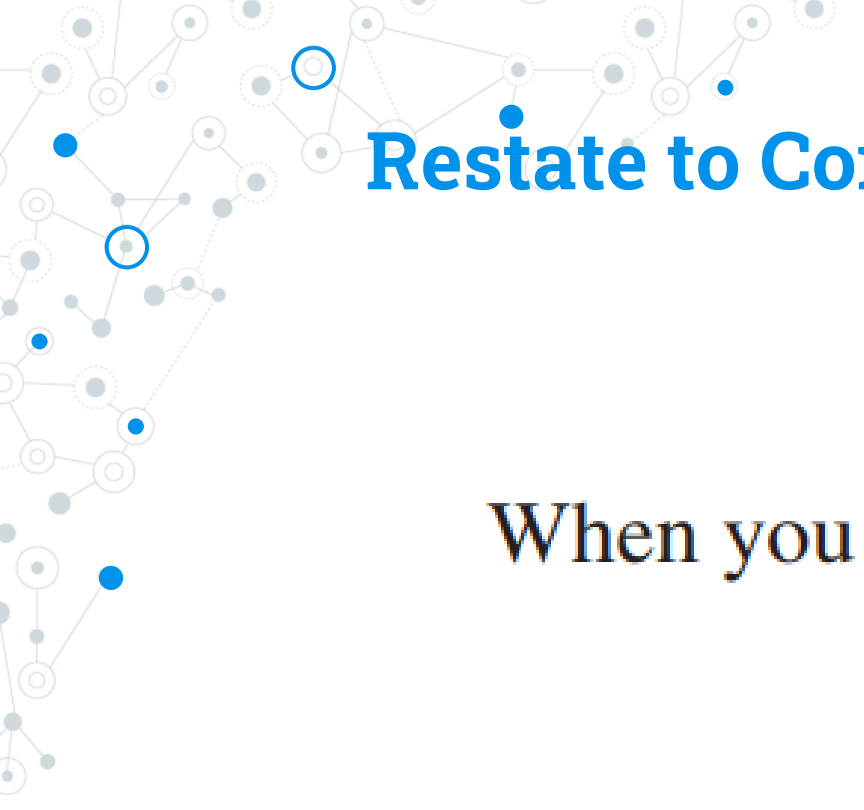


# Restate to Conditional Proposition

John takes calculus only if he has sophomore, junior, or senior standing.

If John takes calculus, then he has sophomore, junior, or senior standing.

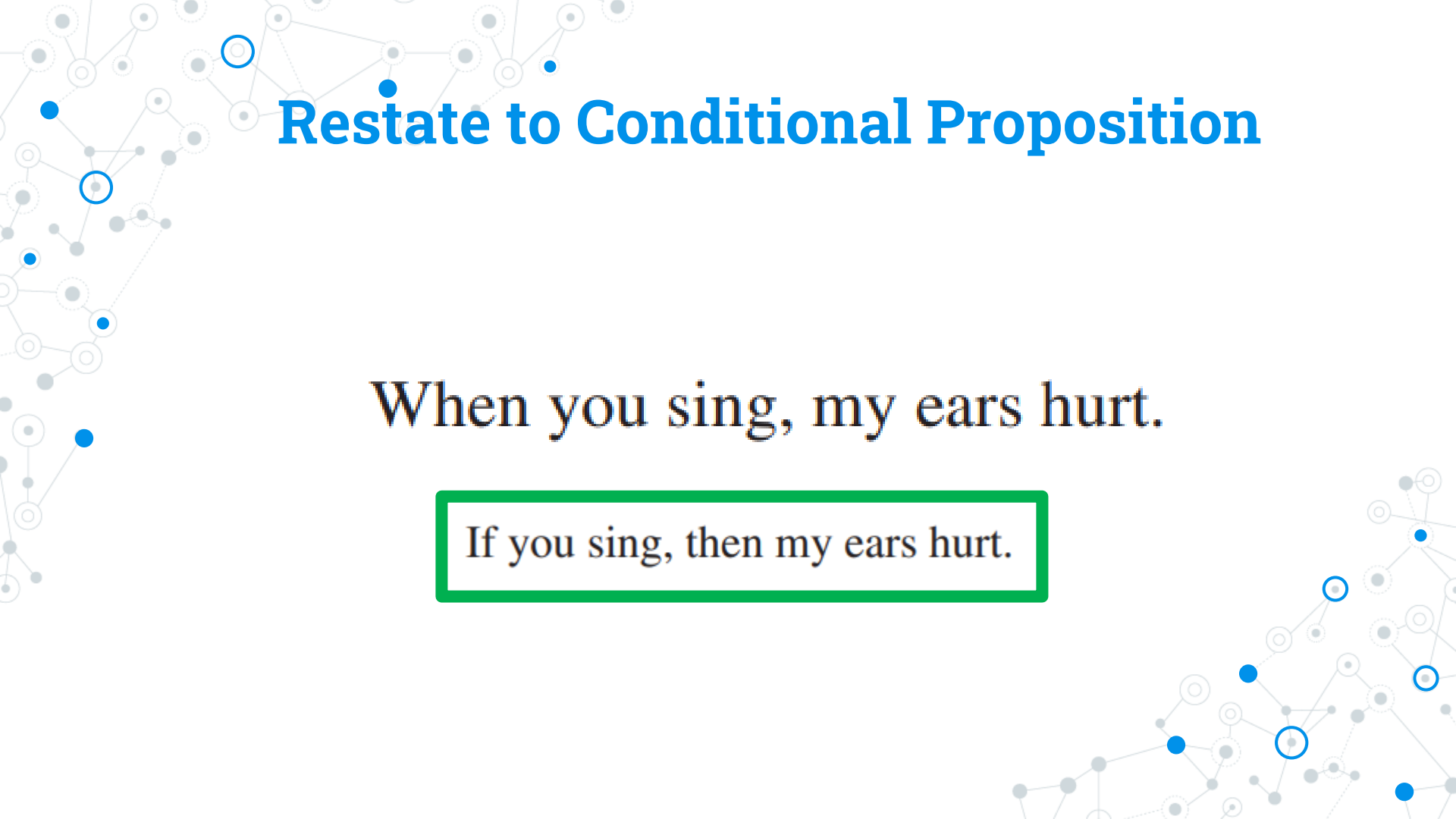
If John has sophomore, junior, or senior standing, then he takes calculus,

A decorative network graph pattern in the top-left corner, consisting of interconnected nodes (some solid blue, some hollow blue, some solid grey) and edges (solid and dashed grey lines).

# Restate to Conditional Proposition

When you sing, my ears hurt.

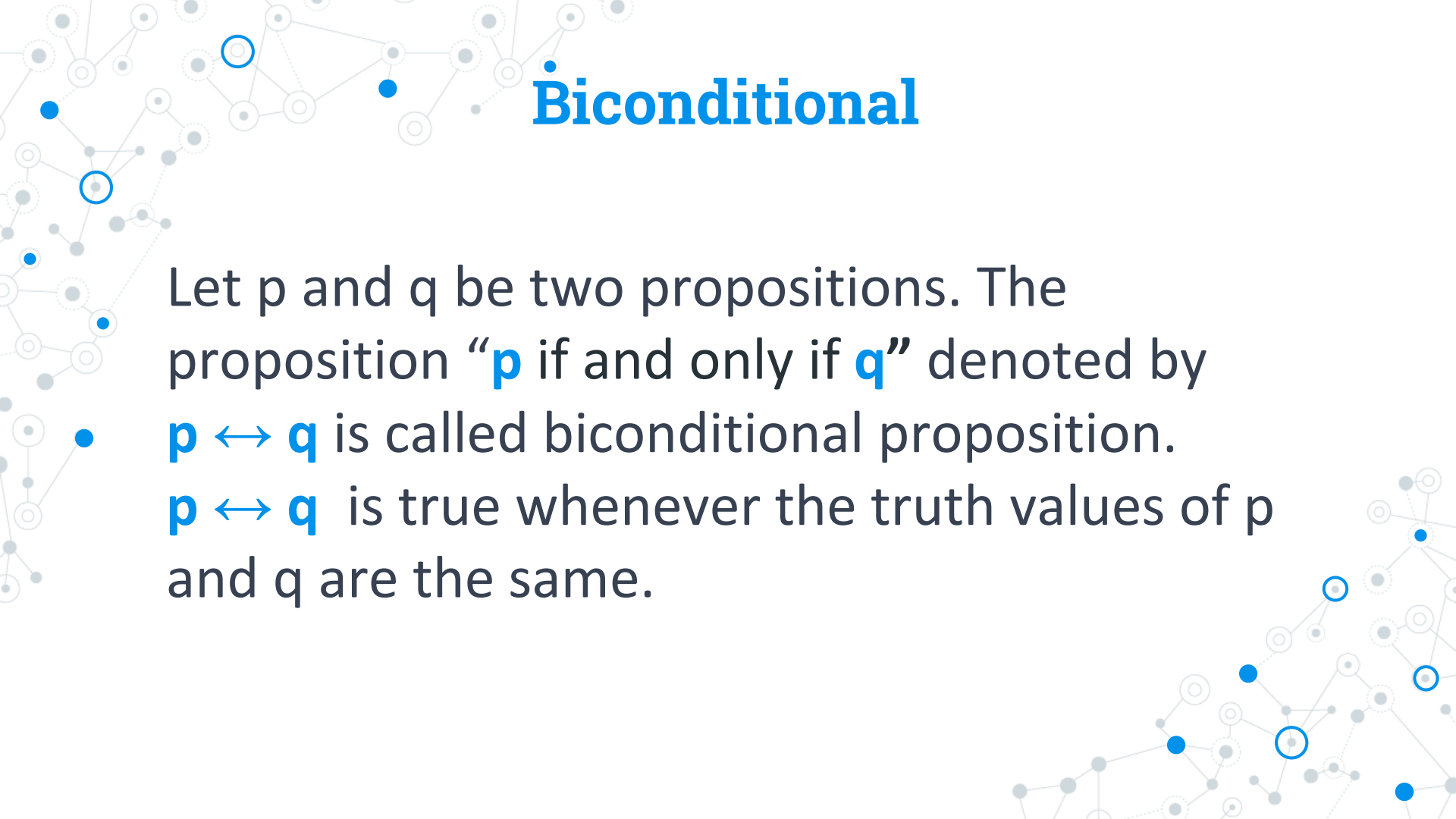
A decorative network graph pattern in the bottom-right corner, consisting of interconnected nodes (some solid blue, some hollow blue, some solid grey) and edges (solid and dashed grey lines).



# Restate to Conditional Proposition

When you sing, my ears hurt.

If you sing, then my ears hurt.



# Biconditional

Let  $p$  and  $q$  be two propositions. The proposition “ $p$  if and only if  $q$ ” denoted by

$p \leftrightarrow q$  is called biconditional proposition.

$p \leftrightarrow q$  is true whenever the truth values of  $p$  and  $q$  are the same.

# Biconditional : Truth Table

| $p$ | $q$ | $p \leftrightarrow q$ |
|-----|-----|-----------------------|
| T   | T   | T                     |
| T   | F   | F                     |
| F   | T   | F                     |
| F   | F   | T                     |



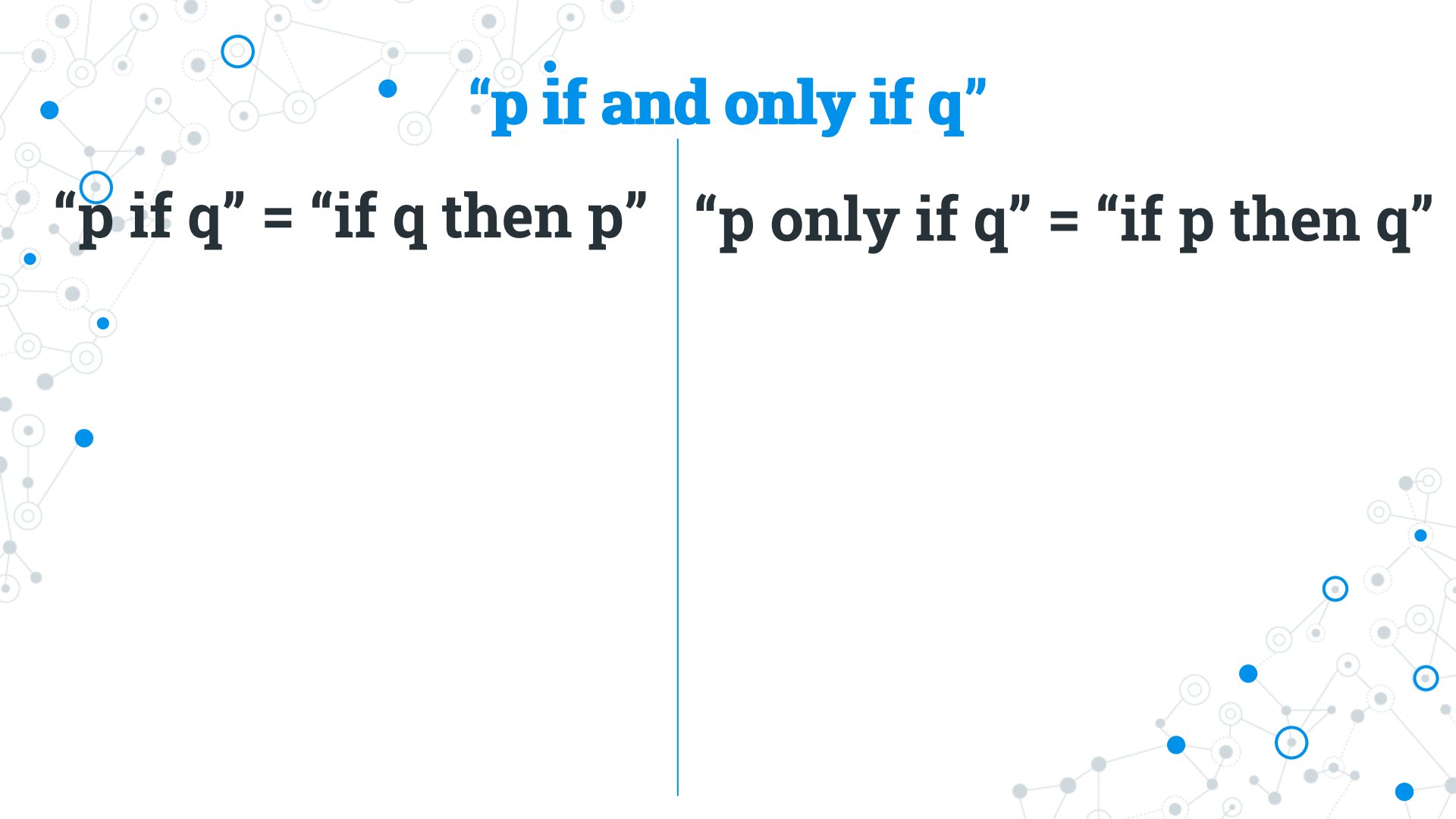
**“p if and only if q”**

**“p if q”**

**“p only if q”**



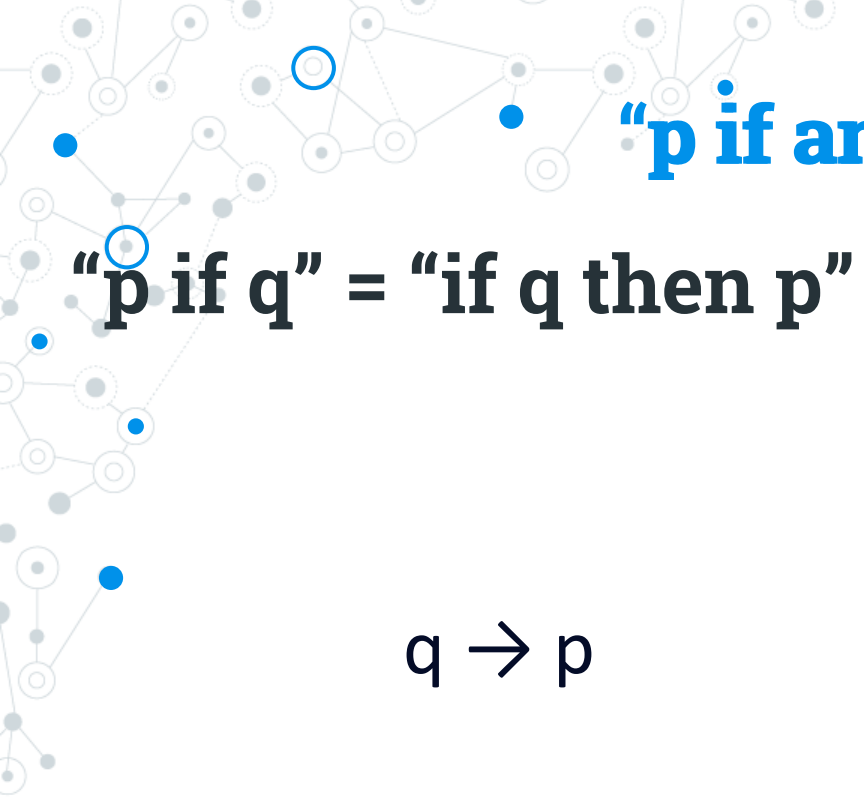




**“p if and only if q”**

**“p if q” = “if q then p”**

**“p only if q” = “if p then q”**



**“p if and only if q”**

**“p if q” = “if q then p”**

$q \rightarrow p$

**“p only if q” = “if p then q”**

$p \rightarrow q$



**“p if and only if q”**

**“p if q” = “if q then p”**

**“p only if q” = “if p then q”**

$$q \rightarrow p$$

$$p \rightarrow q$$

$$(q \rightarrow p) \wedge (p \rightarrow q)$$

**“p if and only if q”**

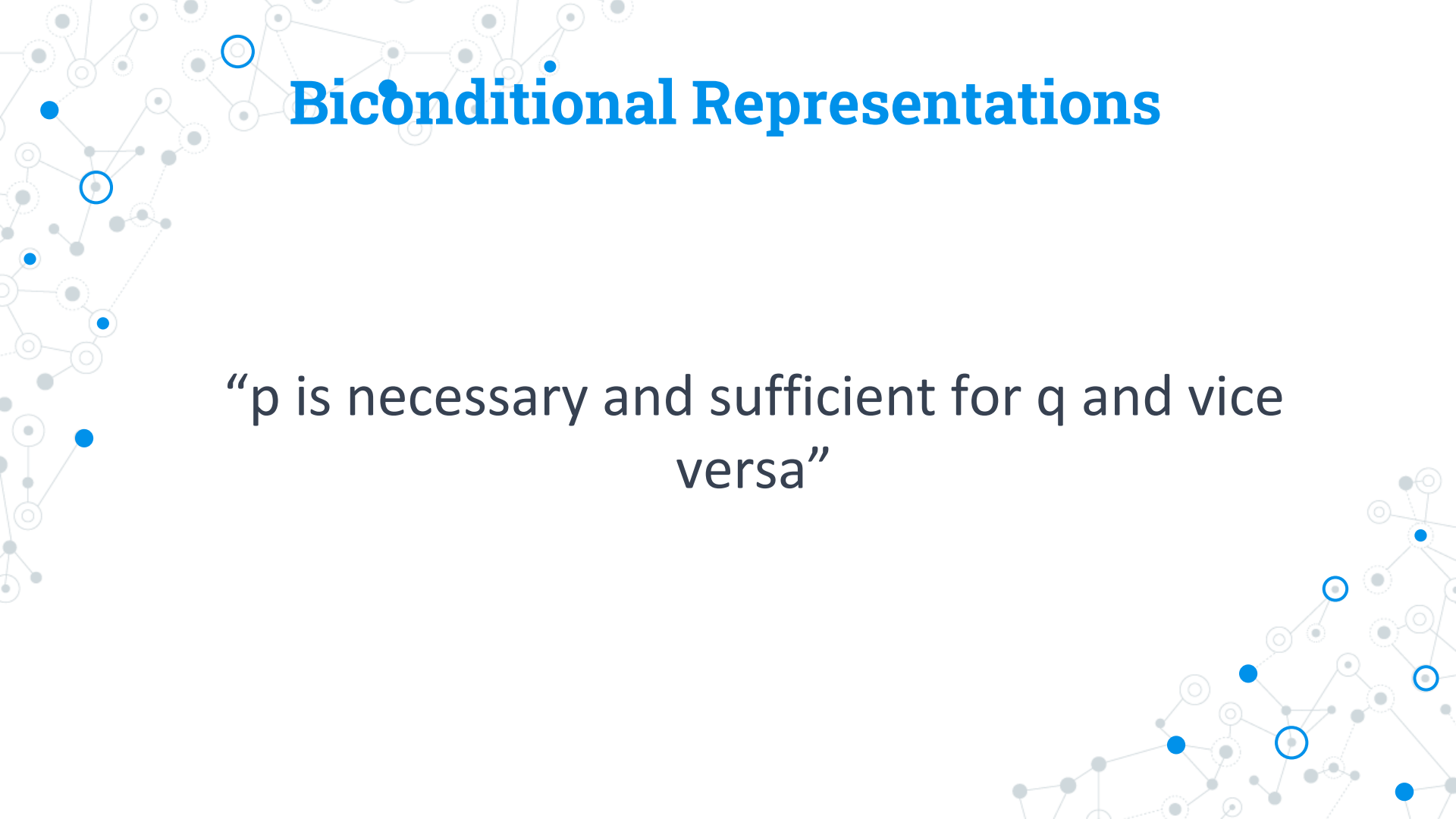
**“p if q” = “if q then p”**

**“p only if q” = “if p then q”**

$$q \rightarrow p$$

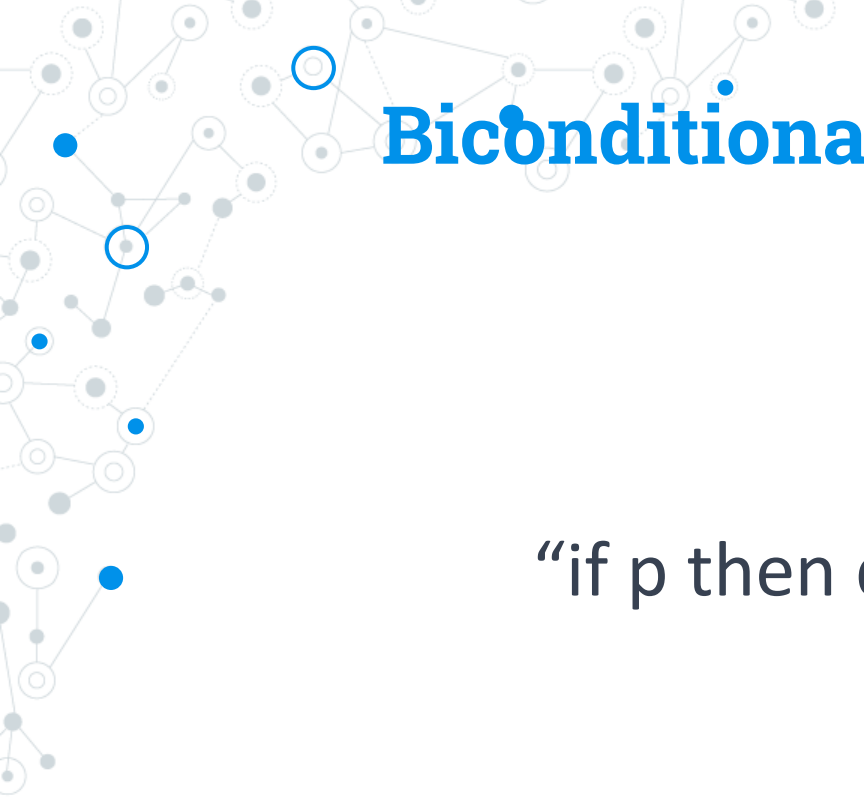
$$p \rightarrow q$$

$$(q \rightarrow p) \wedge (p \rightarrow q) \equiv p \leftrightarrow q$$



# Biconditional Representations

“p is necessary and sufficient for q and vice versa”

A decorative network graph in the top-left corner, featuring a complex web of nodes and edges. Some nodes are highlighted with blue circles, and others with blue dots. The graph is composed of various shapes, including circles and squares, connected by thin lines.

# Biconditional Representations

“if  $p$  then  $q$ , and conversely”

A decorative network graph in the bottom-right corner, similar to the one in the top-left. It features a complex web of nodes and edges, with some nodes highlighted by blue circles and others by blue dots. The graph is composed of various shapes, including circles and squares, connected by thin lines.

A decorative network graph in the top-left corner, featuring a complex web of nodes and edges. Some nodes are highlighted with blue circles, and others with blue dots. The graph is composed of various shapes, including circles and squares, connected by thin lines.

# Biconditional Representations

“ $p \text{ iff } q$ ”

A decorative network graph in the bottom-right corner, similar to the one in the top-left. It features a complex web of nodes and edges, with some nodes highlighted by blue circles and others by blue dots. The graph is composed of various shapes, including circles and squares, connected by thin lines.

# Precedence of Logical Operators

| <i>Operator</i>                    | <i>Precedence</i> |
|------------------------------------|-------------------|
| $\neg$                             | 1                 |
| $\wedge$<br>$\vee$                 | 2<br>3            |
| $\rightarrow$<br>$\leftrightarrow$ | 4<br>5            |



A decorative network graph in the top-left corner, featuring a complex web of interconnected nodes. Some nodes are solid blue circles, while others are white circles with blue outlines. The nodes are connected by thin, light gray lines, creating a dense, organic structure.

**Evaluate :  $p = T$  ,  $q = F$**

$$p \rightarrow q \wedge \neg p$$
A decorative network graph in the bottom-right corner, similar to the one in the top-left. It consists of a cluster of nodes, some solid blue and some white with blue outlines, connected by light gray lines. The structure is more spread out than the one in the top-left.

A decorative network graph in the top-left corner, consisting of a complex web of interconnected nodes and edges. Some nodes are highlighted with blue circles, and some edges are highlighted with blue lines.

Evaluate :  $p = T$  ,  $q = F$

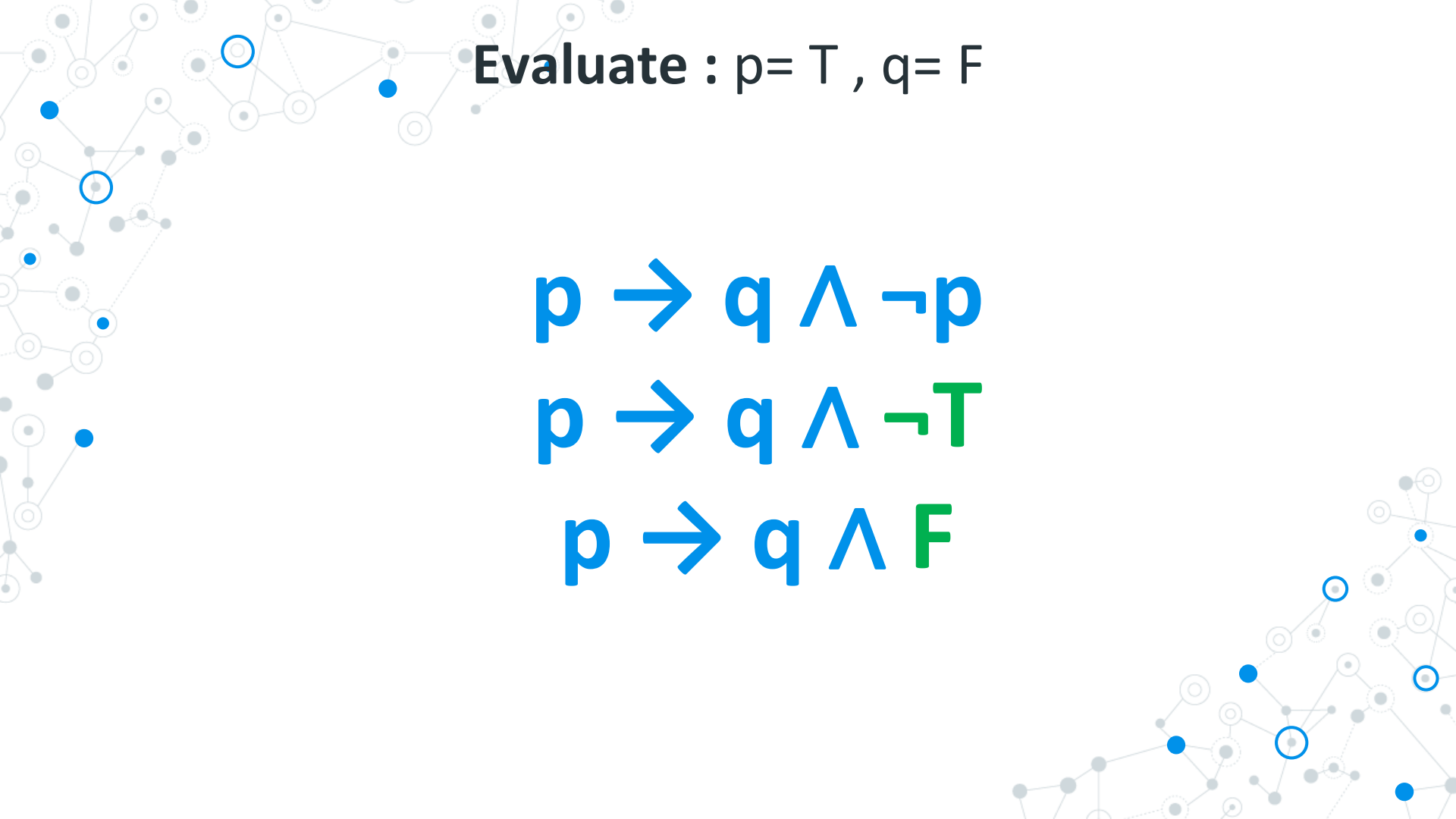
$$p \rightarrow q \wedge \neg p$$
A decorative network graph in the bottom-right corner, consisting of a complex web of interconnected nodes and edges. Some nodes are highlighted with blue circles, and some edges are highlighted with blue lines.

A decorative network graph in the top-left corner, consisting of a complex web of interconnected nodes and edges. Some nodes are highlighted with blue circles or dots.

Evaluate :  $p = T$  ,  $q = F$

$$p \rightarrow q \wedge \neg p$$

$$p \rightarrow q \wedge \neg T$$
A decorative network graph in the bottom-right corner, consisting of a complex web of interconnected nodes and edges. Some nodes are highlighted with blue circles or dots.

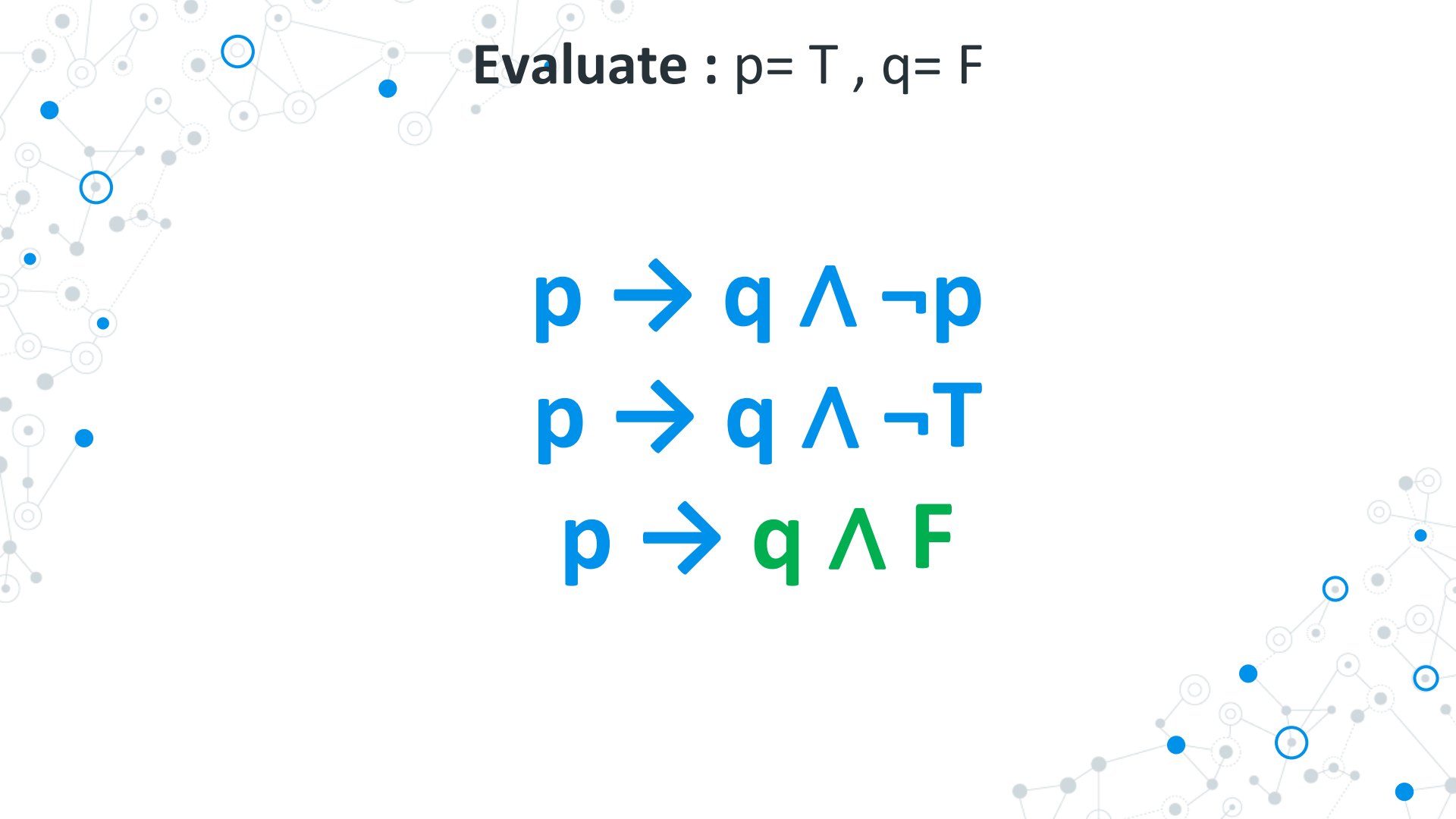


Evaluate :  $p = T$  ,  $q = F$

$$p \rightarrow q \wedge \neg p$$

$$p \rightarrow q \wedge \neg T$$

$$p \rightarrow q \wedge F$$

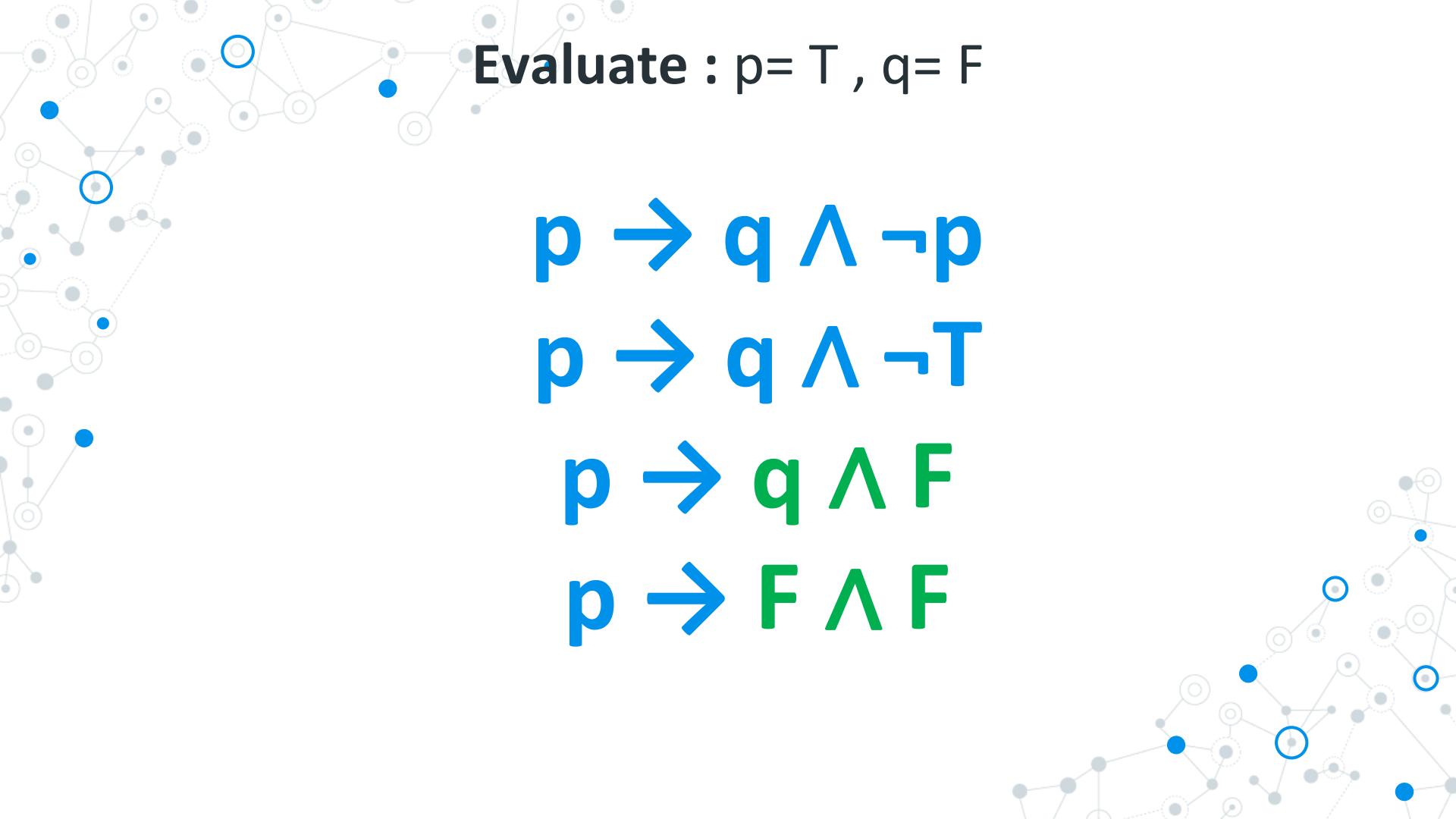


Evaluate :  $p = T, q = F$

$$p \rightarrow q \wedge \neg p$$

$$p \rightarrow q \wedge \neg T$$

$$p \rightarrow q \wedge F$$



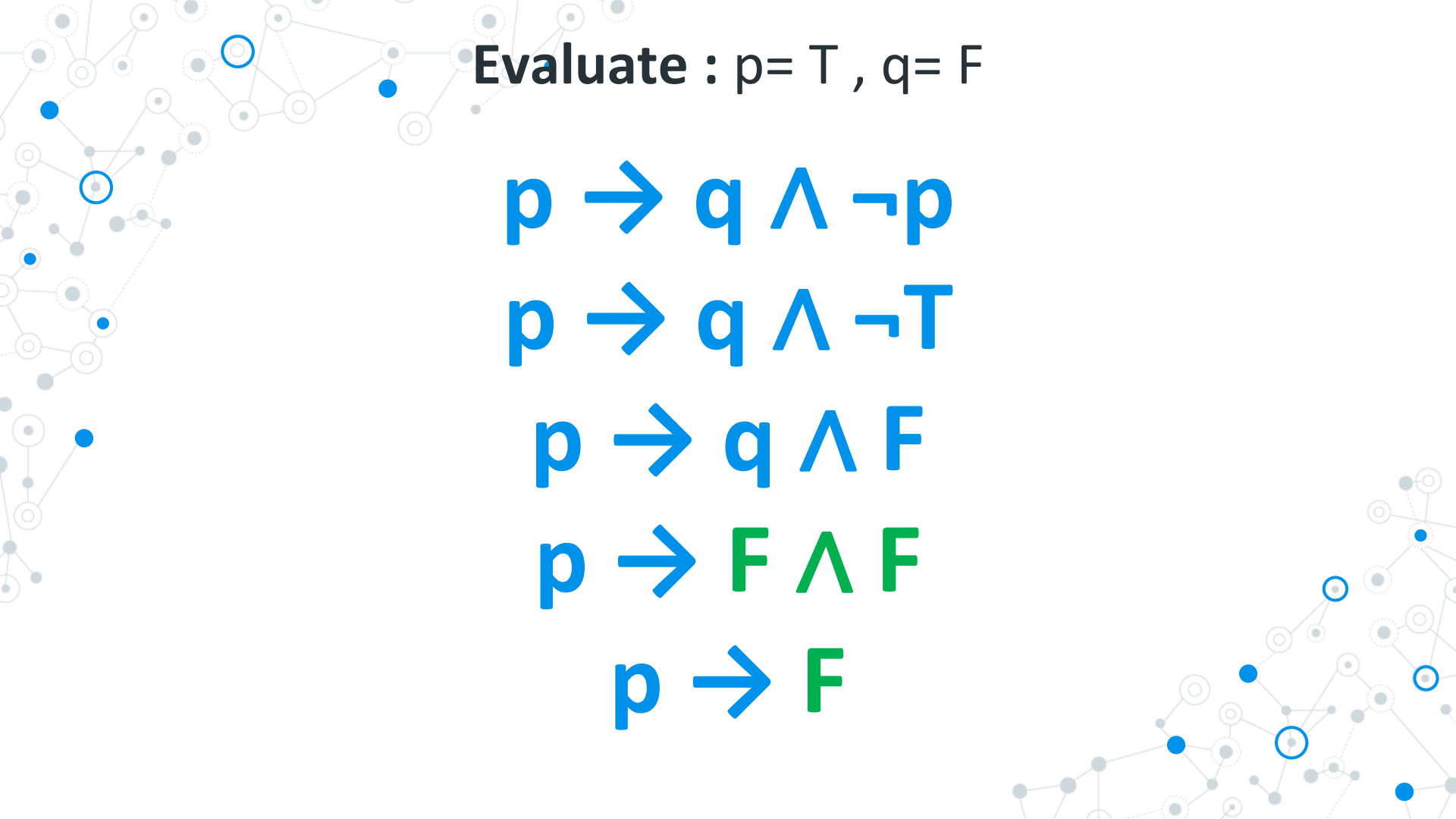
Evaluate :  $p = T$  ,  $q = F$

$$p \rightarrow q \wedge \neg p$$

$$p \rightarrow q \wedge \neg T$$

$$p \rightarrow q \wedge F$$

$$p \rightarrow F \wedge F$$



Evaluate :  $p = T$  ,  $q = F$

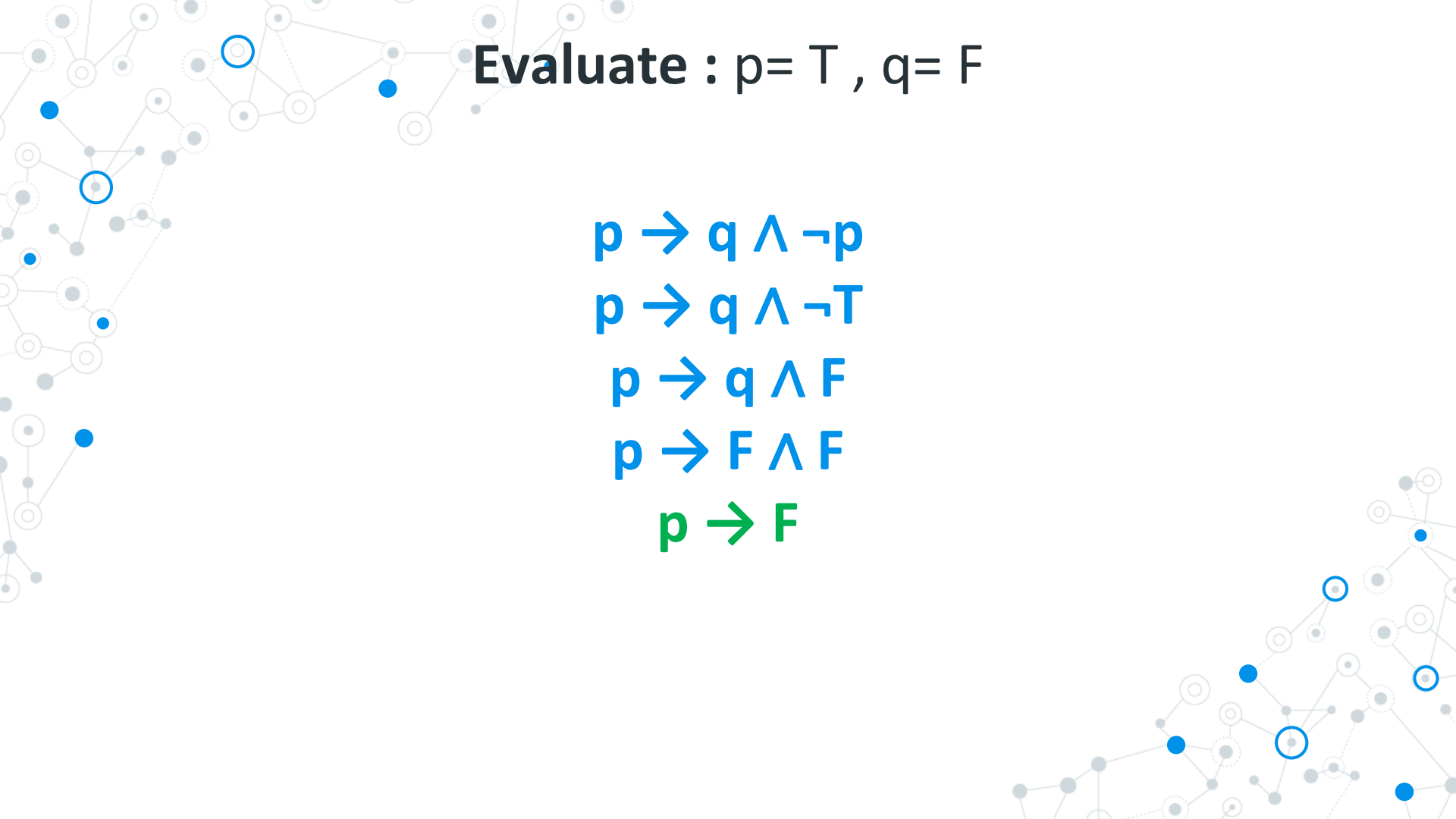
$$p \rightarrow q \wedge \neg p$$

$$p \rightarrow q \wedge \neg T$$

$$p \rightarrow q \wedge F$$

$$p \rightarrow F \wedge F$$

$$p \rightarrow F$$



Evaluate :  $p = T$  ,  $q = F$

$$p \rightarrow q \wedge \neg p$$

$$p \rightarrow q \wedge \neg T$$

$$p \rightarrow q \wedge F$$

$$p \rightarrow F \wedge F$$

$$p \rightarrow F$$





**Evaluate :  $p = T$  ,  $q = F$**

$$p \rightarrow q \wedge \neg p$$

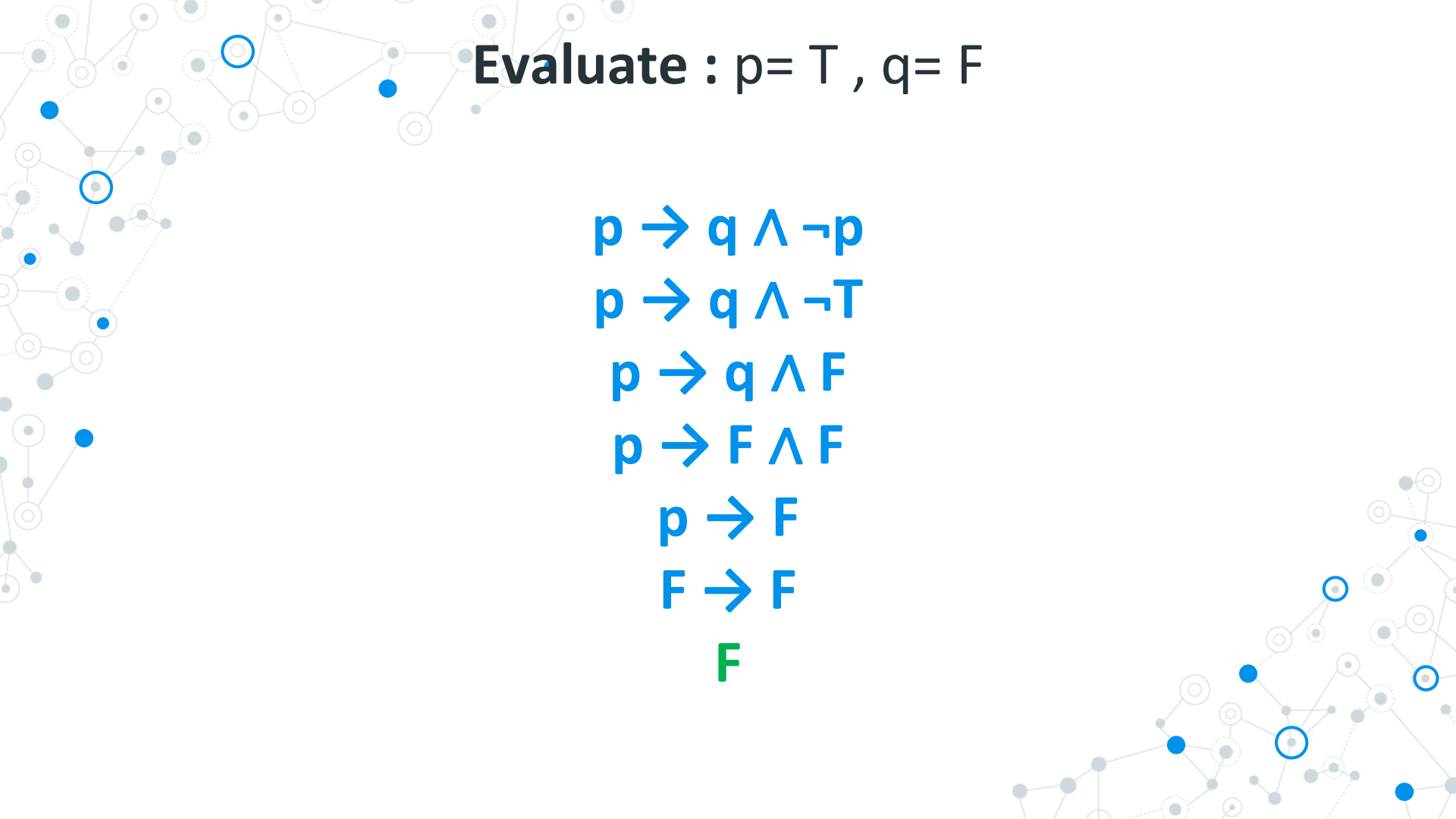
$$p \rightarrow q \wedge \neg T$$

$$p \rightarrow q \wedge F$$

$$p \rightarrow F \wedge F$$

$$p \rightarrow F$$

$$T \rightarrow F$$

Evaluate :  $p = T$  ,  $q = F$

$$p \rightarrow q \wedge \neg p$$

$$p \rightarrow q \wedge \neg T$$

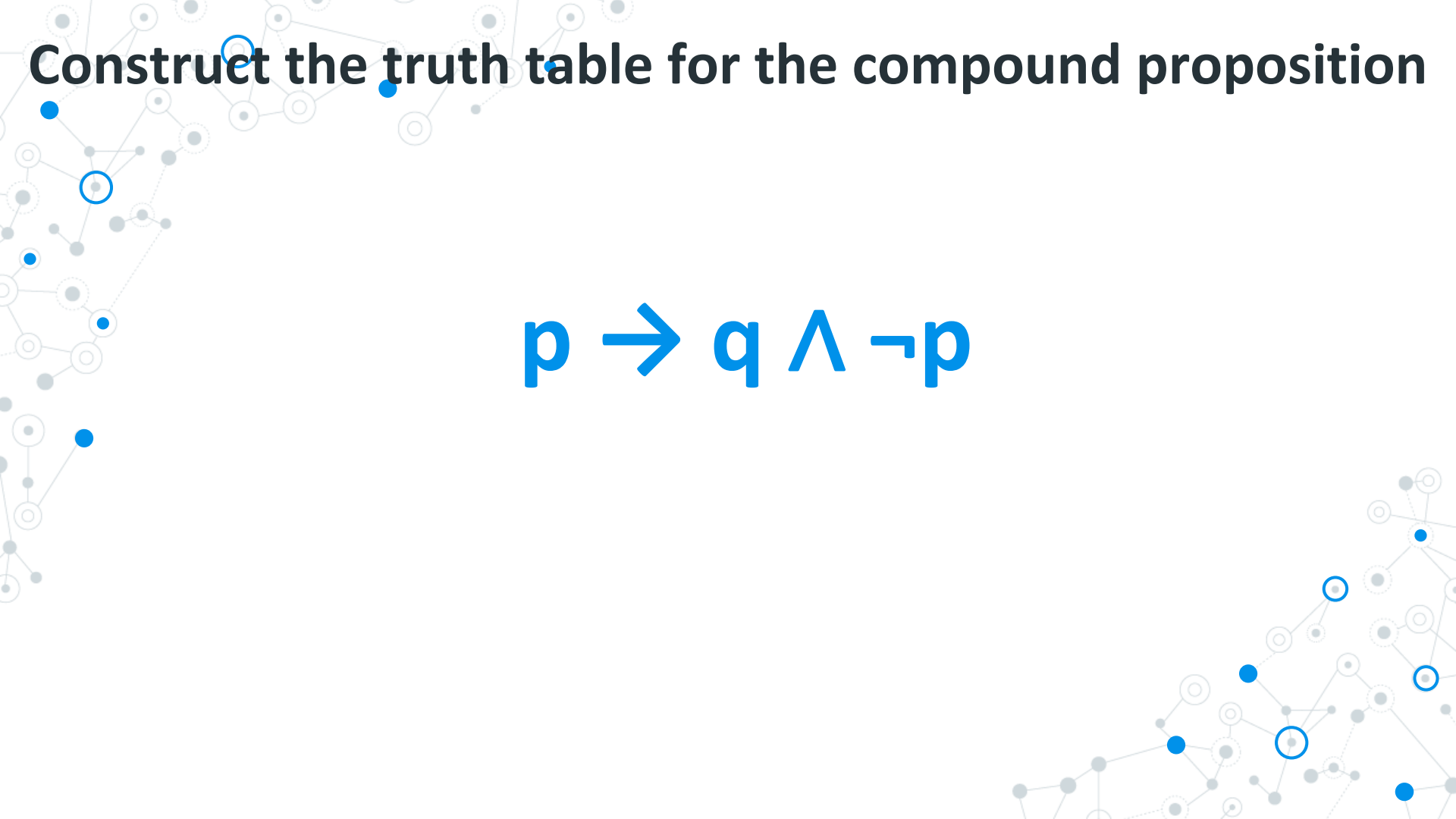
$$p \rightarrow q \wedge F$$

$$p \rightarrow F \wedge F$$

$$p \rightarrow F$$

$$F \rightarrow F$$

**F**



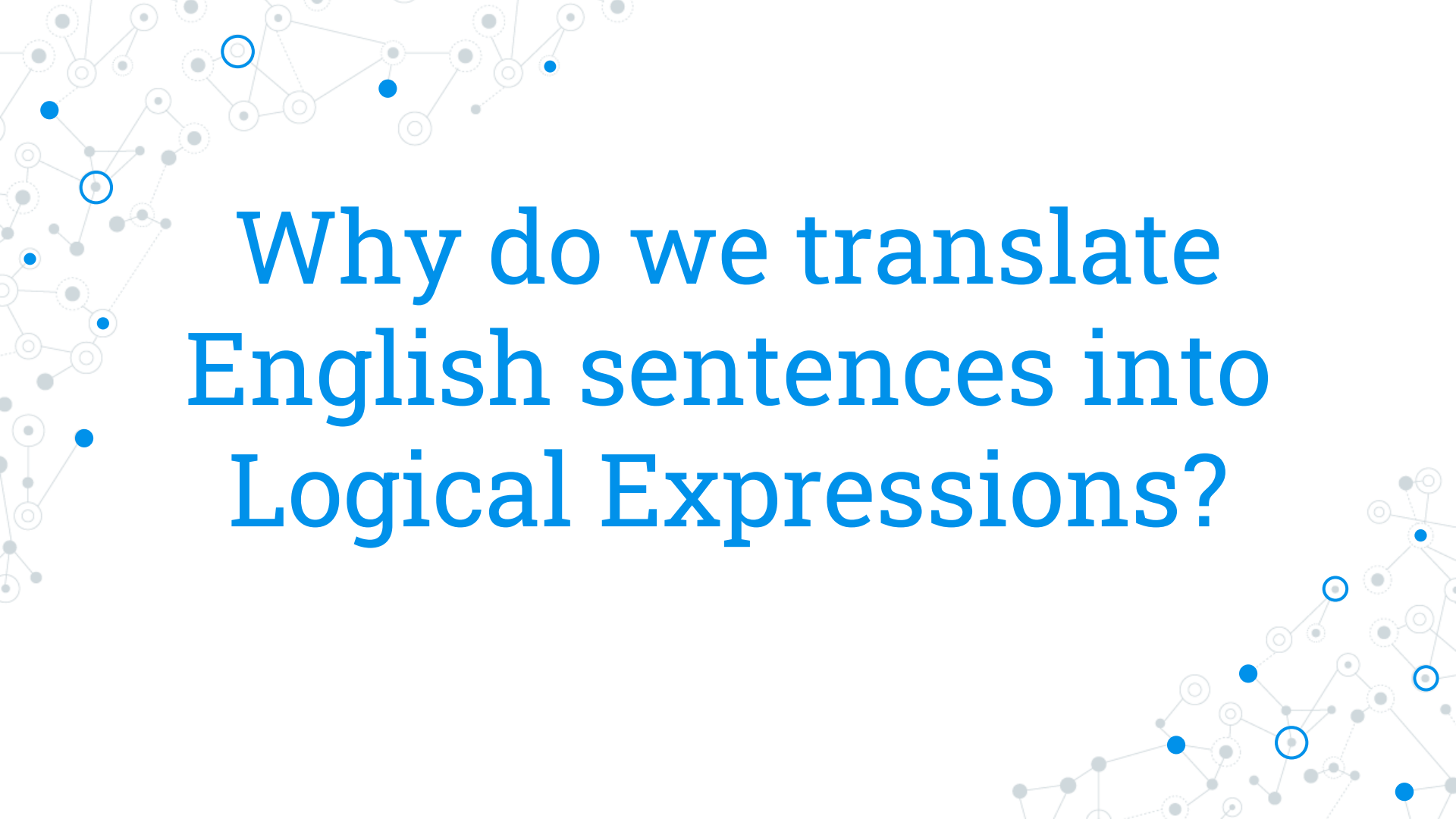
**Construct the truth table for the compound proposition**

$$p \rightarrow q \wedge \neg p$$

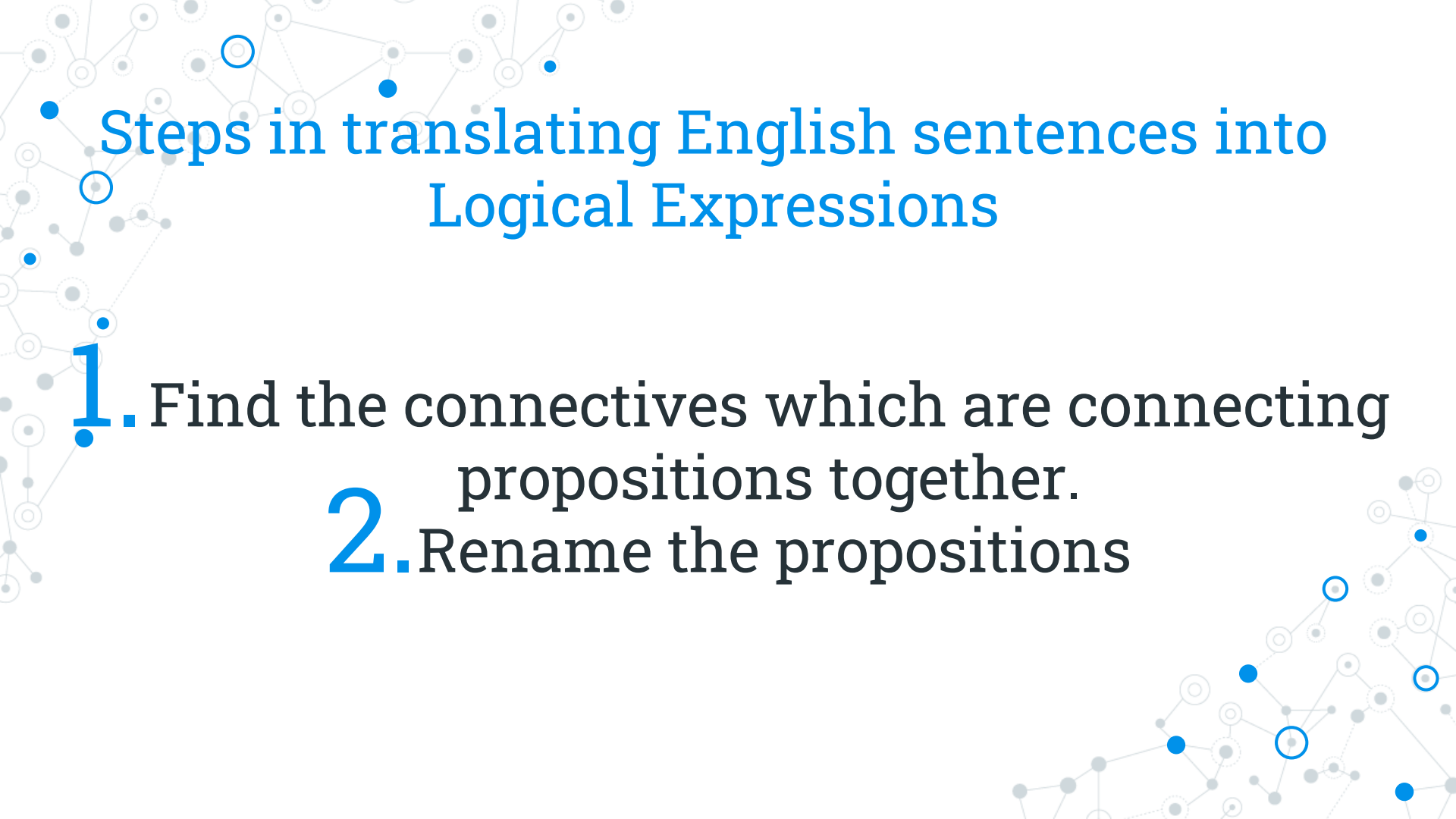
Construct the truth table for the compound proposition

$$p \rightarrow q \wedge \neg p$$

| <b>p</b> | <b>q</b> | <b><math>\neg p</math></b> | <b><math>q \wedge \neg p</math></b> | <b><math>p \rightarrow q \wedge \neg p</math></b> |
|----------|----------|----------------------------|-------------------------------------|---------------------------------------------------|
| T        | T        | F                          | F                                   | F                                                 |
| T        | F        | F                          | F                                   | F                                                 |
| F        | T        | T                          | T                                   | T                                                 |
| F        | F        | T                          | F                                   | T                                                 |

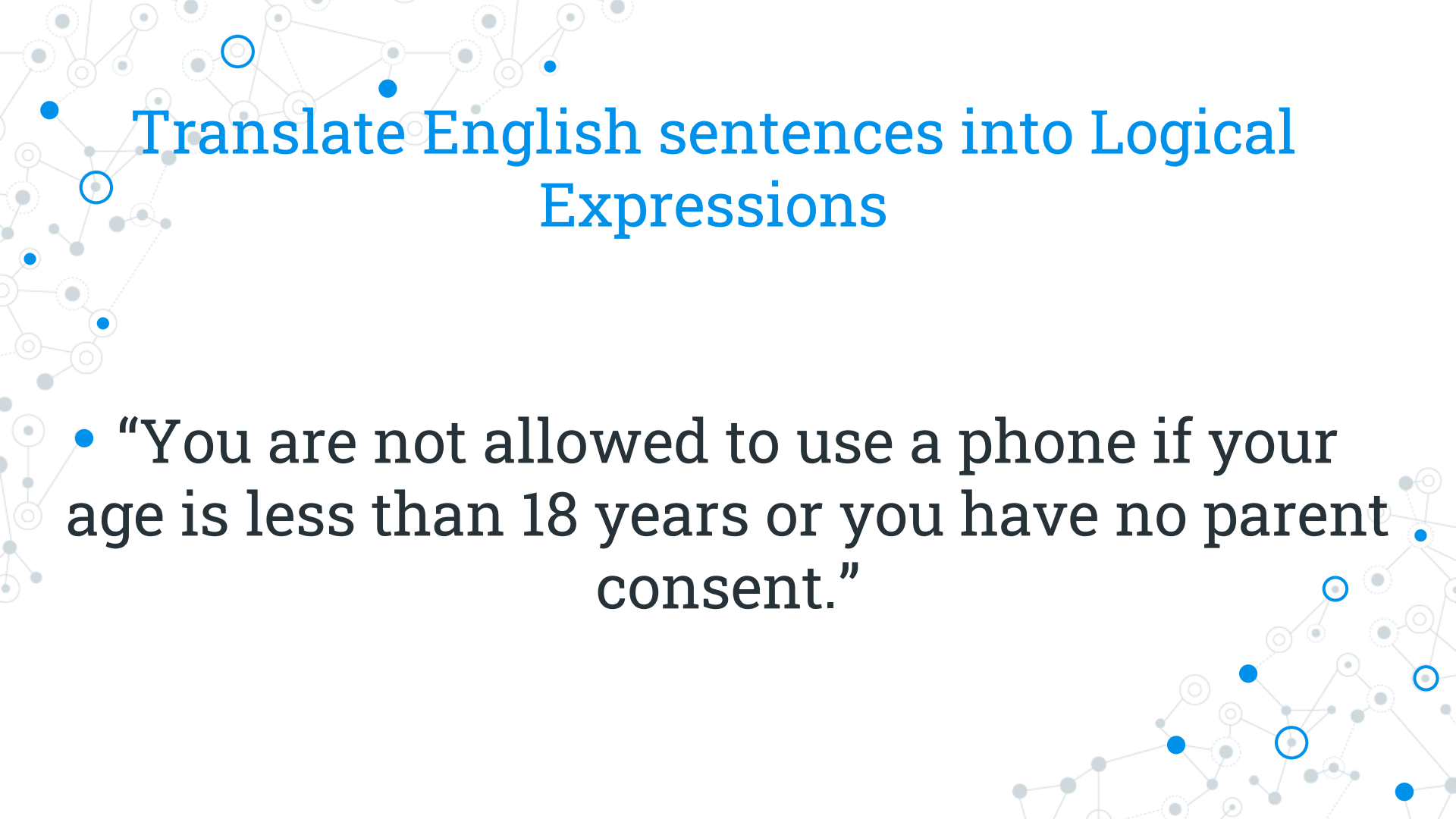
A decorative background pattern consisting of a network graph. It features numerous nodes, represented by small circles in various shades of gray and blue, connected by thin, light gray lines. Some nodes are highlighted with a blue outline. The pattern is more dense on the left and right sides of the slide, framing the central text.

# Why do we translate English sentences into Logical Expressions?



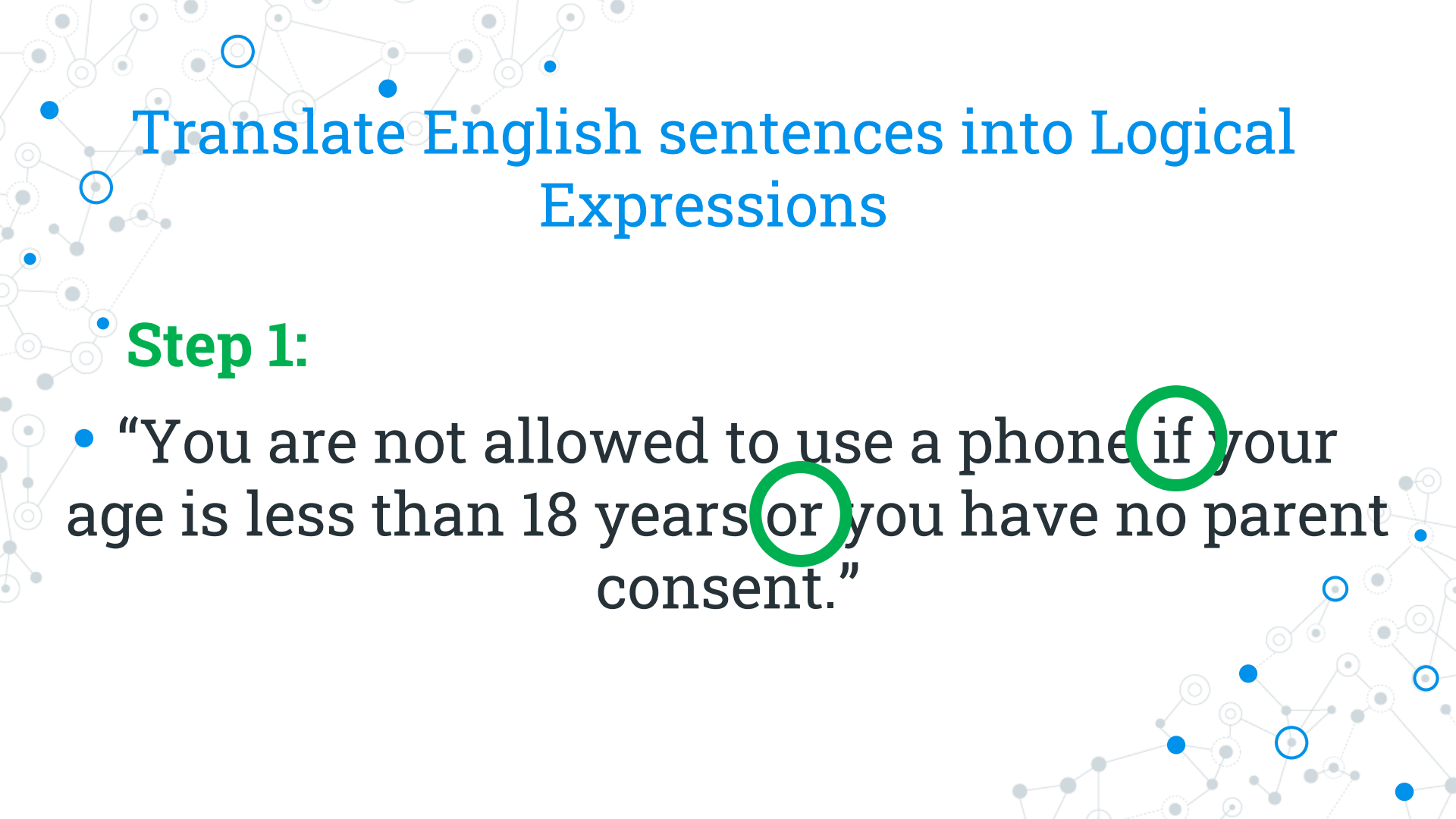
# Steps in translating English sentences into Logical Expressions

1. Find the connectives which are connecting propositions together.
2. Rename the propositions



# Translate English sentences into Logical Expressions

- “You are not allowed to use a phone if your age is less than 18 years or you have no parent consent.”



# Translate English sentences into Logical Expressions

## Step 1:

- “You are not allowed to use a phone **if** your age is less than 18 years **or** you have no parent consent.”



# Translate English sentences into Logical Expressions

## Step 2:

“You are not allowed to use a phone if your age is less than 18 years or you have no parent consent.”

Let:

$q$  = “You are allowed to use a phone”

$r$  = “Your age is less than 18 years”

$s$  = “You have parent consent”

# Translate English sentences into Logical Expressions

## Step 2:

“You are not allowed to use a phone if your age is less than 18 years or you have no parent consent.”

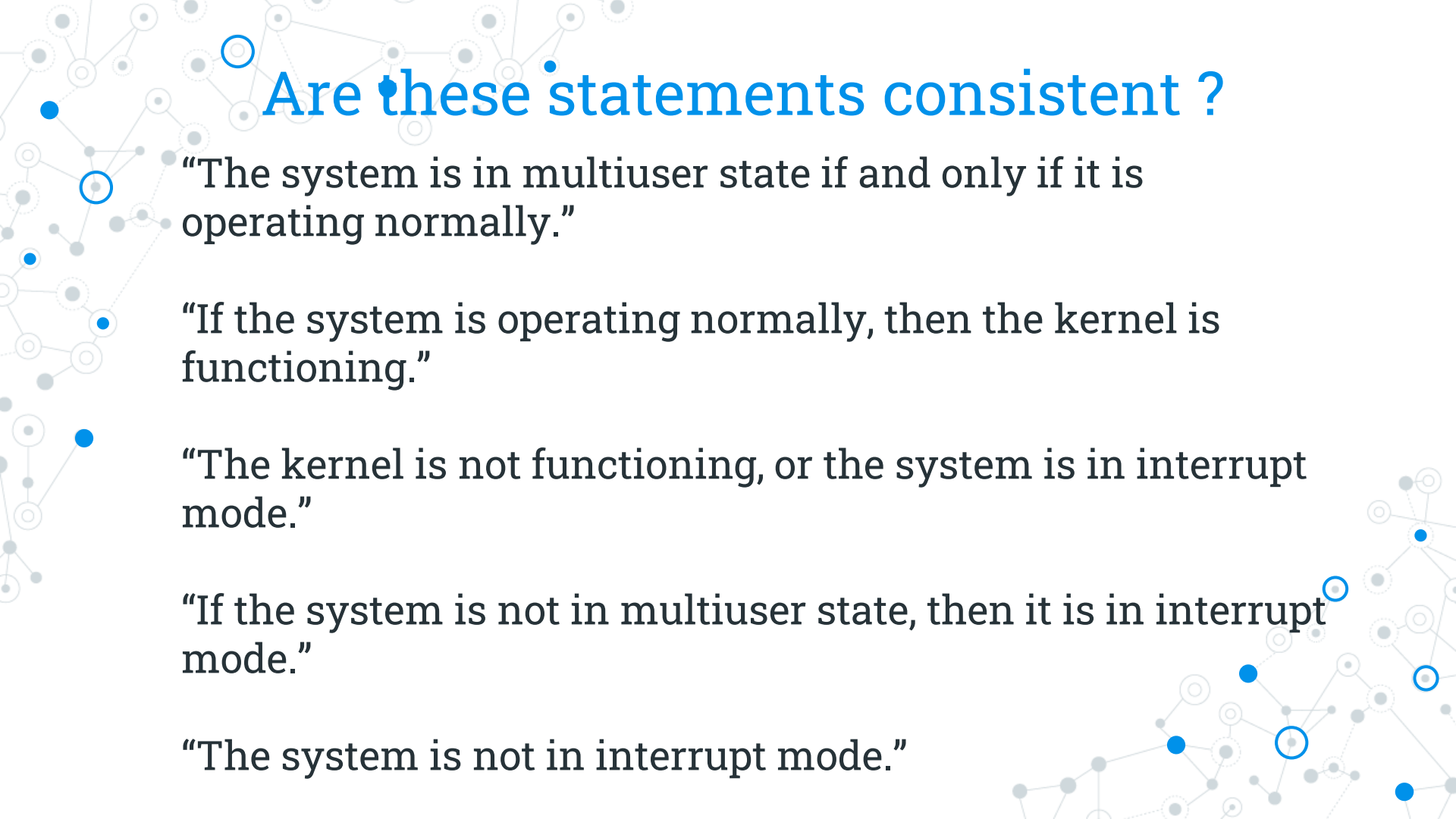
Let:

$q$  = “You are allowed to use a phone”

$r$  = “Your age is less than 18 years”

$s$  = “You have parent consent”

$$(r \vee \neg s) \rightarrow \neg q$$



# Are these statements consistent ?

“The system is in multiuser state if and only if it is operating normally.”

“If the system is operating normally, then the kernel is functioning.”

“The kernel is not functioning, or the system is in interrupt mode.”

“If the system is not in multiuser state, then it is in interrupt mode.”

“The system is not in interrupt mode.”

## Are these statements consistent ?

“The system is in multiuser state if and only if it is operating normally.”

“If the system is operating normally, then the kernel is functioning.”

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“If the system is not in multiuser state, then it is in interrupt mode.”

“The system is not in interrupt mode.”

Let

$p$  = “The system is in multiuser state.”

$q$  = “The system is operating normally.”

$r$  = “the kernel is functioning.”

$s$  = “The system is in interrupt mode.”

Are these statements consistent ?

“The system is in multiuser state if and only if it is operating normally.”

$$p \leftrightarrow q$$

“If the system is operating normally, then the kernel is functioning.”

“The kernel is not functioning, or the system is in interrupt mode.”

“If the system is not in multiuser state, then it is in interrupt mode.”

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Let

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Are these statements consistent ?

“The system is in multiuser state if and only if it is operating normally.”

$$p \leftrightarrow q$$

“If the system is operating normally, then the kernel is functioning.”

$$q \rightarrow r$$

“The kernel is not functioning, or the system is in interrupt mode.”

“If the system is not in multiuser state, then it is in interrupt mode.”

“The system is not in interrupt mode.”

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Are these statements consistent ?

“The system is in multiuser state if and only if it is operating normally.”

$$p \leftrightarrow q$$

“If the system is operating normally, then the kernel is functioning.”

$$q \rightarrow r$$

“The kernel is not functioning, or the system is in interrupt mode.”

$$\neg r \vee s$$

“If the system is not in multiuser state, then it is in interrupt mode.”

“The system is not in interrupt mode.”

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$$q \rightarrow r$$

“The kernel is not functioning, or the system is in interrupt mode.”

$$\neg r \vee s$$

“If the system is not in multiuser state, then it is in interrupt mode.”

$$\neg p \rightarrow s$$

“The system is not in interrupt mode.”

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Are these statements consistent ?

“The system is in multiuser state if and only if it is operating normally.”

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“The kernel is not functioning, or the system is in interrupt mode.”

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“If the system is not in multiuser state, then it is in interrupt mode.”

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$$p \leftrightarrow q$$

“If the system is operating normally, then the kernel is functioning.”

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“The kernel is not functioning, or the system is in interrupt mode.”

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“If the system is not in multiuser state, then it is in interrupt mode.”

$$\neg p \rightarrow s$$

“The system is not in interrupt mode.”



$$\neg s$$

$$s = F$$

Let

$p$  = “The system is in multiuser state.”

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$r$  = “The kernel is functioning.”

$s$  = “The system is in interrupt mode.”

Are these statements consistent ?

“The system is in multiuser state if and only if it is operating normally.”

$$p \leftrightarrow q$$

“If the system is operating normally, then the kernel is functioning.”

$$q \rightarrow r$$

“The kernel is not functioning, or the system is in interrupt mode.”

$$\neg r \vee s$$

“If the system is not in multiuser state, then it is in interrupt mode.”

$$\neg p \rightarrow s^F$$

“The system is not in interrupt mode.”

$$\neg s$$

$$S = F$$



Let

$p$  = “The system is in multiuser state.”

$q$  = “The system is operating normally.”

$r$  = “The kernel is functioning.”

$s$  = “The system is in interrupt mode.”

Are these statements consistent ?

“The system is in multiuser state if and only if it is operating normally.”

$$p \leftrightarrow q$$

“If the system is operating normally, then the kernel is functioning.”

$$q \rightarrow r$$

“The kernel is not functioning, or the system is in interrupt mode.”

$$\neg r \vee s$$

“If the system is not in multiuser state, then it is in interrupt mode.”

$$\neg p \rightarrow s^F \quad P = T$$

“The system is not in interrupt mode.”

$$\neg s \quad S = F$$



Let

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“If the system is not in multiuser state, then it is in interrupt mode.”

$$\neg p \rightarrow s^F \quad P = T$$

“The system is not in interrupt mode.”

$$\neg s \quad S = F$$

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“The kernel is not functioning, or the system is in interrupt mode.”

$$\neg r \vee s$$

“If the system is not in multiuser state, then it is in interrupt mode.”

$$\neg p \rightarrow s$$

$$P = T$$

“The system is not in interrupt mode.”

$$\neg s$$

$$S = F$$

Let

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$s$  = “The system is in interrupt mode.”

Are these statements consistent ?

“The system is in multiuser state if and only if it is operating normally.”

$$p \leftrightarrow q$$

“If the system is operating normally, then the kernel is functioning.”

$$q \rightarrow r$$

“The kernel is not functioning, or the system is in interrupt mode.”

$$T \neg r \vee s F$$

“If the system is not in multiuser state, then it is in interrupt mode.”

$$\neg p \rightarrow s F \quad P = T$$

“The system is not in interrupt mode.”

$$\neg s \quad S = F$$

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$$T \neg r \vee S^F \quad r = F$$

“If the system is not in multiuser state, then it is in interrupt mode.”

$$\neg p \rightarrow S^F \quad P = T$$

“The system is not in interrupt mode.”

$$\neg S \quad S = F$$

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$$T \neg r \vee S^F \quad r = F$$

“If the system is not in multiuser state, then it is in interrupt mode.”

$$\neg p \rightarrow S^F \quad P = T$$

“The system is not in interrupt mode.”

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Are these statements consistent ?

“The system is in multiuser state if and only if it is operating normally.”

$$p \leftrightarrow q$$

“If the system is operating normally, then the kernel is functioning.”

$$q \rightarrow r^F$$

“The kernel is not functioning, or the system is in interrupt mode.”

$$T \neg r \vee s^F \quad r = F$$

“If the system is not in multiuser state, then it is in interrupt mode.”

$$\neg p \rightarrow s^F \quad P = T$$

“The system is not in interrupt mode.”

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“If the system is operating normally, then the kernel is functioning.”

$$q \rightarrow r^F \quad q = F$$

“The kernel is not functioning, or the system is in interrupt mode.”

$$T \neg r \vee s^F \quad r = F$$

“If the system is not in multiuser state, then it is in interrupt mode.”

$$\neg p \rightarrow s^F \quad p = T$$

“The system is not in interrupt mode.”

$$\neg s \quad s = F$$

Let

$p$  = “The system is in multiuser state.”

$q$  = “The system is operating normally.”

$r$  = “The kernel is functioning.”

$s$  = “The system is in interrupt mode.”

Are these statements consistent ?

“The system is in multiuser state if and only if it is operating normally.”

$$p \leftrightarrow q$$

“If the system is operating normally, then the kernel is functioning.”

$$q \rightarrow r^F \quad q = F$$

“The kernel is not functioning, or the system is in interrupt mode.”

$$T \neg r \vee s^F \quad r = F$$

“If the system is not in multiuser state, then it is in interrupt mode.”

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“The kernel is not functioning, or the system is in interrupt mode.”

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Are these statements consistent ?

“The system is in multiuser state if and only if it is operating normally.”

$\times \quad T p \leftrightarrow q \quad F$

“If the system is operating normally, then the kernel is functioning.”

$q \rightarrow r \quad F \quad q = F$

“The kernel is not functioning, or the system is in interrupt mode.”

$T \neg r \vee s \quad F \quad r = F$

“If the system is not in multiuser state, then it is in interrupt mode.”

$\neg p \rightarrow s \quad F \quad p = T$

“The system is not in interrupt mode.”

$\neg s \quad S = F$

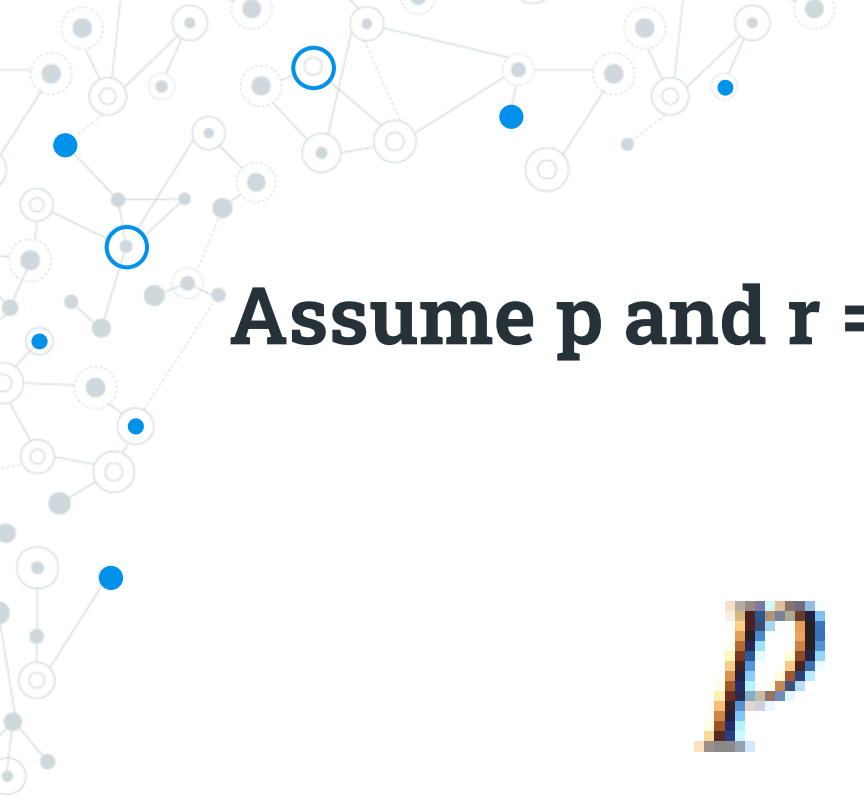
Let

p = “The system is in multiuser state.”

q = “The system is operating normally.”

r = “The kernel is functioning.”

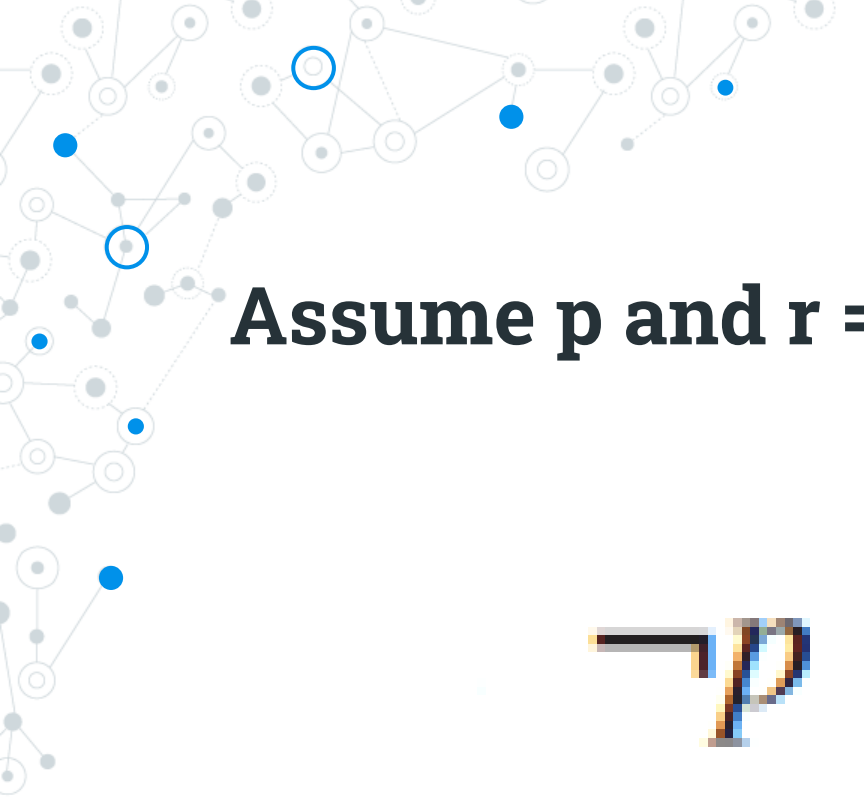
s = “The system is in interrupt mode.”



#1.

**Assume p and r = false, q and s = true**

$$p \rightarrow q$$

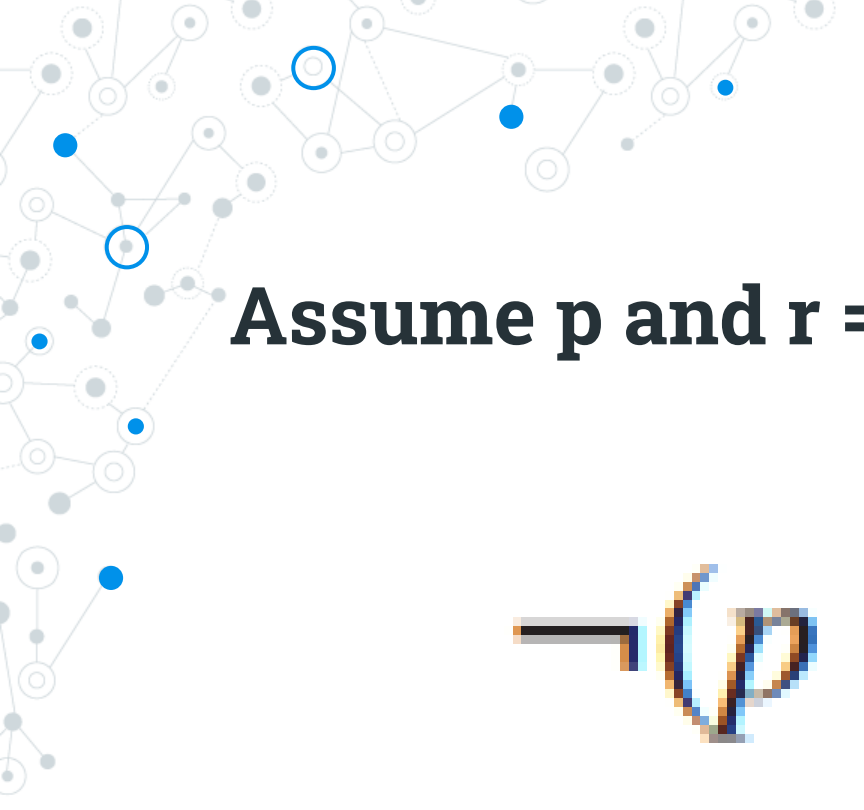



#2.

**Assume p and r = false, q and s = true**

$$\neg p \rightarrow \neg q$$

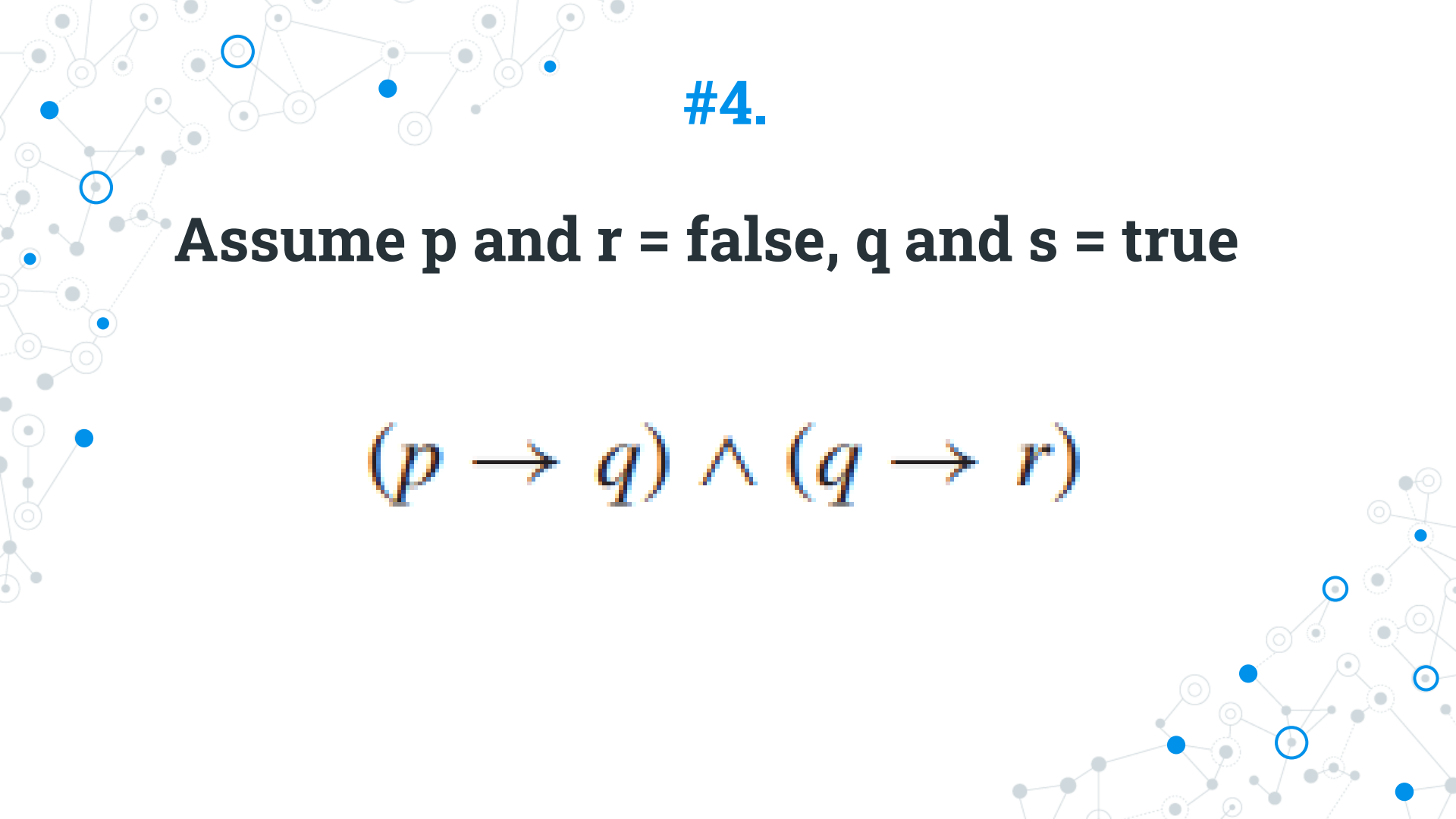


#3.

**Assume p and r = false, q and s = true**

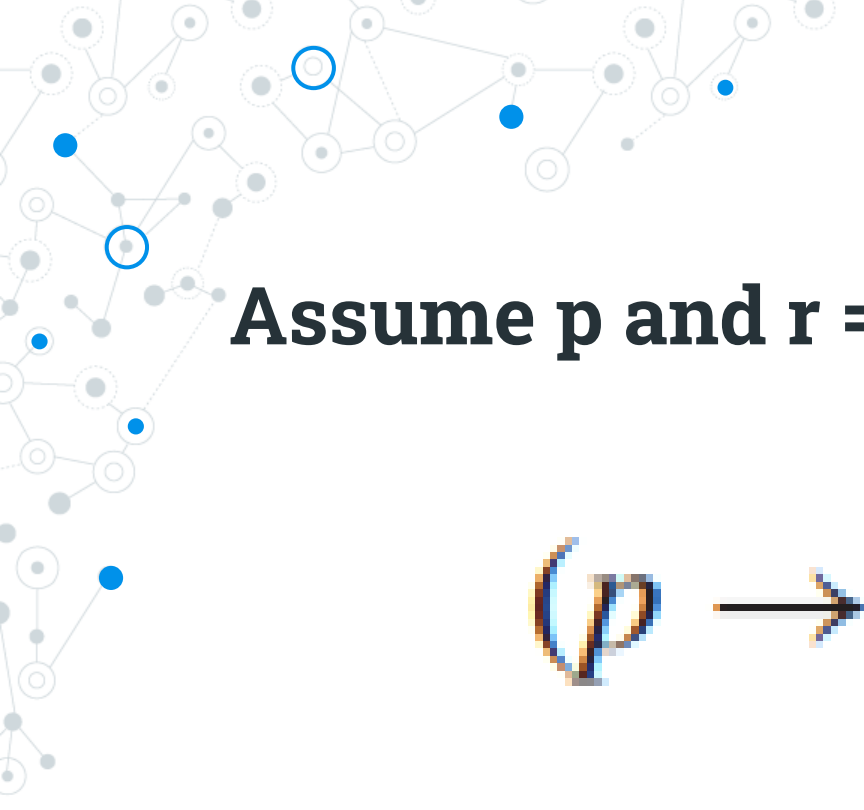
$$\neg(p \rightarrow q)$$

#4.

**Assume  $p$  and  $r$  = false,  $q$  and  $s$  = true**

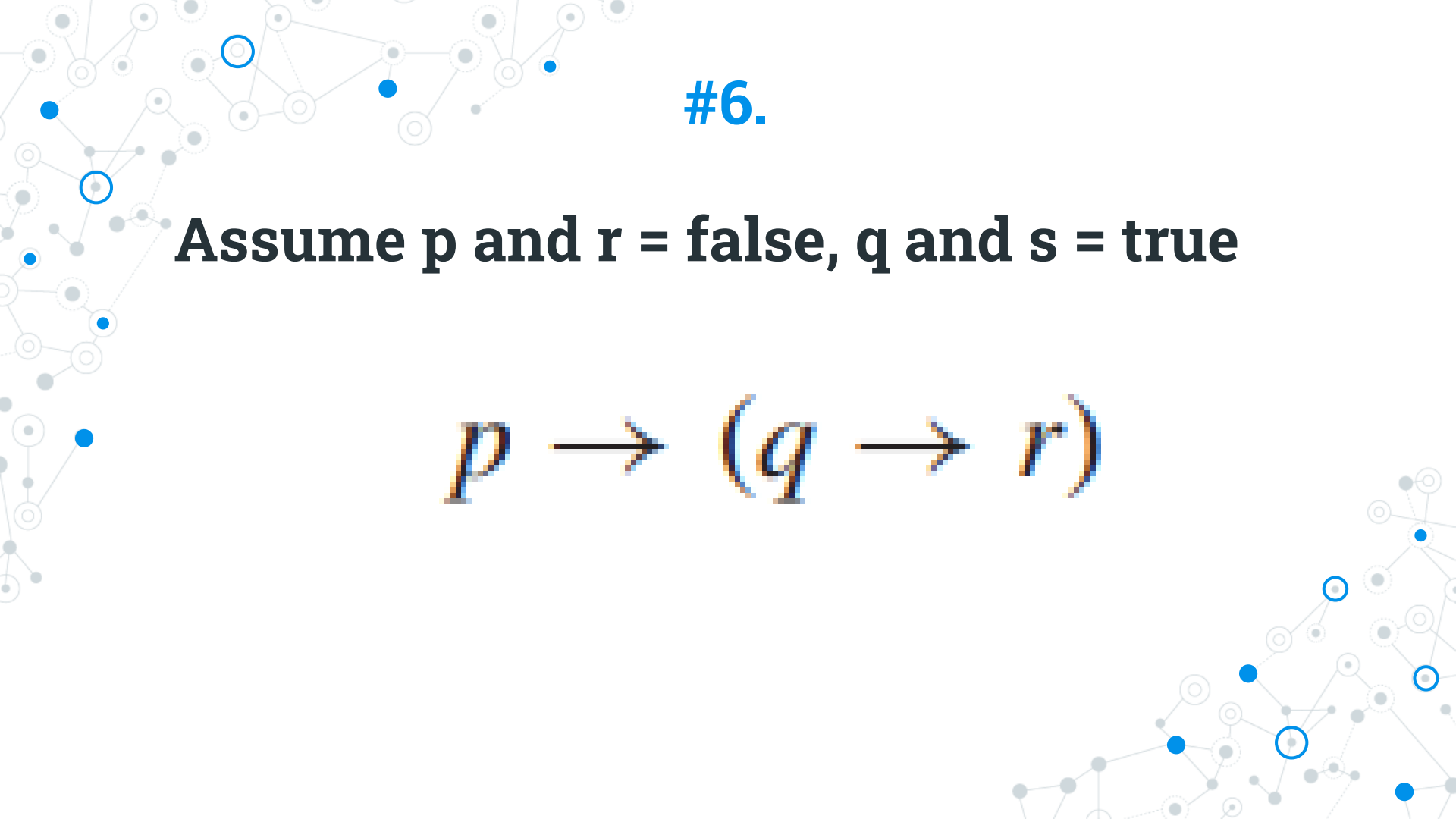
$$(p \rightarrow q) \wedge (q \rightarrow r)$$



#5.

**Assume p and r = false, q and s = true**

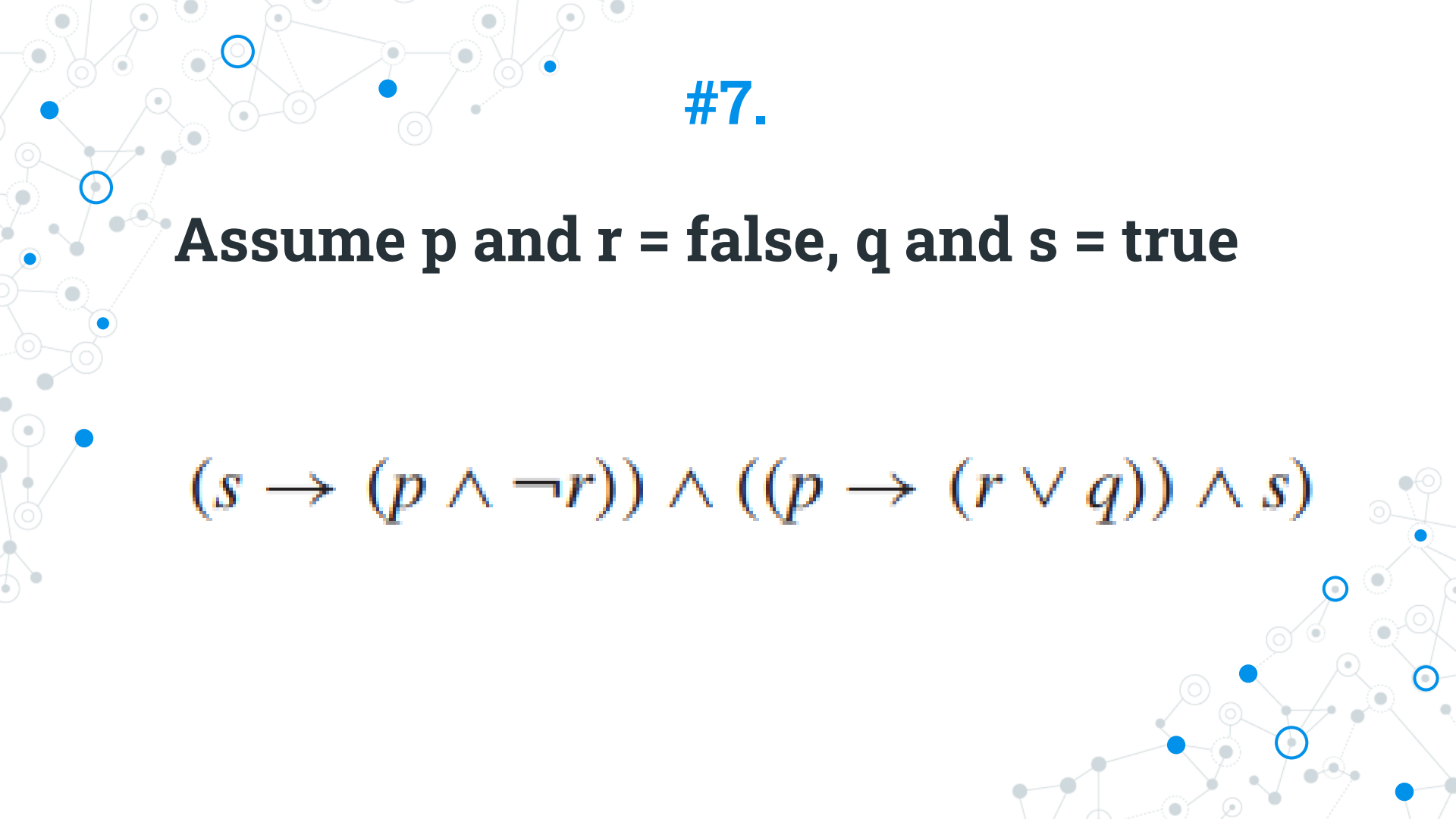
$$(p \rightarrow q) \rightarrow r$$

#6.

**Assume  $p$  and  $r$  = false,  $q$  and  $s$  = true**

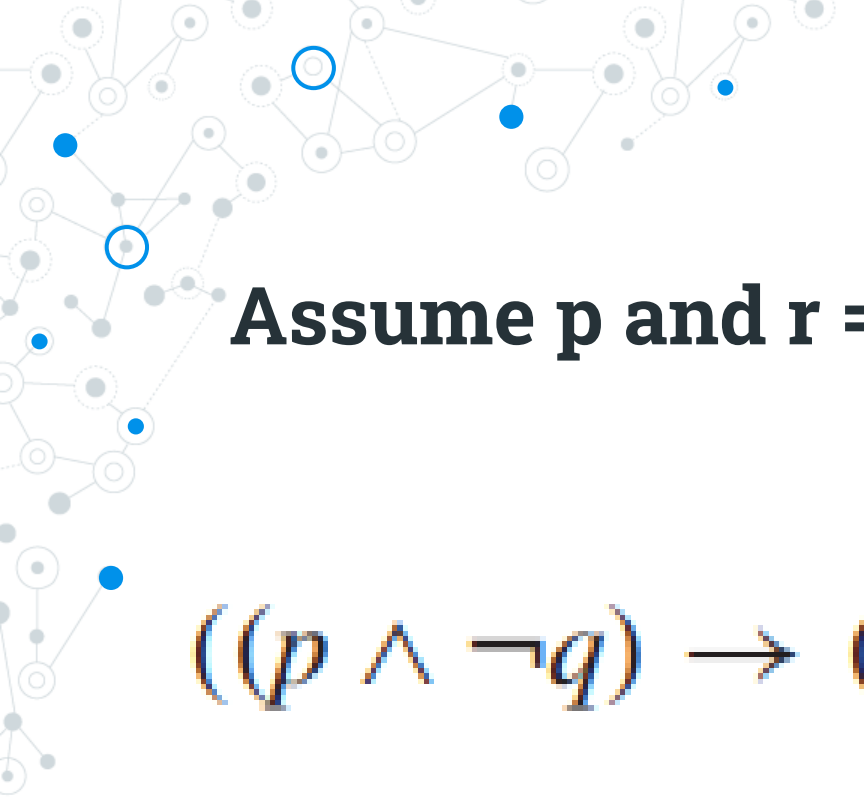
$$p \rightarrow (q \rightarrow r)$$



#7.

**Assume p and r = false, q and s = true**

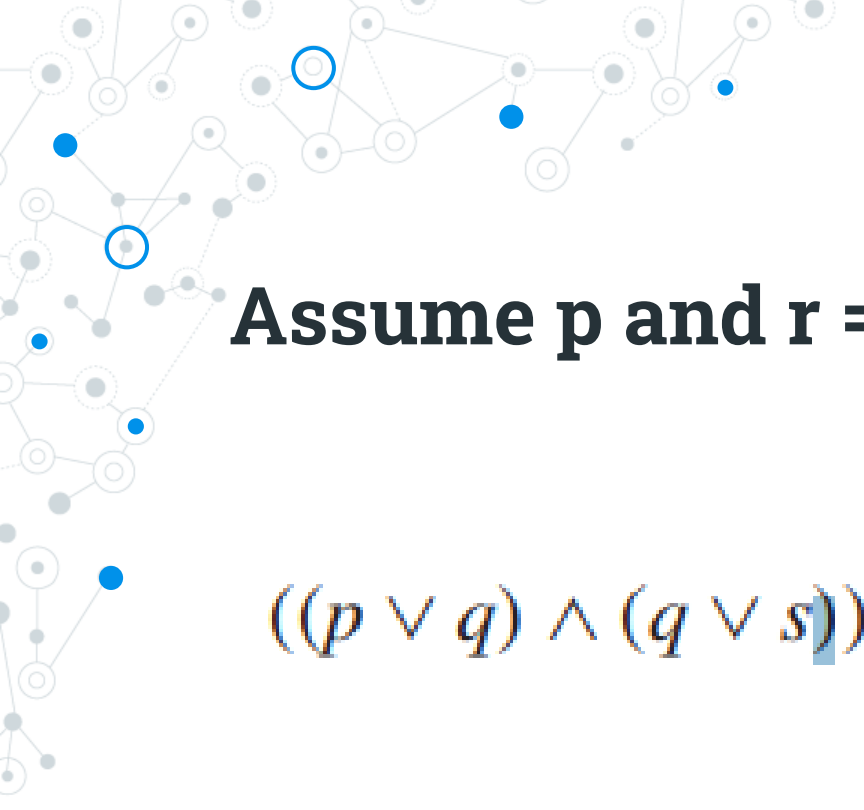
$$(s \rightarrow (p \wedge \neg r)) \wedge ((p \rightarrow (r \vee q)) \wedge s)$$



#8.

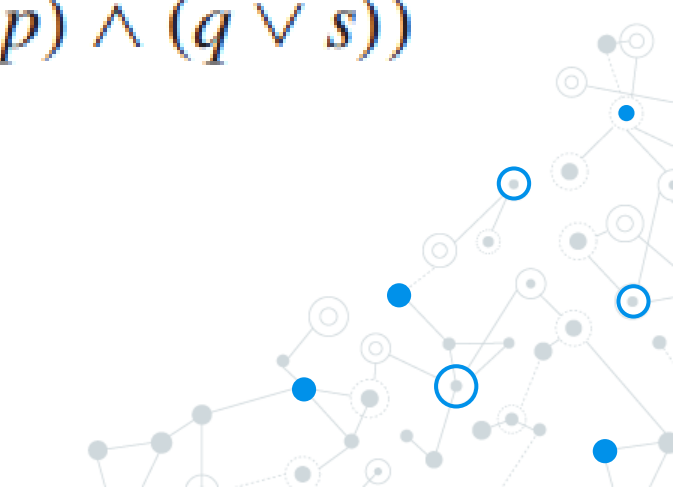
**Assume p and r = false, q and s = true**

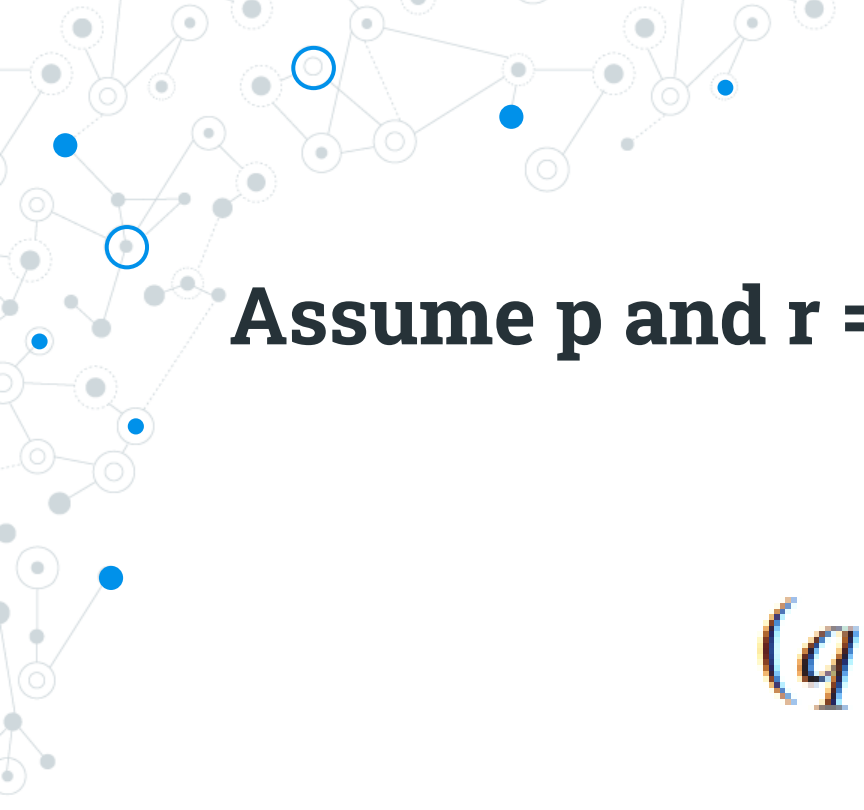
$$((p \wedge \neg q) \rightarrow (q \wedge r)) \rightarrow (s \vee \neg q)$$

#9.

**Assume  $p$  and  $r$  = false,  $q$  and  $s$  = true**

$$((p \vee q) \wedge (q \vee s)) \rightarrow ((\neg r \vee p) \wedge (q \vee s))$$


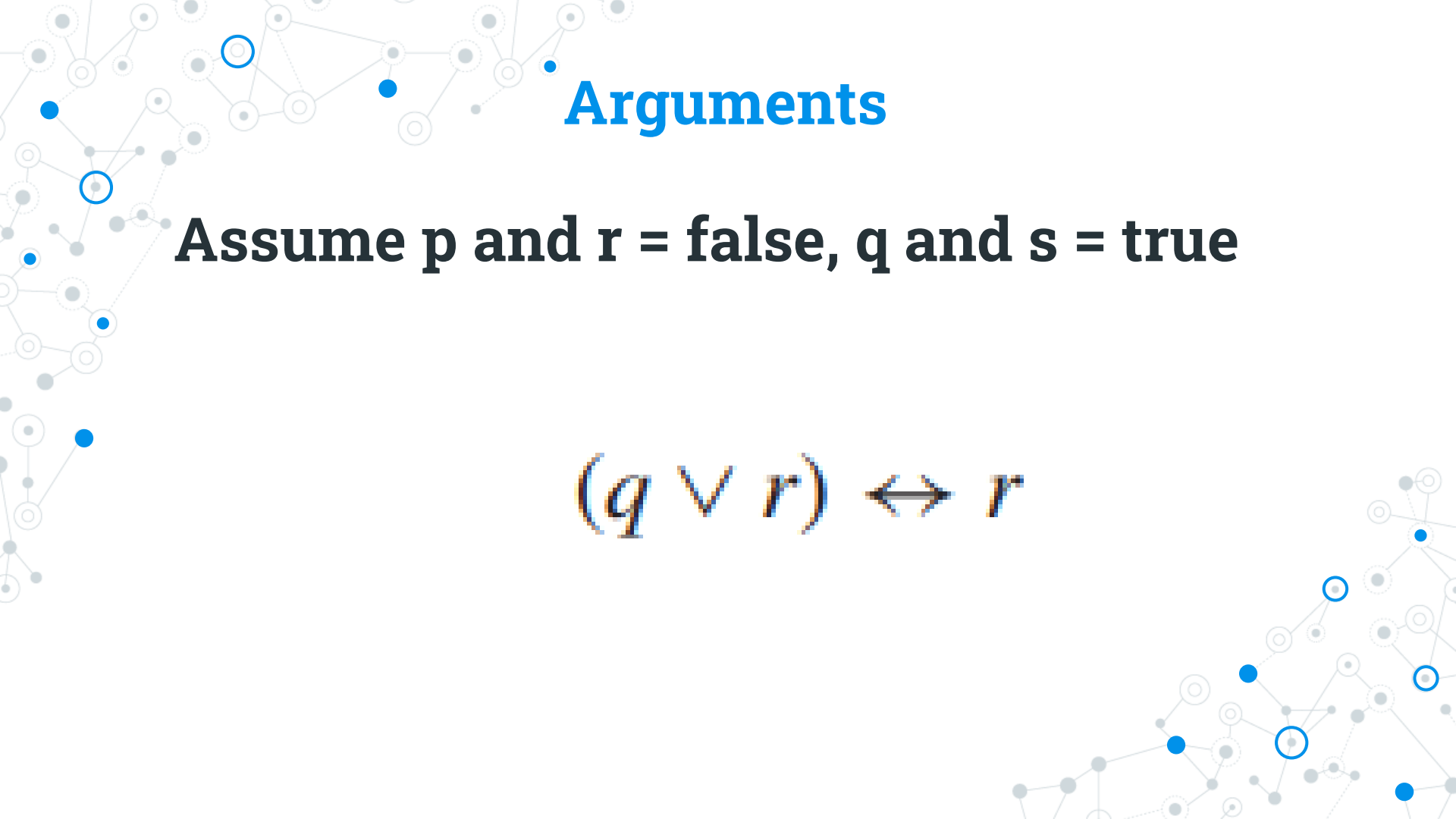


#10.

**Assume  $p$  and  $r$  = false,  $q$  and  $s$  = true**

$$(q \vee r) \leftrightarrow r$$



# Arguments

**Assume  $p$  and  $r = \text{false}$ ,  $q$  and  $s = \text{true}$**

$$(q \vee r) \leftrightarrow r$$

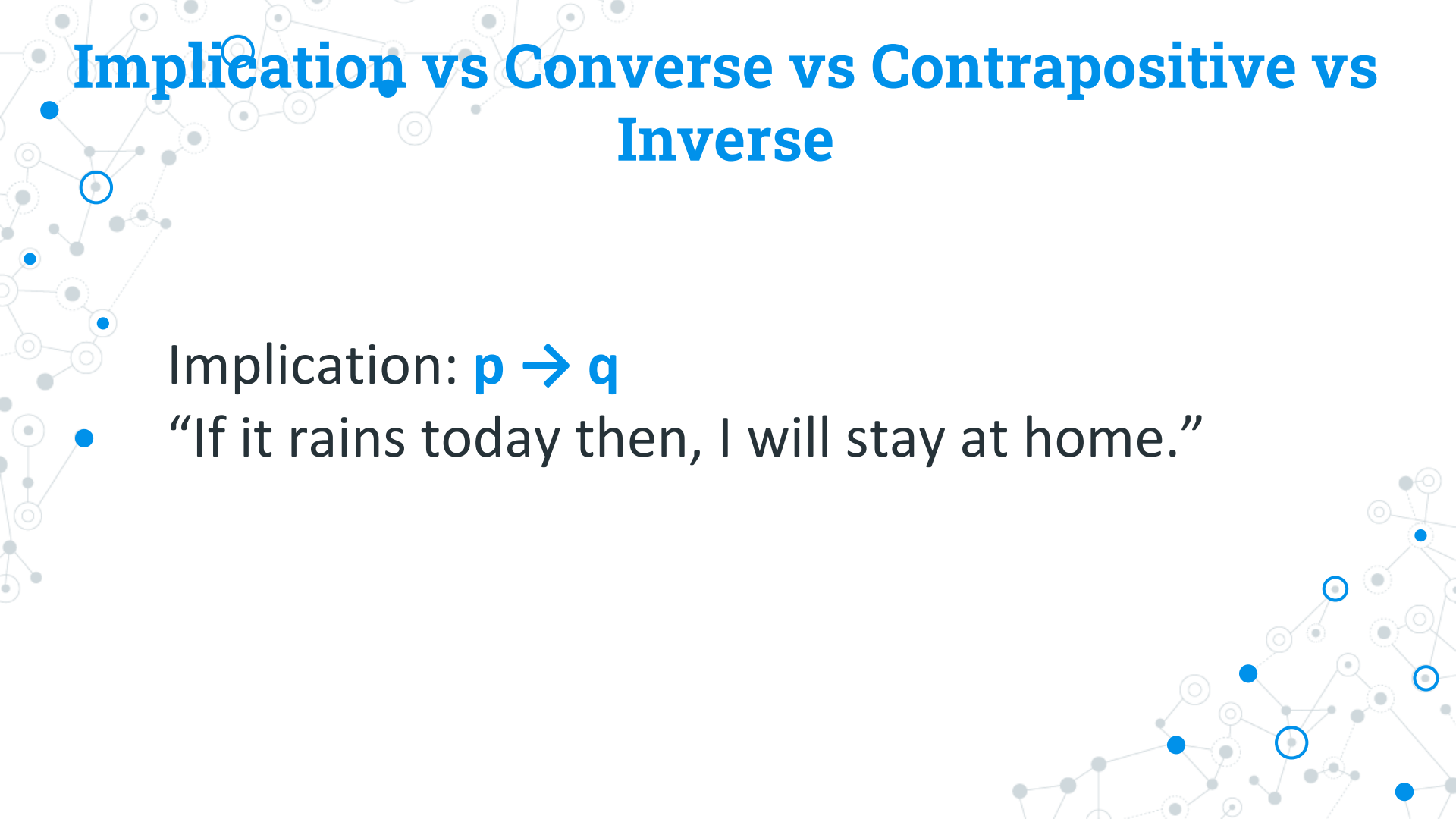
# Implication vs Converse vs Contrapositive vs Inverse

Implication:  $p \rightarrow q$

Converse:  $q \rightarrow p$

Contrapositive:  $\neg q \rightarrow \neg p$

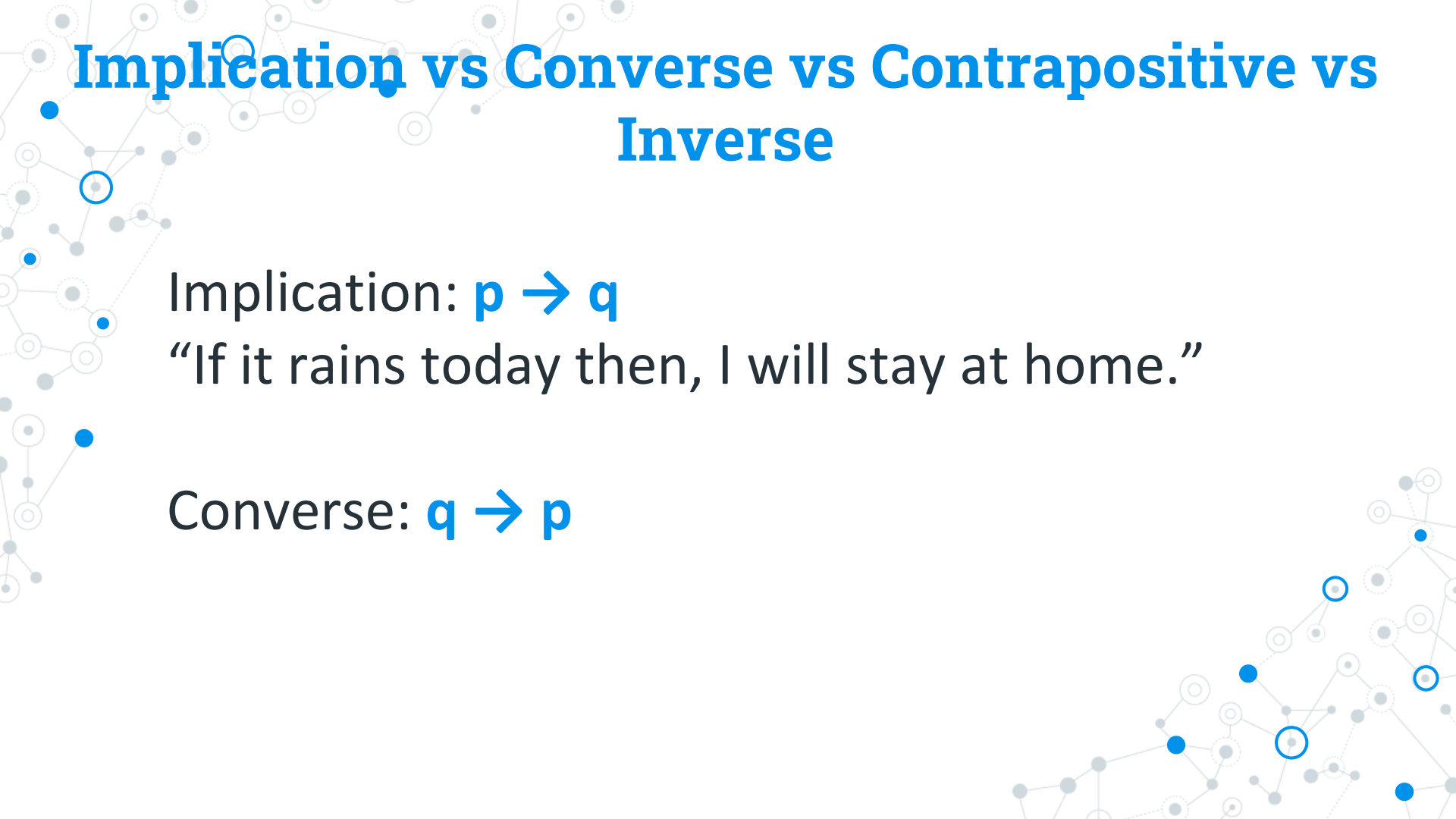
Inverse:  $\neg p \rightarrow \neg q$



# Implication vs Converse vs Contrapositive vs Inverse

Implication:  $p \rightarrow q$

“If it rains today then, I will stay at home.”



# Implication vs Converse vs Contrapositive vs Inverse

Implication:  $p \rightarrow q$

“If it rains today then, I will stay at home.”

Converse:  $q \rightarrow p$

# Implication vs Converse vs Contrapositive vs Inverse

Implication:  $p \rightarrow q$

“If it rains today then, I will stay at home.”

Converse:  $q \rightarrow p$

“If I will stay at home, then it rains today.”

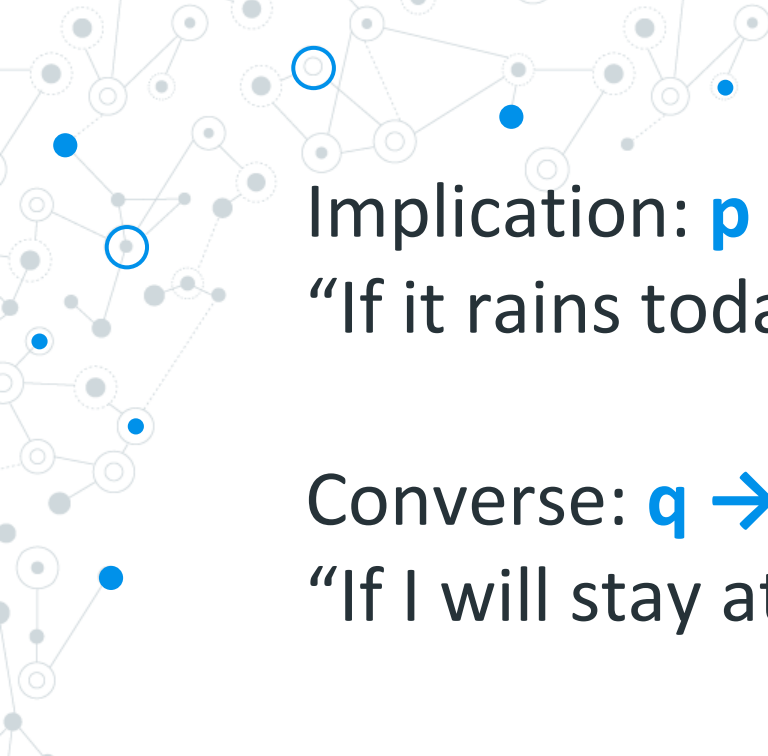
# Implication vs Converse vs Contrapositive vs Inverse

Implication:  $p \rightarrow q$

“If it rains today then, I will stay at home.”

Converse:  $q \rightarrow p$

“If I will stay at home, then it rains today.”



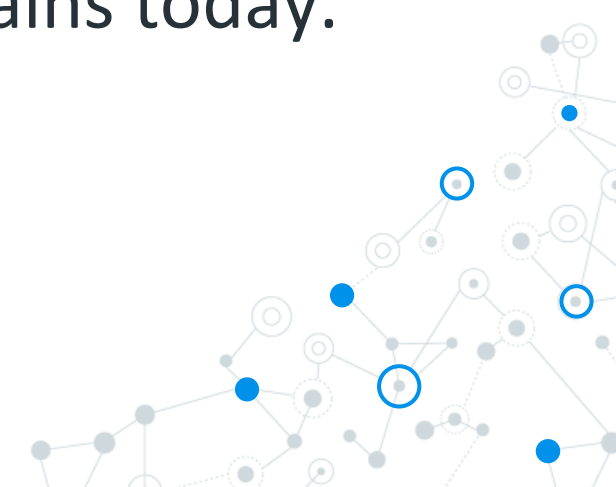
Implication:  $p \rightarrow q$

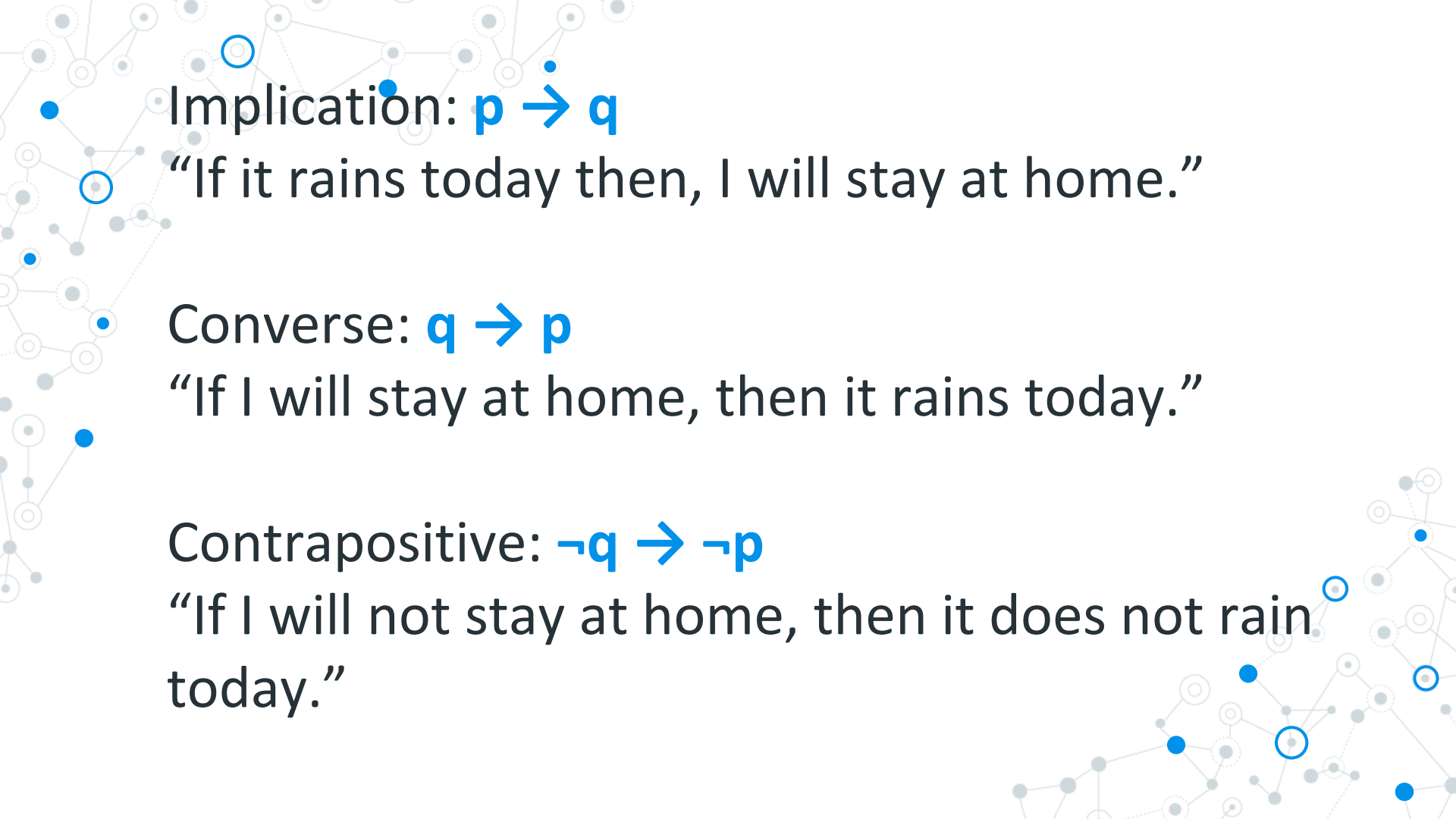
“If it rains today then, I will stay at home.”

Converse:  $q \rightarrow p$

“If I will stay at home, then it rains today.”

Contrapositive:  $\neg q \rightarrow \neg p$





Implication:  $p \rightarrow q$

“If it rains today then, I will stay at home.”

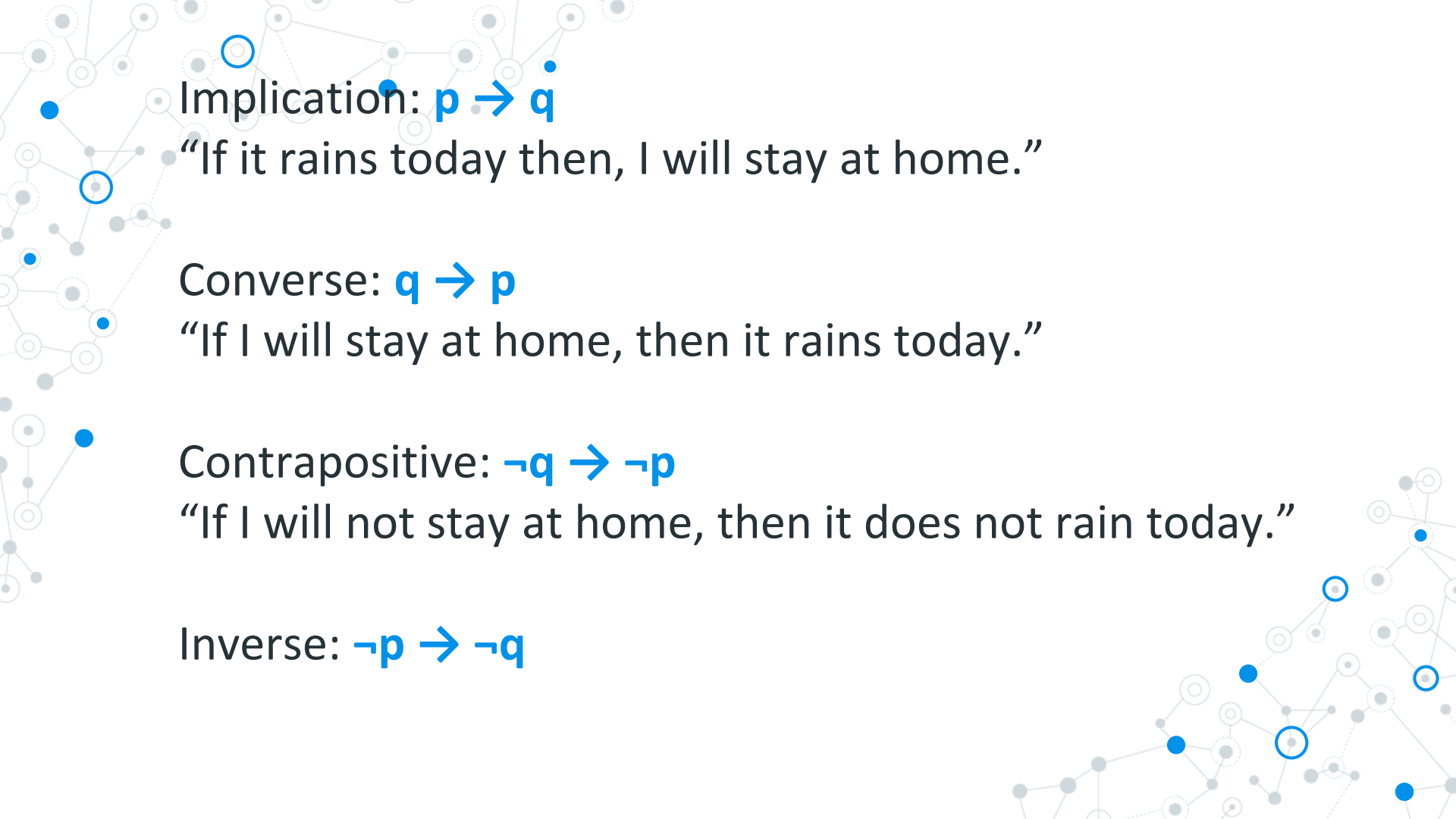
Converse:  $q \rightarrow p$

“If I will stay at home, then it rains today.”

Contrapositive:  $\neg q \rightarrow \neg p$

“If I will not stay at home, then it does not rain today.”





Implication:  $p \rightarrow q$

“If it rains today then, I will stay at home.”

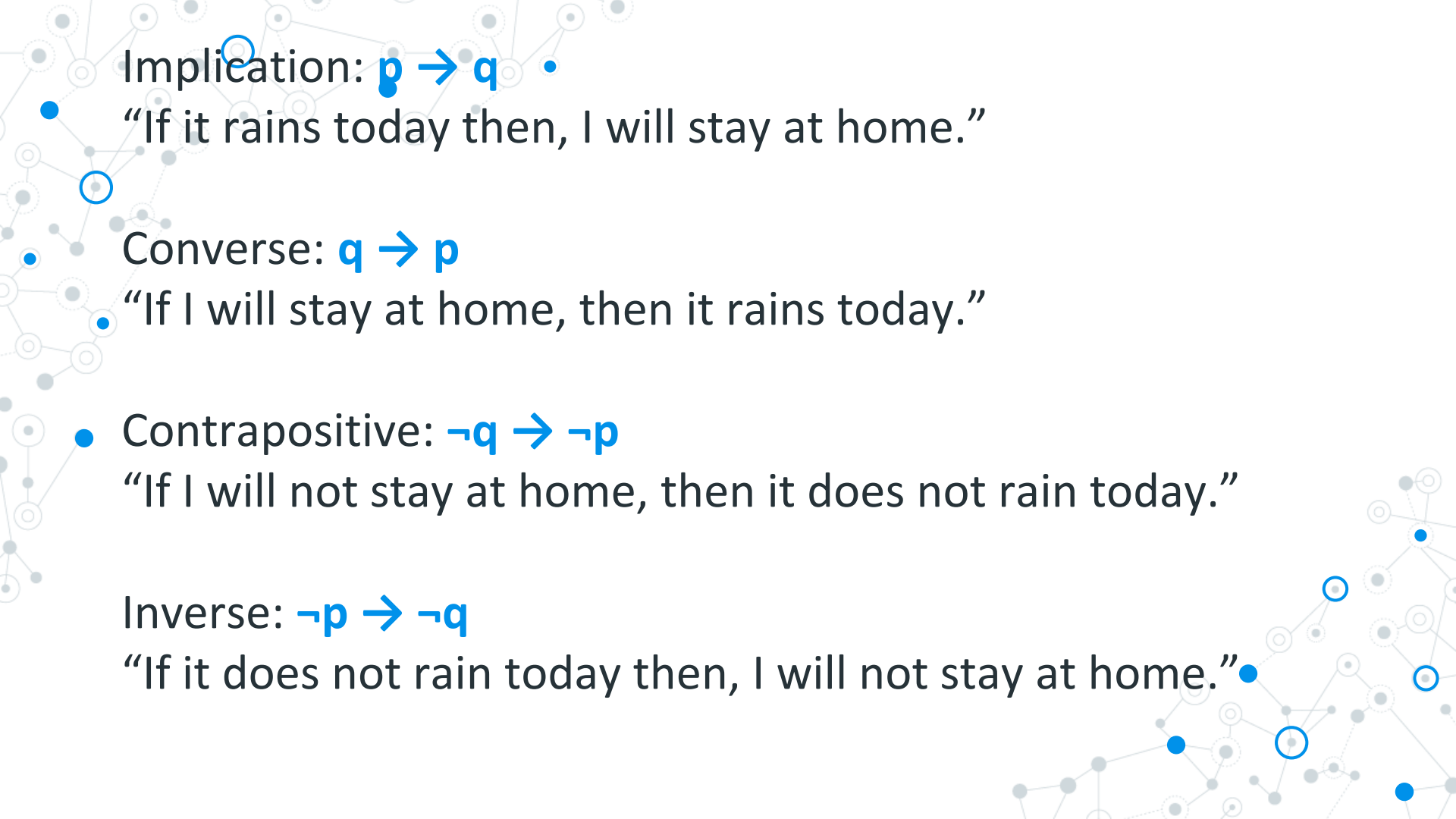
Converse:  $q \rightarrow p$

“If I will stay at home, then it rains today.”

Contrapositive:  $\neg q \rightarrow \neg p$

“If I will not stay at home, then it does not rain today.”

Inverse:  $\neg p \rightarrow \neg q$



Implication:  $p \rightarrow q$

“If it rains today then, I will stay at home.”

Converse:  $q \rightarrow p$

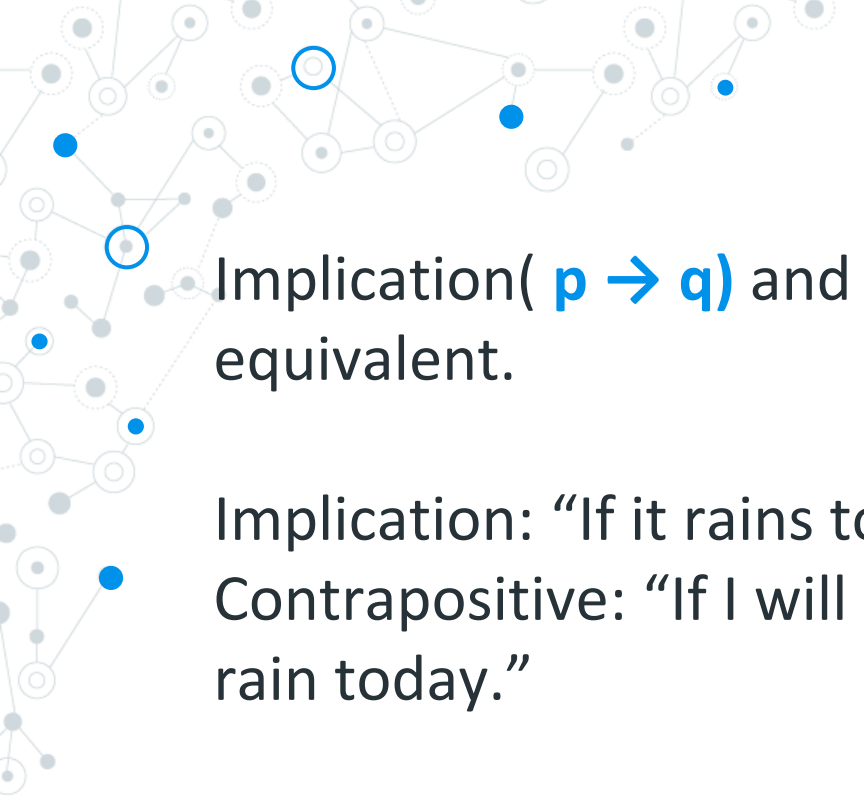
“If I will stay at home, then it rains today.”

• Contrapositive:  $\neg q \rightarrow \neg p$

“If I will not stay at home, then it does not rain today.”

Inverse:  $\neg p \rightarrow \neg q$

“If it does not rain today then, I will not stay at home.”



Implication(  $p \rightarrow q$ ) and Contrapositive( $\neg q \rightarrow \neg p$ ) are equivalent.

Implication: “If it rains today then, I will stay at home.”

Contrapositive: “If I will not stay at home, then it does not rain today.”



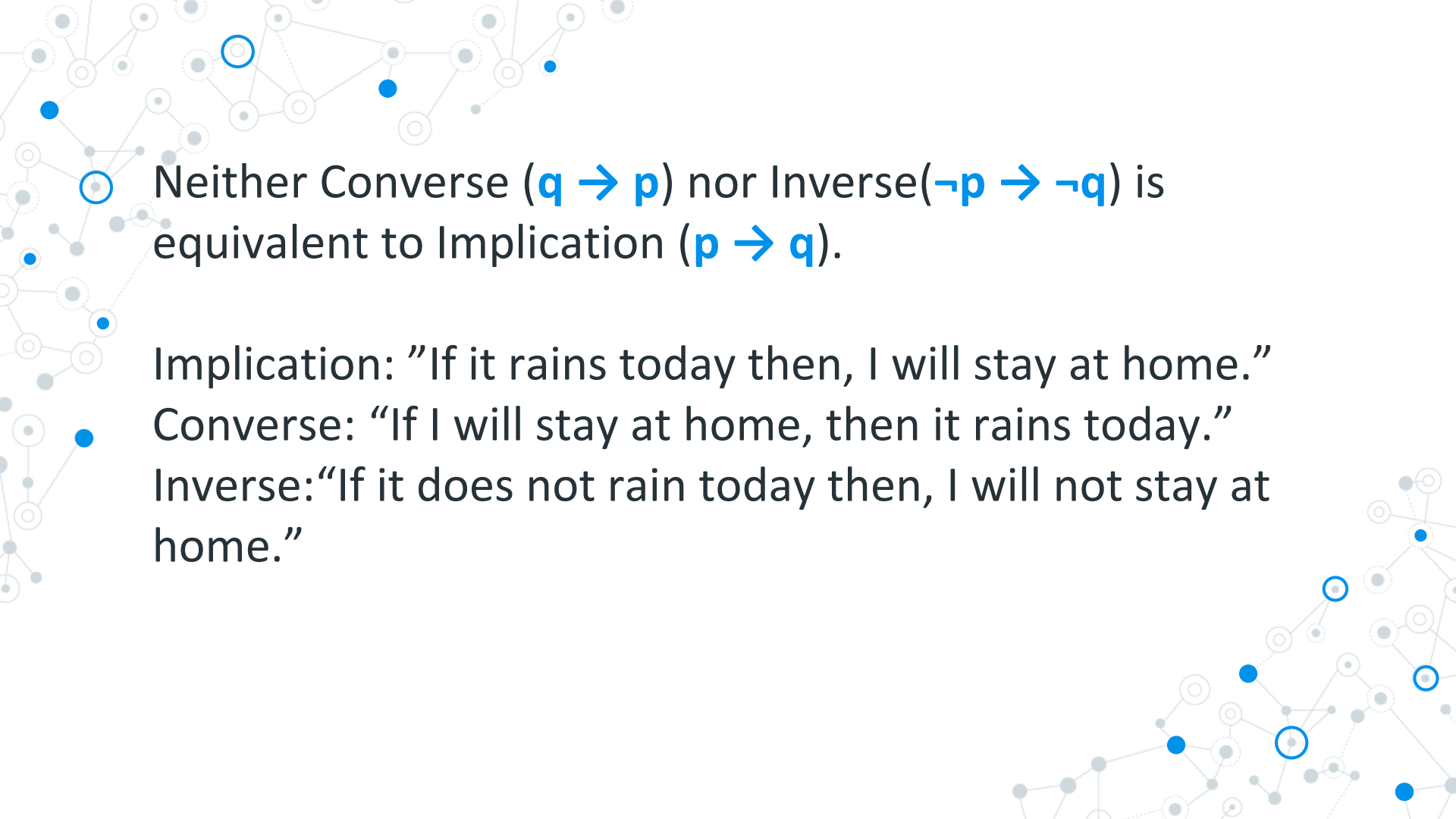


Converse ( $q \rightarrow p$ ) and Inverse( $\neg p \rightarrow \neg q$ ) are equivalent.

Converse: “If I will stay at home, then it rains today.”

Inverse: “If it does not rain today then, I will not stay at home.”



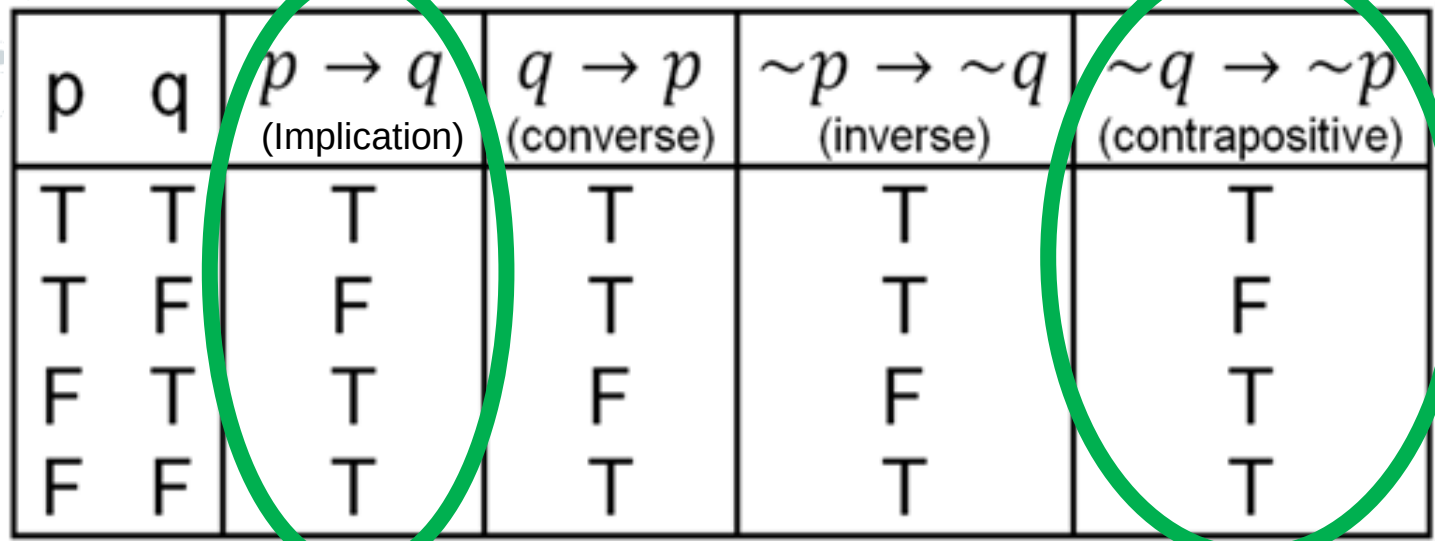


Neither Converse ( $q \rightarrow p$ ) nor Inverse ( $\neg p \rightarrow \neg q$ ) is equivalent to Implication ( $p \rightarrow q$ ).

Implication: "If it rains today then, I will stay at home."

Converse: "If I will stay at home, then it rains today."

Inverse: "If it does not rain today then, I will not stay at home."

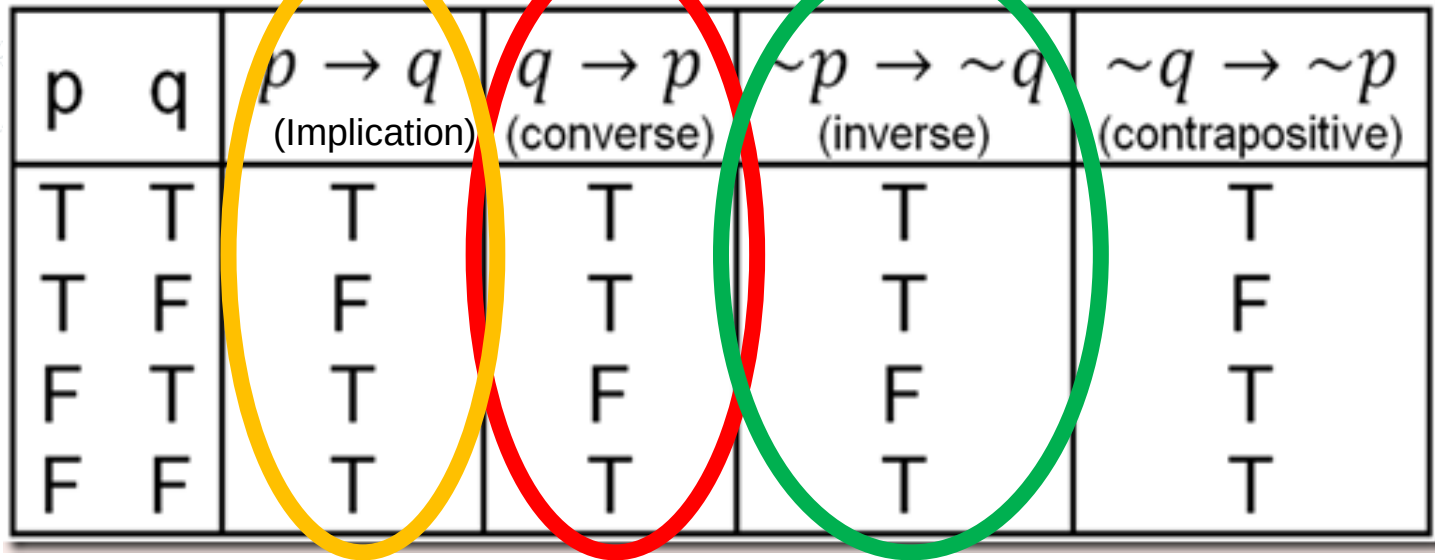


| p | q | $p \rightarrow q$<br>(Implication) | $q \rightarrow p$<br>(converse) | $\sim p \rightarrow \sim q$<br>(inverse) | $\sim q \rightarrow \sim p$<br>(contrapositive) |
|---|---|------------------------------------|---------------------------------|------------------------------------------|-------------------------------------------------|
| T | T | T                                  | T                               | T                                        | T                                               |
| T | F | F                                  | T                               | T                                        | F                                               |
| F | T | T                                  | F                               | F                                        | T                                               |
| F | F | T                                  | T                               | T                                        | T                                               |

Implication(  $p \rightarrow q$  ) and Contrapositive( $\neg q \rightarrow \neg p$ ) are equivalent.

| p | q | $p \rightarrow q$<br>(Implication) | $q \rightarrow p$<br>(converse) | $\sim p \rightarrow \sim q$<br>(inverse) | $\sim q \rightarrow \sim p$<br>(contrapositive) |
|---|---|------------------------------------|---------------------------------|------------------------------------------|-------------------------------------------------|
| T | T | T                                  | T                               | T                                        | T                                               |
| T | F | F                                  | T                               | T                                        | F                                               |
| F | T | T                                  | F                               | F                                        | T                                               |
| F | F | T                                  | T                               | T                                        | T                                               |

Converse ( $q \rightarrow p$ ) and Inverse ( $\sim p \rightarrow \sim q$ ) are equivalent.



The image shows a truth table for logical implications. The table has five columns: 'p', 'q', ' $p \rightarrow q$  (Implication)', ' $q \rightarrow p$  (converse)', ' $\sim p \rightarrow \sim q$  (inverse)', and ' $\sim q \rightarrow \sim p$  (contrapositive)'. The rows represent the four possible combinations of truth values for p and q: (T, T), (T, F), (F, T), and (F, F). Three large, overlapping ovals are drawn over the table: a yellow oval around the 'Implication' column, a red oval around the 'converse' column, and a green oval around the 'inverse' column. The 'contrapositive' column is not highlighted.

| p | q | $p \rightarrow q$<br>(Implication) | $q \rightarrow p$<br>(converse) | $\sim p \rightarrow \sim q$<br>(inverse) | $\sim q \rightarrow \sim p$<br>(contrapositive) |
|---|---|------------------------------------|---------------------------------|------------------------------------------|-------------------------------------------------|
| T | T | T                                  | T                               | T                                        | T                                               |
| T | F | F                                  | T                               | T                                        | F                                               |
| F | T | T                                  | F                               | F                                        | T                                               |
| F | F | T                                  | T                               | T                                        | T                                               |

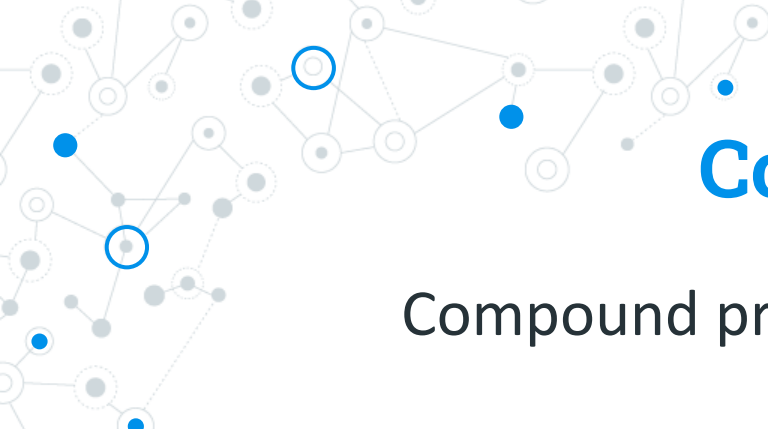
Neither Converse ( $q \rightarrow p$ ) nor Inverse ( $\sim p \rightarrow \sim q$ ) is equivalent to Implication ( $p \rightarrow q$ ).



# Tautology

Compound proposition which is **always true**.

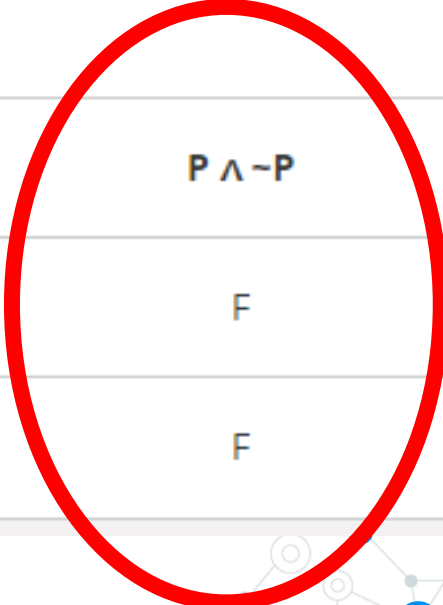
| P | $\neg P$ | $P \vee \neg P$ |
|---|----------|-----------------|
| T | F        | T               |
| F | T        | T               |



# Contradiction

Compound proposition which is **always false**.

| P | $\neg P$ | $P \wedge \neg P$ |
|---|----------|-------------------|
| T | F        | F                 |
| F | T        | F                 |



# Contingency

Compound proposition which is **sometimes true** and **sometimes false**.

| x | y | $x \wedge y$ |
|---|---|--------------|
| T | T | T            |
| T | F | F            |
| F | T | F            |
| F | F | F            |

# Satisfiable

Compound proposition which has at least **one true** result.

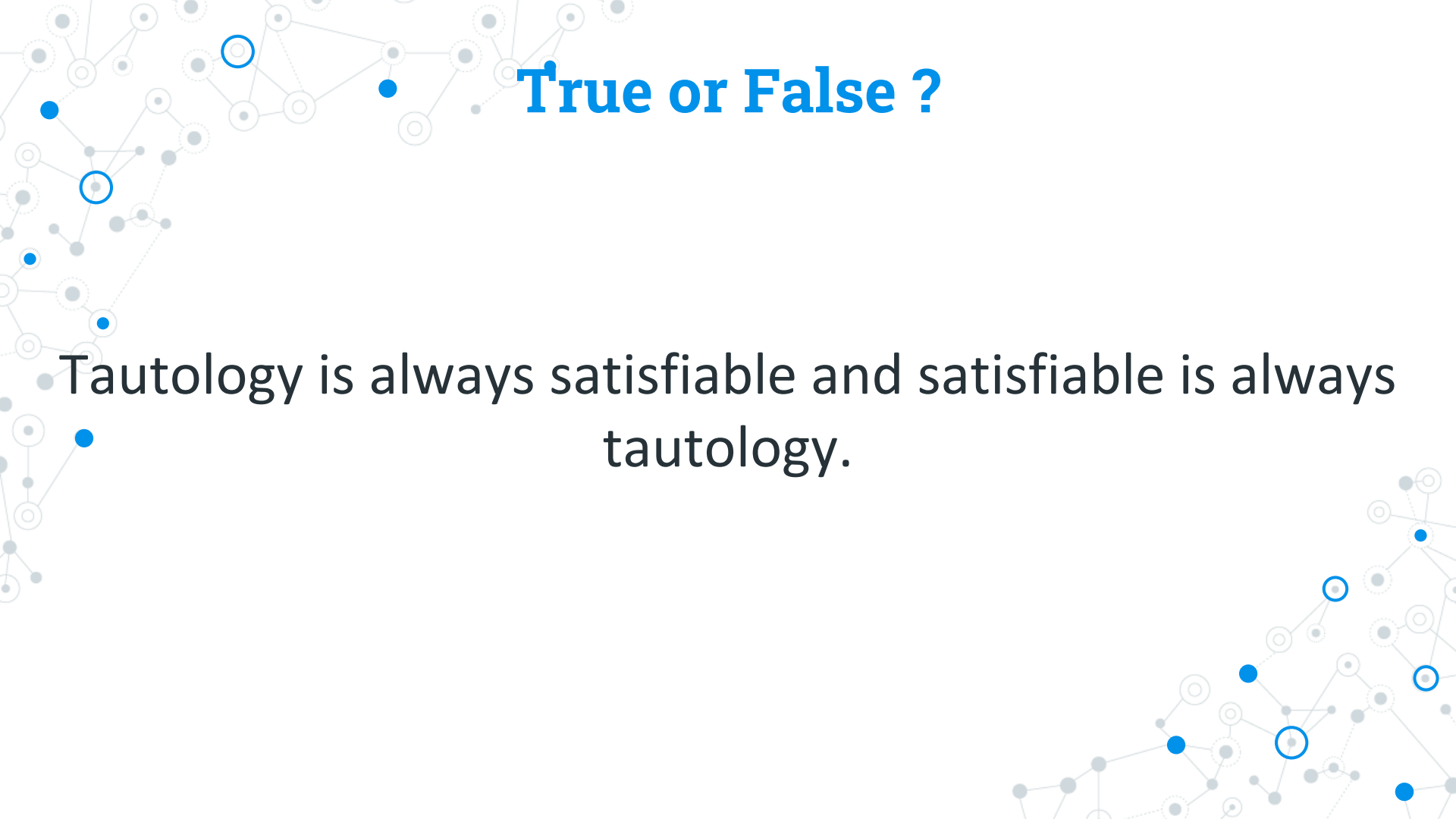
| $x$ | $y$ | $x \wedge y$ |
|-----|-----|--------------|
| T   | T   | T            |
| T   | F   | F            |
| F   | T   | F            |
| F   | F   | F            |

A decorative network graph in the top-left corner, featuring a complex web of interconnected nodes and edges. Some nodes are highlighted with blue circles, and others with blue dots.

**True or False ?**

Tautology is always satisfiable.

A decorative network graph in the bottom-right corner, featuring a complex web of interconnected nodes and edges. Some nodes are highlighted with blue circles, and others with blue dots.

A decorative background featuring a network diagram with nodes and edges. The nodes are represented by small circles, some of which are highlighted with blue outlines or filled with blue. The edges are thin lines connecting the nodes, creating a complex web-like structure. The overall color scheme is light gray and blue.

# True or False ?

Tautology is always satisfiable and satisfiable is always tautology.

A decorative network graph in the top-left corner, featuring a complex web of nodes and edges. Some nodes are highlighted with blue circles, and others with blue dots. The graph is composed of solid and dashed lines.

# True or False ?

Contradiction is always unsatisfiable.

A decorative network graph in the bottom-right corner, similar to the one in the top-left, with nodes and edges, some highlighted with blue circles and others with blue dots.

# Valid

Compound proposition is valid when it is a tautology.

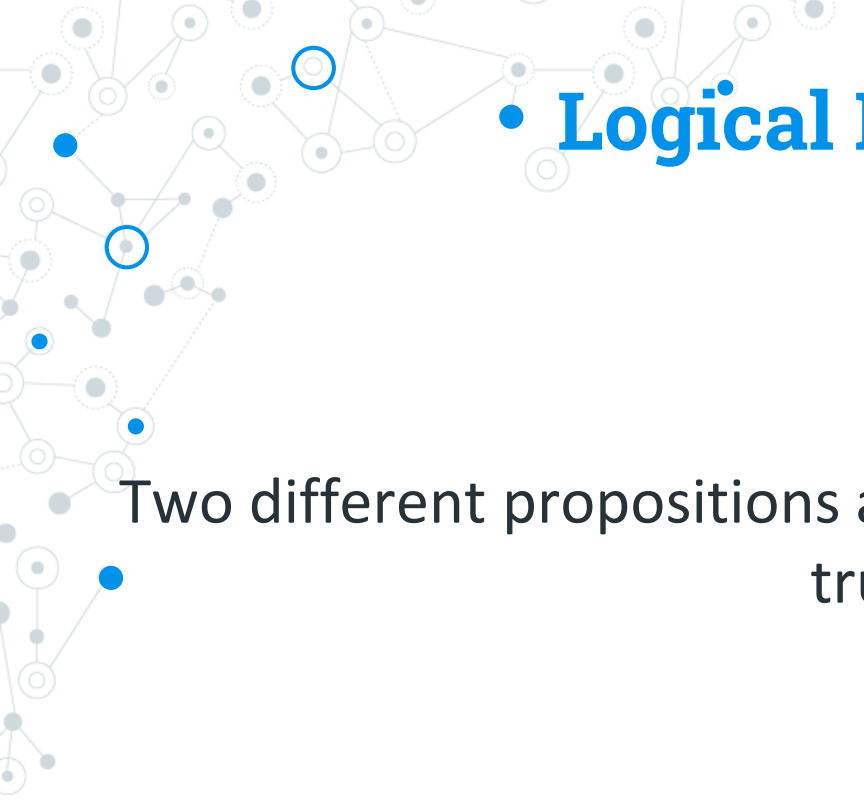
| P | $\neg P$ | $P \vee \neg P$ |
|---|----------|-----------------|
| T | F        | T               |
| F | T        | T               |



# Invalid

Compound proposition which is either a contradiction or contingency.

| P | $\neg P$ | $P \wedge \neg P$ |
|---|----------|-------------------|
| T | F        | F                 |
| F | T        | F                 |

A decorative network diagram in the top-left corner consisting of a web of interconnected nodes. Some nodes are solid blue circles, while others are white circles with blue outlines. The nodes are connected by thin, light gray lines.

# • Logical Equivalences

Two different propositions are equivalent if they have the same truth values.

A decorative network diagram in the bottom-right corner, similar to the one in the top-left, featuring a cluster of interconnected nodes with some solid blue circles and some white circles with blue outlines, connected by thin gray lines.

# • Logical Equivalences

Two different propositions are equivalent if they have the same truth values.

| $p$ | $q$ | $\neg(p \rightarrow q)$ | $p \wedge \neg q$ |
|-----|-----|-------------------------|-------------------|
| T   | T   | F                       | F                 |
| T   | F   | T                       | T                 |
| F   | T   | F                       | F                 |
| F   | F   | F                       | F                 |

# • Logical Equivalences

Two different propositions are equivalent if they have the same truth values.

| $p$ | $q$ | $\neg(p \rightarrow q)$ | $p \wedge \neg q$ |
|-----|-----|-------------------------|-------------------|
| T   | T   | F                       | F                 |
| T   | F   | T                       | T                 |
| F   | T   | F                       | F                 |
| F   | F   | F                       | F                 |

A decorative network graph pattern in the top-left corner, consisting of various nodes (some solid blue, some solid grey, some hollow blue, some hollow grey) connected by thin grey lines.

# • Logical Equivalences

$$p \equiv q$$
A decorative network graph pattern in the bottom-right corner, consisting of various nodes (some solid blue, some solid grey, some hollow blue, some hollow grey) connected by thin grey lines.



**Table 4-1    Laws of the algebra of propositions**

|                           |                                                                |                                                                  |
|---------------------------|----------------------------------------------------------------|------------------------------------------------------------------|
| <b>Idempotent laws:</b>   | (1a) $p \vee p \equiv p$                                       | (1b) $p \wedge p \equiv p$                                       |
| <b>Associative laws:</b>  | (2a) $(p \vee q) \vee r \equiv p \vee (q \vee r)$              | (2b) $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$        |
| <b>Commutative laws:</b>  | (3a) $p \vee q \equiv q \vee p$                                | (3b) $p \wedge q \equiv q \wedge p$                              |
| <b>Distributive laws:</b> | (4a) $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ | (4b) $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$ |
| <b>Identity laws:</b>     | (5a) $p \vee F \equiv p$<br>(6a) $p \vee T \equiv T$           | (5b) $p \wedge T \equiv p$<br>(6b) $p \wedge F \equiv F$         |
| <b>Involution law:</b>    | (7) $\neg\neg p \equiv p$                                      |                                                                  |
| <b>Complement laws:</b>   | (8a) $p \vee \neg p \equiv T$<br>(9a) $\neg T \equiv F$        | (8b) $p \wedge \neg p \equiv F$<br>(9b) $\neg F \equiv T$        |
| <b>DeMorgan's laws:</b>   | (10a) $\neg(p \vee q) \equiv \neg p \wedge \neg q$             | (10b) $\neg(p \wedge q) \equiv \neg p \vee \neg q$               |





# Rules of Inference

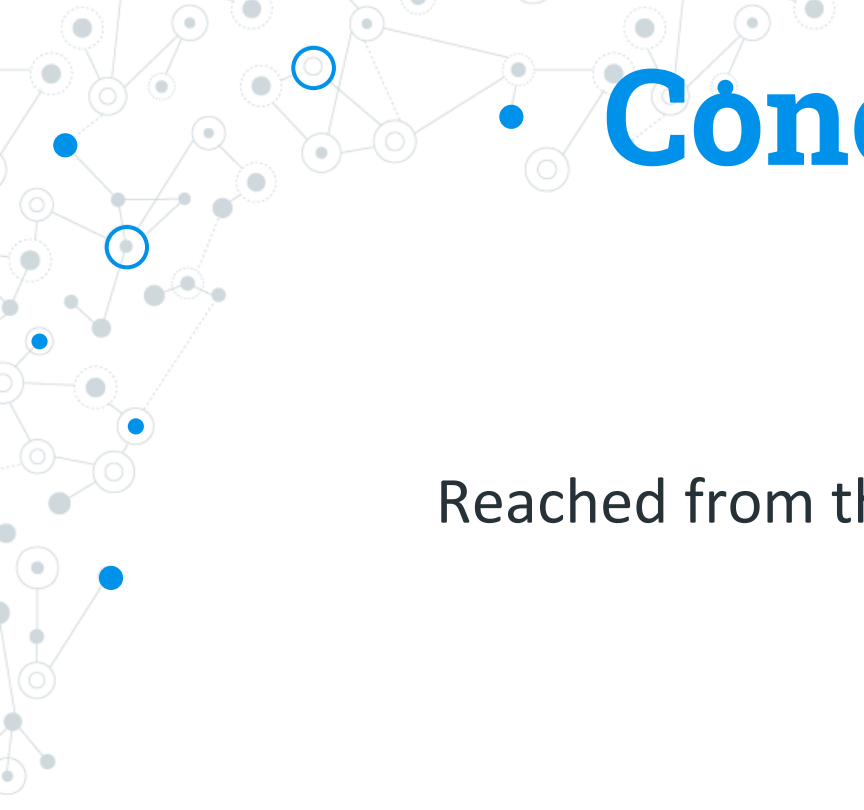
A decorative network diagram in the top-left corner, featuring a complex web of interconnected nodes. Some nodes are solid blue circles, while others are white circles with blue outlines. The nodes are connected by thin, light gray lines, creating a mesh-like structure that extends towards the center of the slide.

# Premise

Basis for the conclusion / evidence / assumption

A decorative network diagram in the bottom-right corner, similar to the one in the top-left. It consists of a cluster of interconnected nodes, with some nodes highlighted as solid blue circles and others as white circles with blue outlines, all connected by a network of thin gray lines.



A decorative network diagram in the top-left corner, featuring a complex web of interconnected nodes. Some nodes are solid blue circles, while others are white circles with blue outlines. The nodes are connected by thin, light gray lines, creating a dense, organic structure.

# Conclusion

Reached from the given set of premises.

A decorative network diagram in the bottom-right corner, similar to the one in the top-left. It shows a cluster of interconnected nodes, with some highlighted in blue and others in white with blue outlines, connected by light gray lines.

A decorative network diagram in the top-left corner, featuring a cluster of interconnected nodes. Some nodes are solid blue circles, while others are white circles with blue outlines. They are connected by thin grey lines.

If premise then conclusion

A decorative network diagram in the bottom-right corner, similar to the one in the top-left. It shows a cluster of interconnected nodes, with some being solid blue circles and others being white circles with blue outlines, connected by thin grey lines.

A decorative network diagram in the top-left corner, featuring a complex web of interconnected nodes. Some nodes are solid blue circles, while others are white circles with blue outlines. The nodes are connected by thin, light gray lines, creating a mesh-like structure.

# Argument

Set of one or more premises and a conclusion.

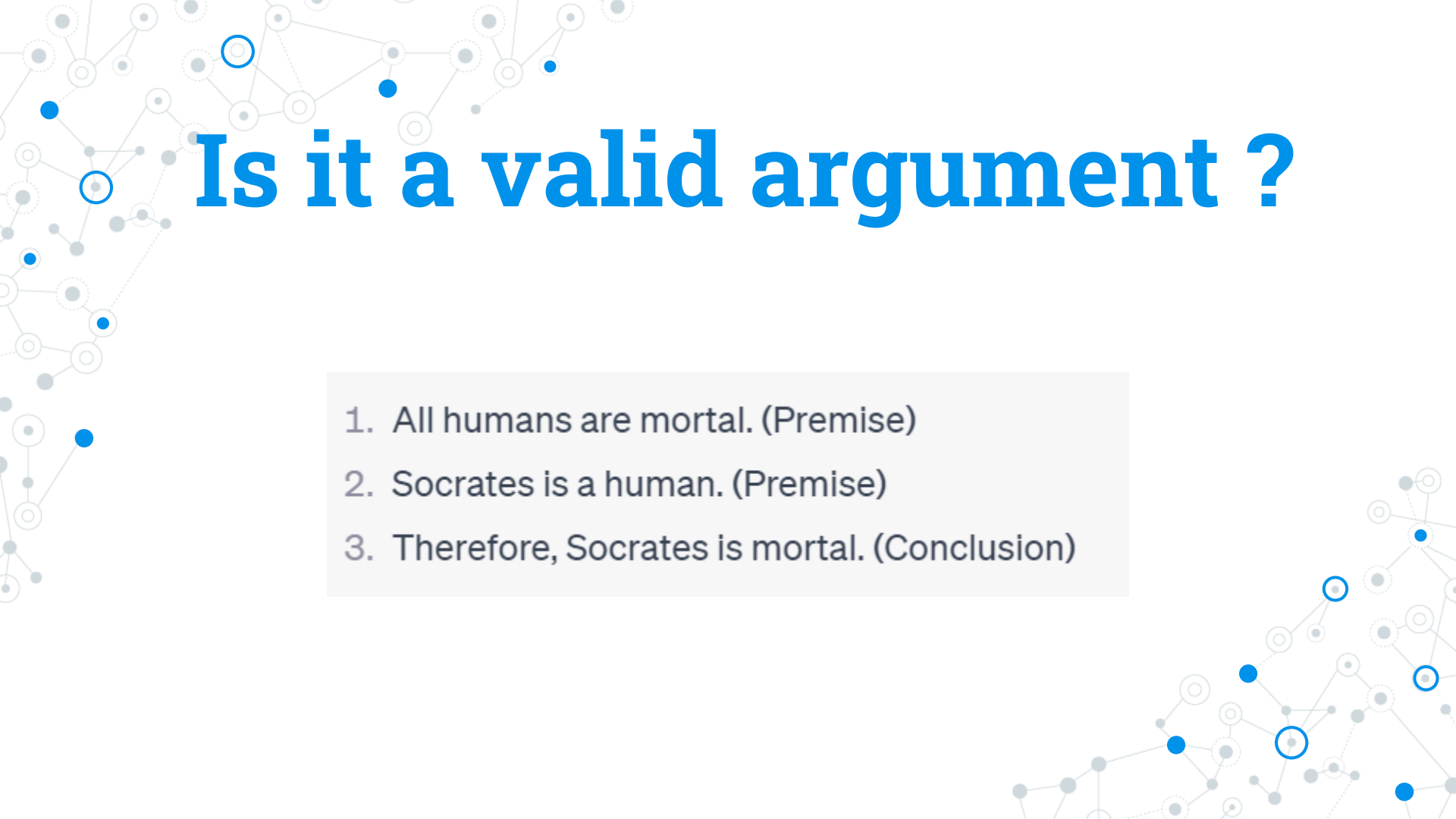
A decorative network diagram in the bottom-right corner, similar to the one in the top-left. It shows a cluster of nodes, with some highlighted in blue and others in white with blue outlines, connected by a network of light gray lines.

A decorative network diagram in the top-left corner, featuring a complex web of interconnected nodes and lines. Some nodes are highlighted with blue circles, and others with blue dots. The lines are thin and grey, creating a mesh-like structure.

# Valid Argument

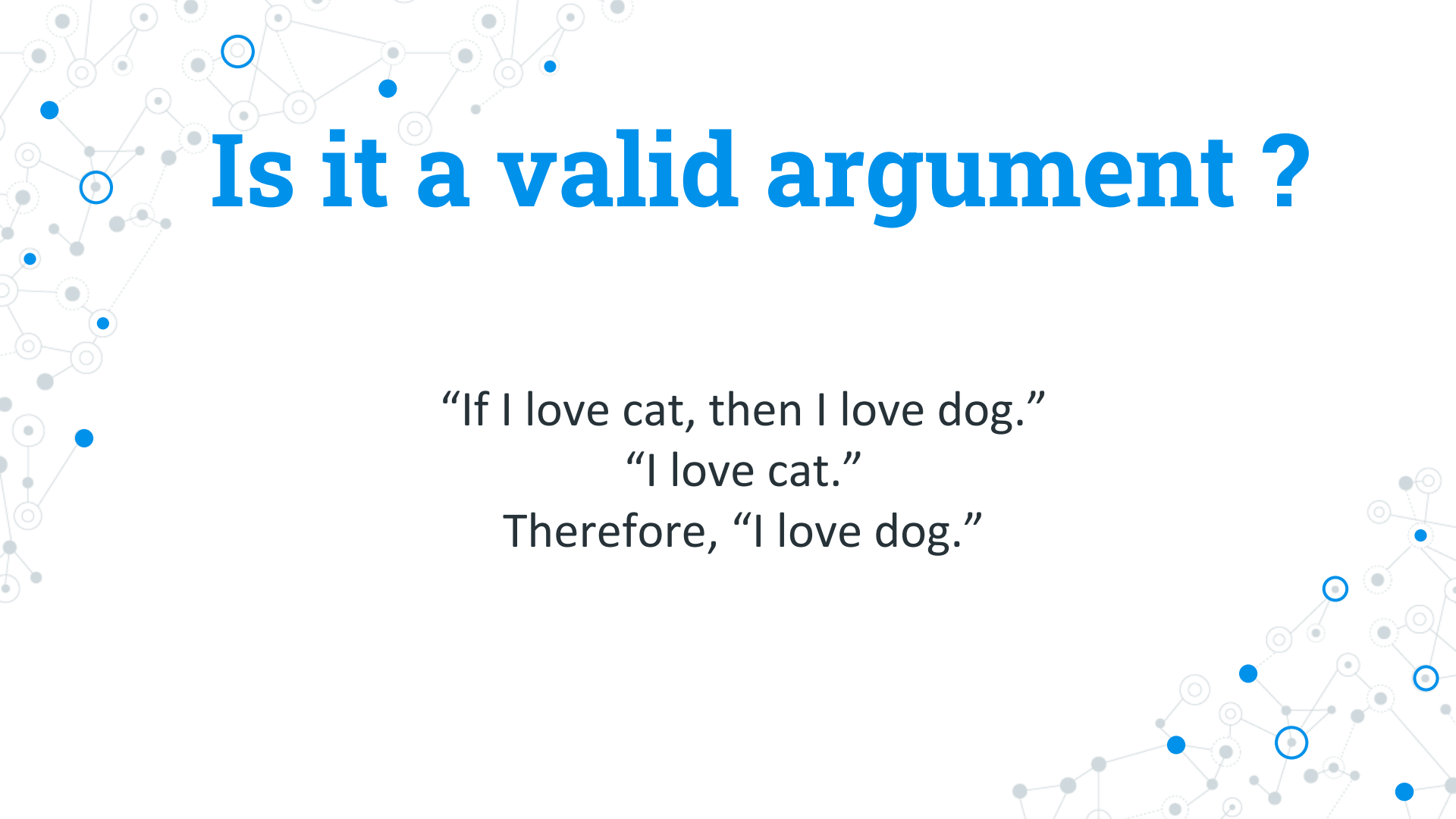
If premises are all true, then conclusion must also be true.

A decorative network diagram in the bottom-right corner, similar to the one in the top-left. It shows a cluster of nodes connected by lines, with several nodes highlighted in blue.

A decorative background featuring a network diagram with various nodes (circles) and connecting lines. Some nodes are highlighted with blue outlines or solid blue dots. The diagram is more prominent on the left and right sides of the slide.

# Is it a valid argument ?

1. All humans are mortal. (Premise)
2. Socrates is a human. (Premise)
3. Therefore, Socrates is mortal. (Conclusion)

A decorative background featuring a network diagram with nodes and lines. The nodes are represented by circles of varying sizes and colors (blue, grey, white), connected by thin grey lines. Some nodes are highlighted with a blue outline. The network is distributed across the slide, with a denser concentration on the left side and a more sparse arrangement on the right.

# Is it a valid argument ?

“If I love cat, then I love dog.”

“I love cat.”

Therefore, “I love dog.”

# Translate into Logical Expressions

“If I love cat, then I love dog.”  
“I love cat.”  
Therefore, “I love dog.”

$$p \rightarrow q$$

# Translate into Logical Expressions

“If I love cat, then I love dog.”  
“I love cat.”  
Therefore, “I love dog.”

$p \rightarrow q$   
 $p$



# Translate into Logical Expressions

“If I love cat, then I love dog.”  
“I love cat.”  
Therefore, “I love dog.”

$p \rightarrow q$

$p$

---

$\therefore q$

# Translate into Logical Expressions

“If I love cat, then I love dog.”  
    “I love cat.”  
Therefore, “I love dog.”

$p \rightarrow q$

$p$

---

$\therefore q$

$$((p \rightarrow q) \wedge p) \rightarrow q$$

## Identify if it is a Valid Argument

“If I love cat, then I love dog.”  
“I love cat.”  
Therefore, “I love dog.”

$p \rightarrow q$

T

$p$

T

---

$\therefore q$

Assume all premises are true, we need to make  
conclusion true.

# Identify if it is a Valid Argument

“If I love cat, then I love dog.”  
“I love cat.”  
Therefore, “I love dog.”

$p \rightarrow q$

T

$p$  T

T

---

$\therefore q$

T

# Identify if it is a Valid Argument

“If I love cat, then I love dog.”  
“I love cat.”  
Therefore, “I love dog.”

$p^T \rightarrow q$

T

$p^T$

T

$\therefore q$

T

# Identify if it is a Valid Argument

“If I love cat, then I love dog.”  
“I love cat.”  
Therefore, “I love dog.”

$p^T \rightarrow q$

T

$p^T$

T

---

$\therefore q$

T

What is the value of  $q$ , that will make the first premise true ?

## Identify if it is a Valid Argument

“If I love cat, then I love dog.”  
“I love cat.”  
Therefore, “I love dog.”

|                       |   |
|-----------------------|---|
| $p^T \rightarrow q^T$ | T |
| $p^T$                 | T |
| <hr/>                 |   |
| $\therefore q$        | T |

What is the value of  $q$ , that will make the first premise true ?

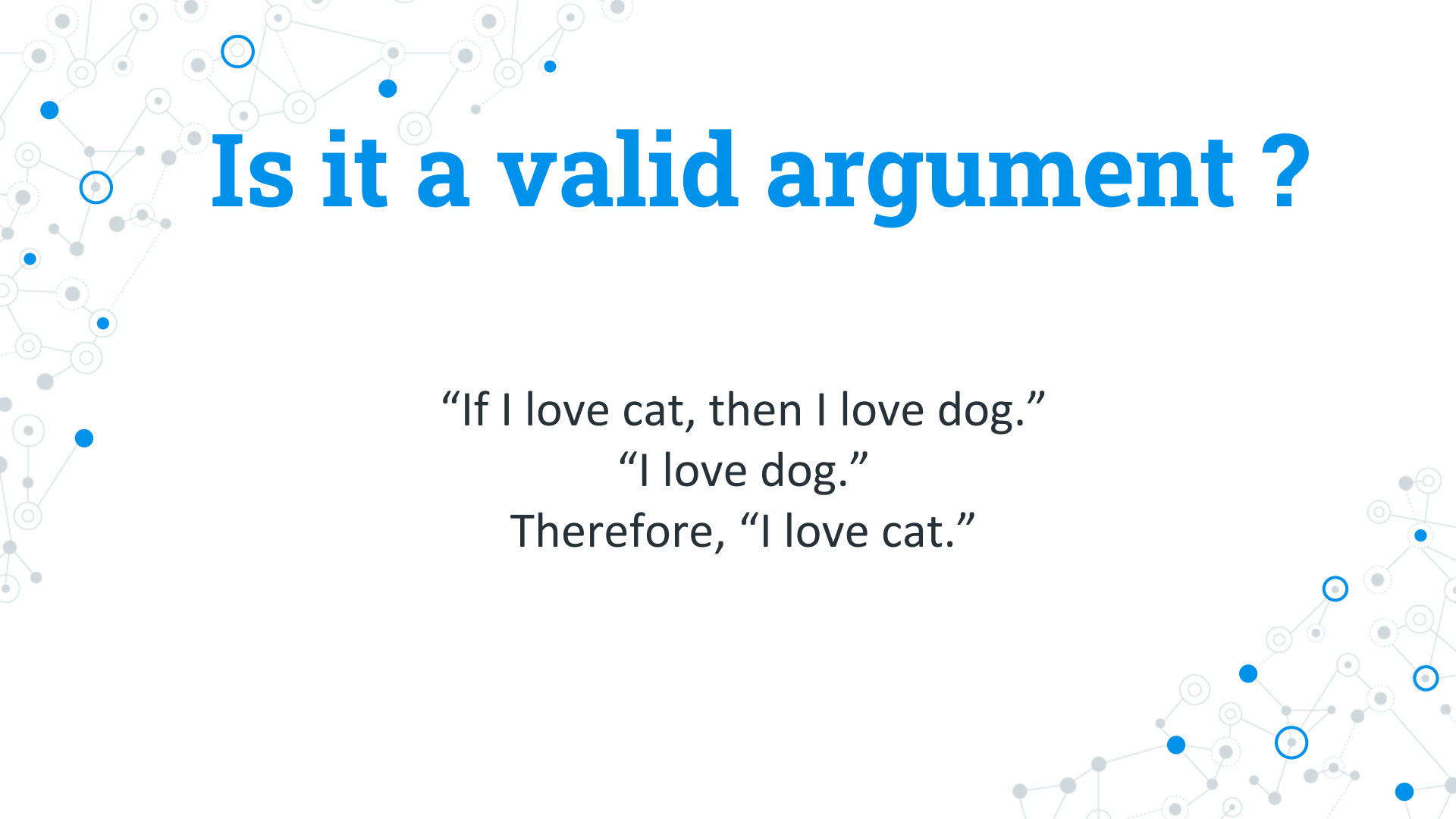
# Identify if it is a Valid Argument

“If I love cat, then I love dog.”  
“I love cat.”  
Therefore, “I love dog.”

|                       |   |
|-----------------------|---|
| $p^T \rightarrow q^T$ | T |
| $p^T$                 | T |
| <hr/>                 |   |
| $\therefore q^T$      | T |

If all premises and conclusion are true, then it is a valid argument.



A decorative background featuring a network diagram with nodes and lines. The nodes are represented by circles of varying sizes and colors (blue, grey, white), connected by thin grey lines. Some nodes are highlighted with a blue border. The network is spread across the top and bottom of the slide.

# Is it a valid argument ?

“If I love cat, then I love dog.”

“I love dog.”

Therefore, “I love cat.”

# Is it a valid argument ?

“If I love cat, then I love dog.”

“I love dog.”

Therefore, “I love cat.”

$p \rightarrow q$

# Is it a valid argument ?

“If I love cat, then I love dog.”  
“I love dog.”  
Therefore, “I love cat.”

$p \rightarrow q$

$q$

# Is it a valid argument ?

“If I love cat, then I love dog.”  
“I love dog.”  
Therefore, “I love cat.”

$p \rightarrow q$

$q$

---

$\therefore p$

# Is it a valid argument ?

“If I love cat, then I love dog.”  
“I love dog.”  
Therefore, “I love cat.”

$p \rightarrow q$

T

$q$

T

---

$\therefore p$

Assume all premises are true, we need to make  
conclusion true.

# Is it a valid argument ?

“If I love cat, then I love dog.”  
“I love dog.”  
Therefore, “I love cat.”

$p \rightarrow q$

T

$q$  T

T

---

$\therefore p$

# Is it a valid argument ?

“If I love cat, then I love dog.”  
“I love dog.”  
Therefore, “I love cat.”

|                     |     |
|---------------------|-----|
| $p \rightarrow q^T$ | $T$ |
| $q^T$               | $T$ |
| <hr/>               |     |
| $\therefore p$      |     |

# Is it a valid argument ?

“If I love cat, then I love dog.”  
“I love dog.”  
Therefore, “I love cat.”

|                     |     |
|---------------------|-----|
| $p \rightarrow q^T$ | $T$ |
| $q^T$               | $T$ |
| <hr/>               |     |
| $\therefore p$      |     |

What is the value of  $p$  that will make the first premise true ?



# Is it a valid argument ?

“If I love cat, then I love dog.”  
“I love dog.”  
Therefore, “I love cat.”

|                     |     |
|---------------------|-----|
| $p \rightarrow q^T$ | $T$ |
| $q^T$               | $T$ |
| <hr/>               |     |
| $\therefore p$      |     |

What is the value of  $p$  that will make the first premise true ? **T / F**

# Is it a valid argument ?

If p is true

“If I love cat, then I love dog.”  
“I love dog.”  
Therefore, “I love cat.”

|                   |   |
|-------------------|---|
| $p \rightarrow q$ | T |
| $q$               | T |
| <hr/>             |   |
| $\therefore p$    |   |

What is the value of p that will make the first premise true ? **T / F**

# Is it a valid argument ?

If p is true

“If I love cat, then I love dog.”  
“I love dog.”  
Therefore, “I love cat.”

|                   |   |
|-------------------|---|
| $p \rightarrow q$ | T |
| $q$               | T |
| <hr/>             |   |
| $\therefore p$    |   |

What is the value of p that will make the first premise true ? **T / F**

# Is it a valid argument ?

If p is false

“If I love cat, then I love dog.”  
“I love dog.”  
Therefore, “I love cat.”

|                   |     |
|-------------------|-----|
| $p \rightarrow q$ | $T$ |
| $q$               | $T$ |
| <hr/>             |     |
| $\therefore p$    | $F$ |

What is the value of p that will make the first premise true ?  $T / F$

# Is it a valid argument ?

If  $p$  is false, then all premises are true and the conclusion is false

“If I love cat, then I love dog.”  
“I love dog.”  
Therefore, “I love cat.”

|                   |     |
|-------------------|-----|
| $p \rightarrow q$ | $T$ |
| $q$               | $T$ |
| <hr/>             |     |
| $\therefore p$    | $F$ |

What is the value of  $p$  that will make the first premise true ?  $T / F$

# Is it a valid argument ?

If p is false, then all premises are true and the conclusion is false



“If I love cat, then I love dog.”  
“I love dog.”  
Therefore, “I love cat.”

|                   |     |
|-------------------|-----|
| $p \rightarrow q$ | $T$ |
| $q$               | $T$ |
| <hr/>             |     |
| $\therefore p$    | $F$ |

What is the value of p that will make the first premise true ?  $T / F$

# Is it a valid argument ?

If  $p$  is false, then all premises are true and the conclusion is false



“If I love cat, then I love dog.”  
“I love dog.”  
Therefore, “I love cat.”

|                   |     |
|-------------------|-----|
| $p \rightarrow q$ | $T$ |
| $q$               | $T$ |
| <hr/>             |     |
| $\therefore p$    | $F$ |

What is the value of  $p$  that will make the first premise true ?  $T / F$

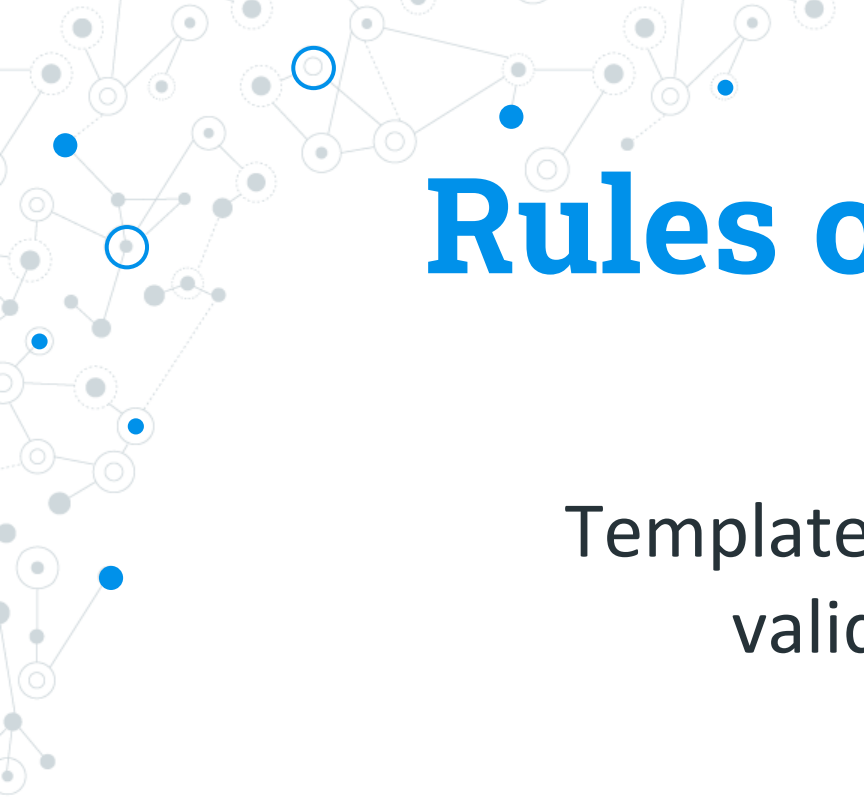
A decorative network diagram in the top-left corner of the slide. It features a complex web of interconnected nodes and edges. The nodes are represented by small circles, some of which are solid blue, some are solid grey, and some are hollow with a blue outline. The edges are thin grey lines connecting these nodes. The overall structure is dense and organic, resembling a molecular or biological network.

# Inference

Deriving conclusions from  
evidences

A decorative network diagram in the bottom-right corner of the slide. It features a complex web of interconnected nodes and edges. The nodes are represented by small circles, some of which are solid blue, some are solid grey, and some are hollow with a blue outline. The edges are thin grey lines connecting these nodes. The overall structure is dense and organic, resembling a molecular or biological network.

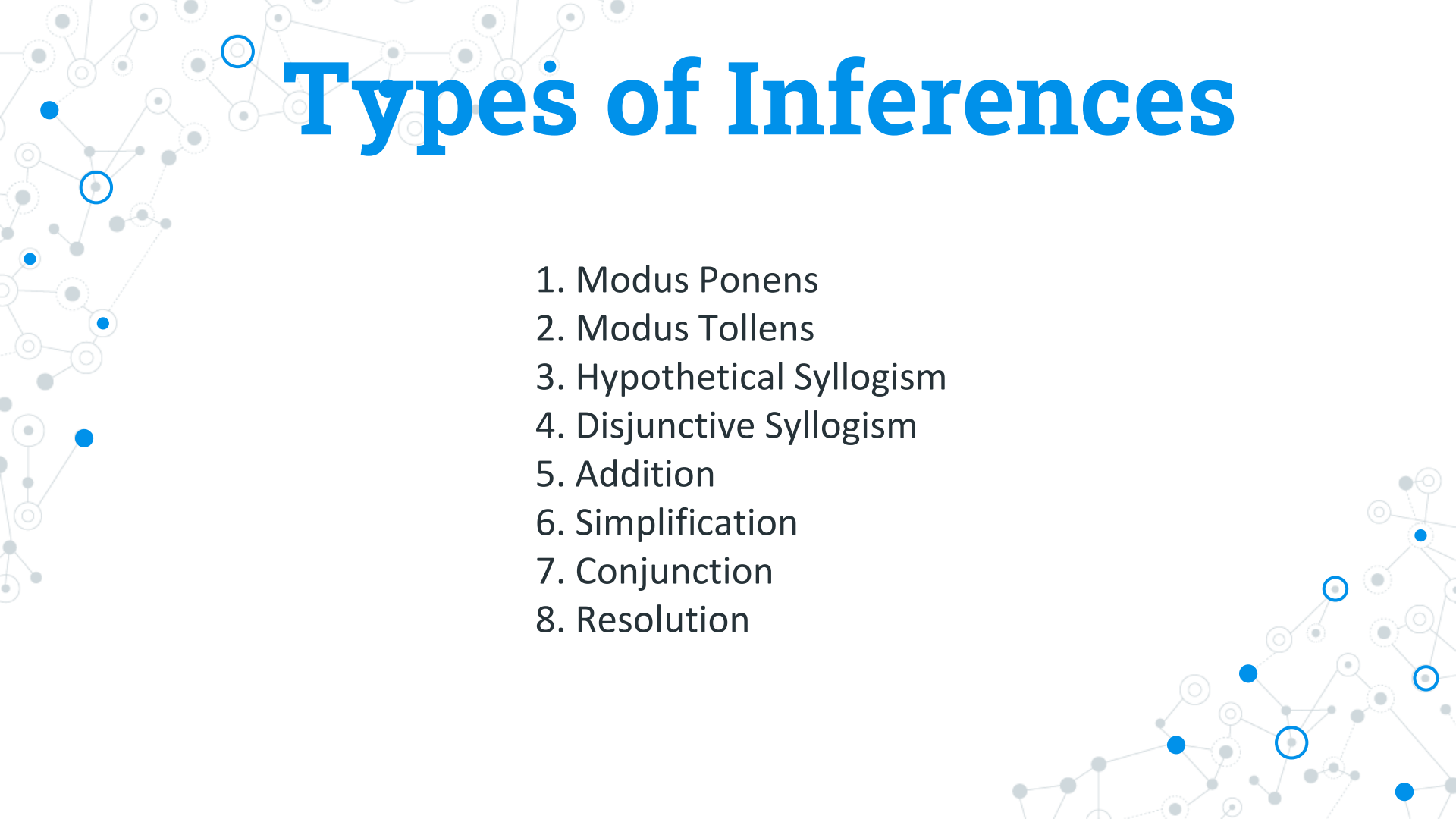


A decorative network diagram in the top-left corner, featuring a complex web of interconnected nodes. Some nodes are solid blue circles, while others are white circles with blue outlines. The connections are represented by thin grey lines.

# Rules of Inference

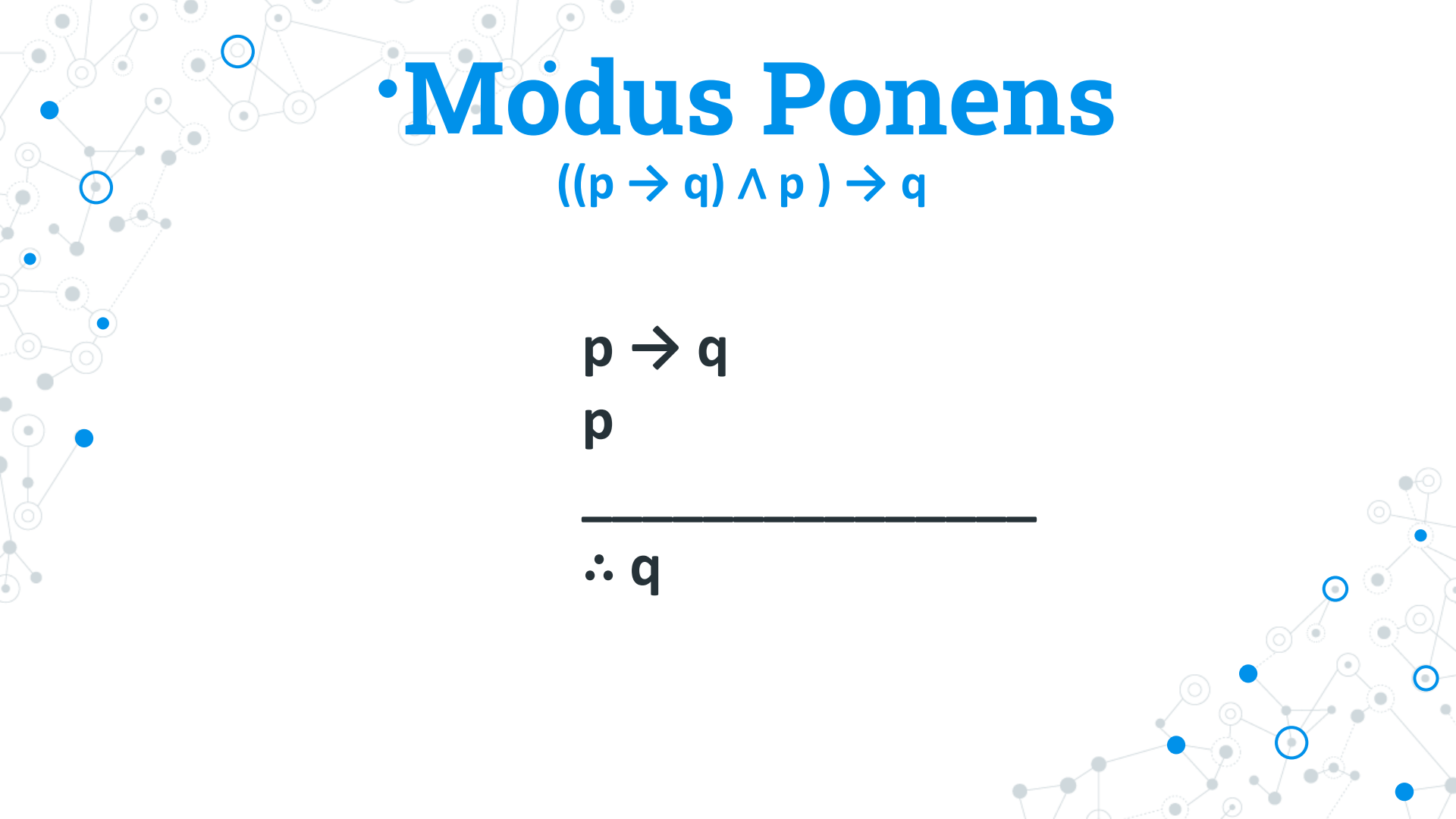
Templates for constructing  
valid arguments.

A decorative network diagram in the bottom-right corner, similar to the one in the top-left, showing a cluster of interconnected nodes with some highlighted in blue and others in white with blue outlines.

A decorative background featuring a network of interconnected nodes and lines. The nodes are represented by small circles, some of which are highlighted with blue outlines or filled with blue. The lines are thin and gray, creating a web-like structure that spans the entire slide.

# Types of Inferences

1. Modus Ponens
2. Modus Tollens
3. Hypothetical Syllogism
4. Disjunctive Syllogism
5. Addition
6. Simplification
7. Conjunction
8. Resolution



# • Modus Ponens

$$((p \rightarrow q) \wedge p) \rightarrow q$$

$$p \rightarrow q$$

$$p$$

---

$$\therefore q$$

# • Modus Ponens

$$((p \rightarrow q) \wedge p) \rightarrow q$$

$p \rightarrow q$  T

$p$  T

---

$\therefore q$

# • Modus Ponens

$$((p \rightarrow q) \wedge p) \rightarrow q$$

$$p \rightarrow q \quad \text{T}$$

$$p^{\text{T}} \quad \text{T}$$

---

$$\therefore q$$

# • Modus Ponens

$$((p \rightarrow q) \wedge p) \rightarrow q$$

$$\boxed{p^T \rightarrow q} \quad T$$

$$\boxed{p^T} \quad T$$

---

$$\therefore q$$

What is the value of  $q$  that will make the first premise true ?

# • Modus Ponens

$$((p \rightarrow q) \wedge p) \rightarrow q$$

$$\boxed{p^T \rightarrow q} \quad T$$

$$\boxed{p^T} \quad T$$

---

$$\therefore q$$

What is the value of  $q$  that will make the first premise true ? **T**

# • Modus Ponens

$$((p \rightarrow q) \wedge p) \rightarrow q$$

$$\begin{array}{|l} p^T \rightarrow q^T \end{array} \quad T$$

$$\begin{array}{|l} p^T \end{array} \quad T$$

---

$$\therefore q$$



# • Modus Ponens

$$((p \rightarrow q) \wedge p) \rightarrow q$$

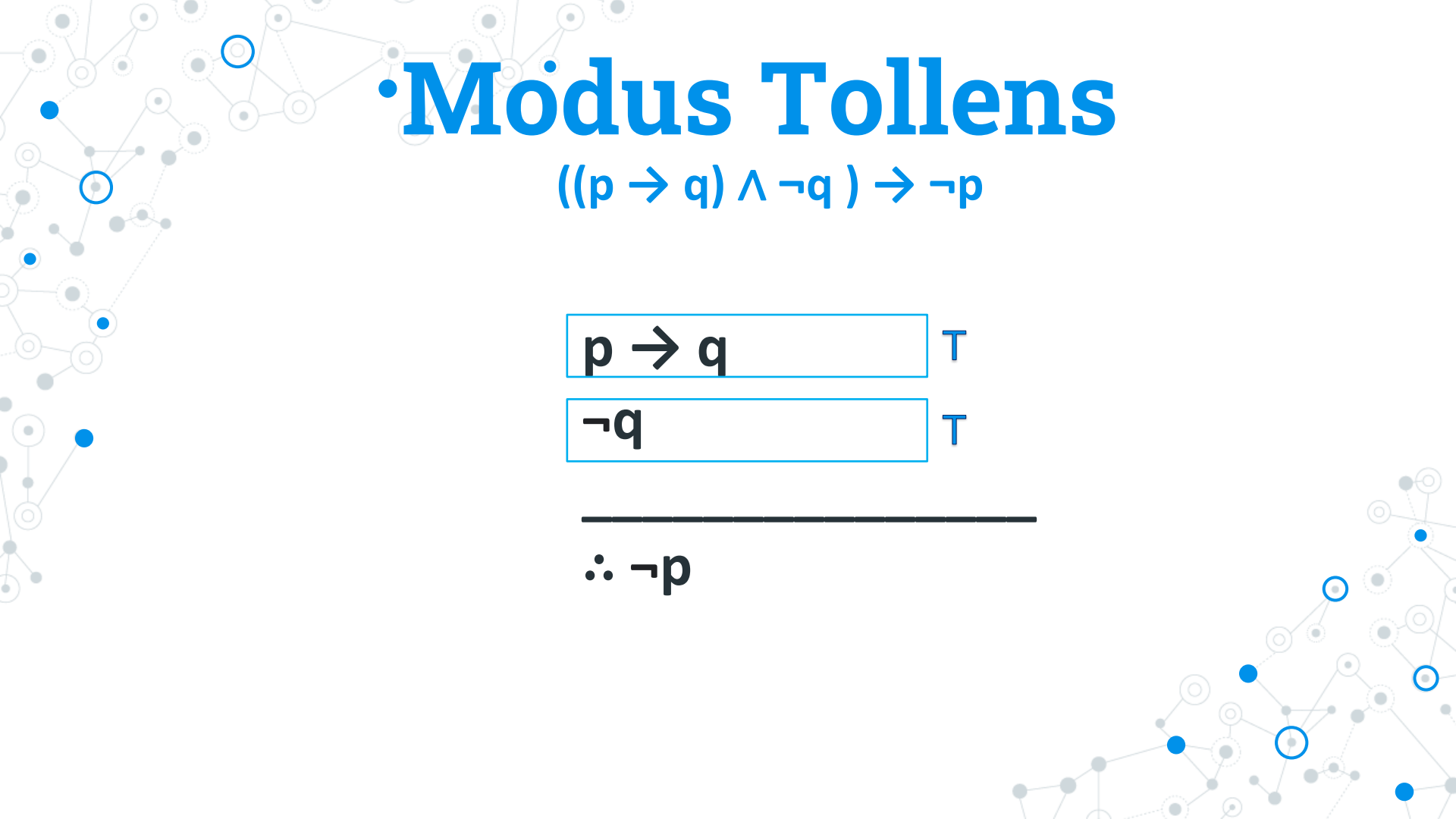
$$\boxed{p^T \rightarrow q^T} \quad T$$

$$\boxed{p^T} \quad T$$

---

$$\therefore q^T$$

We have proven that conclusion is true given all premises are true.



# • Modus Tollens

$$((p \rightarrow q) \wedge \neg q) \rightarrow \neg p$$

|                   |        |
|-------------------|--------|
| $p \rightarrow q$ | $\top$ |
|-------------------|--------|

|          |        |
|----------|--------|
| $\neg q$ | $\top$ |
|----------|--------|

---

$$\therefore \neg p$$

# Hypothetical Syllogism

$$[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow \underline{r})$$

$$p \rightarrow q$$

$$q \rightarrow r$$

---

$$\therefore p \rightarrow r$$

# Disjunctive Syllogism

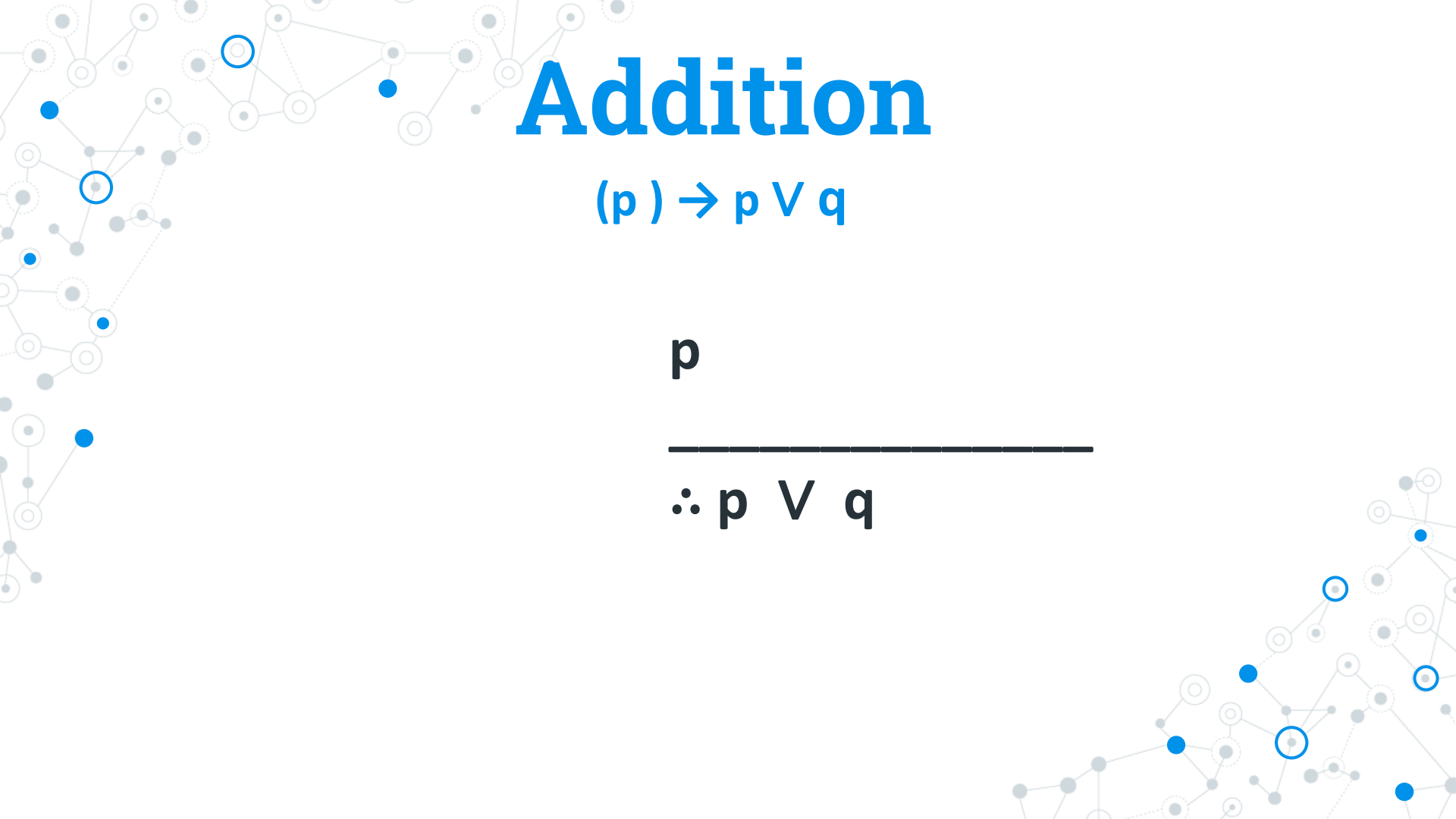
$$((p \vee q) \wedge \neg p) \rightarrow q$$

$p \vee q$

$\neg p$

---

$\therefore q$



# Addition

$$(p) \rightarrow p \vee q$$

$p$

---

$\therefore p \vee q$

# Simplification

$$(p \wedge q) \rightarrow p$$

$$p \wedge q$$

---

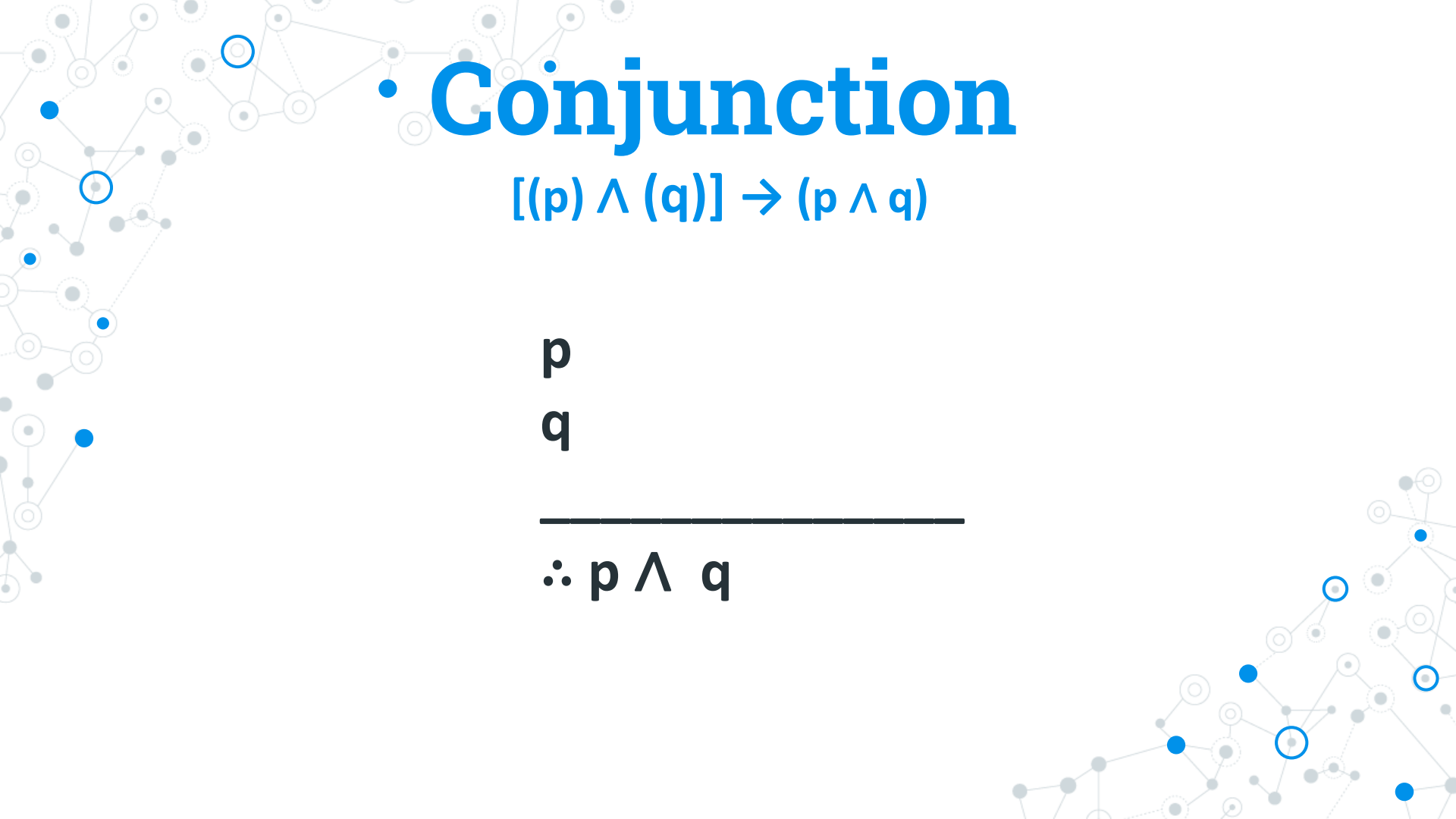
$$\therefore p$$

or

$$p \wedge q$$

---

$$\therefore q$$



# • Conjunction

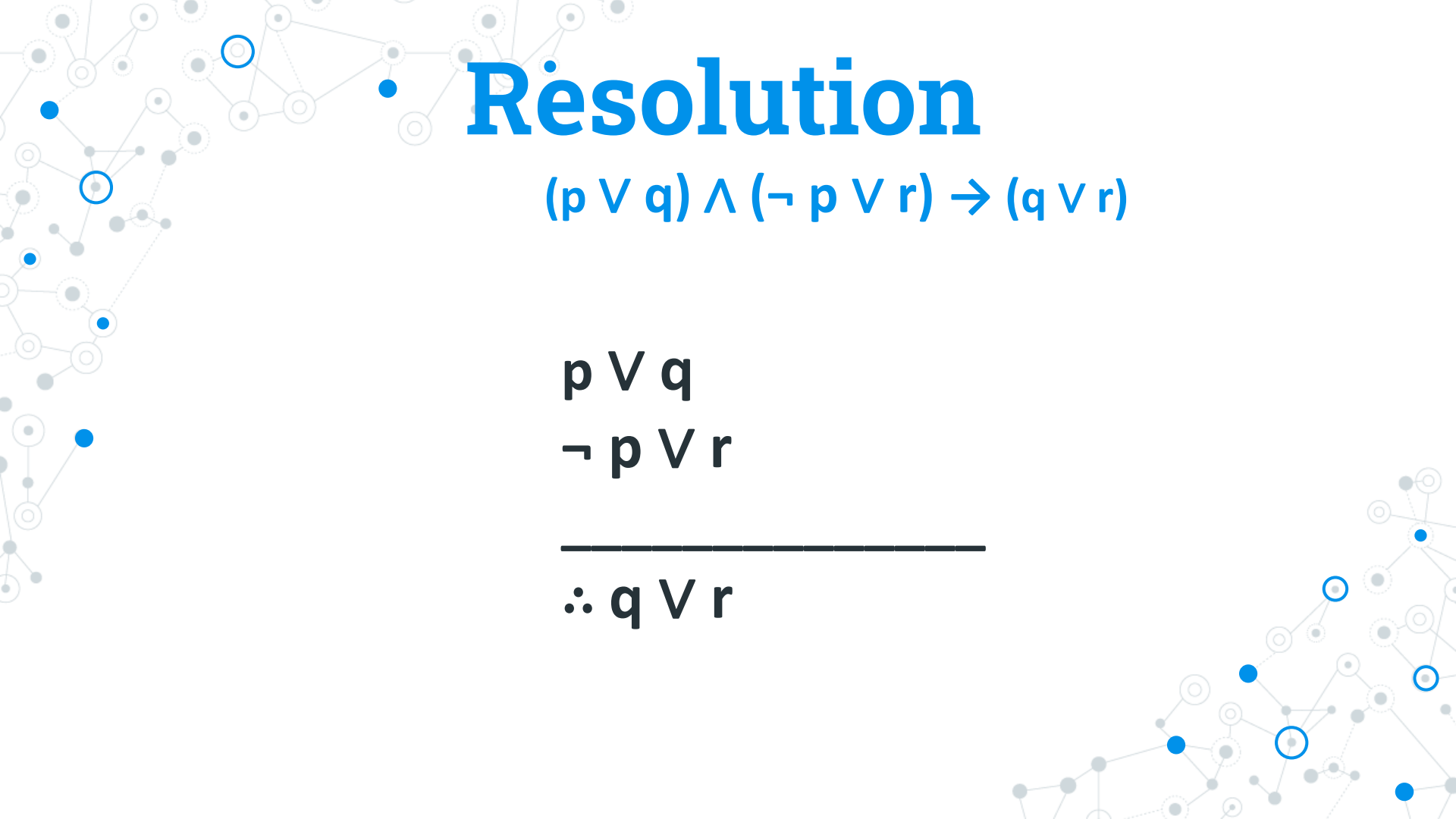
$$[(p) \wedge (q)] \rightarrow (p \wedge q)$$

**p**

**q**

---

**$\therefore p \wedge q$**



# Resolution

$$(p \vee q) \wedge (\neg p \vee r) \rightarrow (q \vee r)$$

$$p \vee q$$

$$\neg p \vee r$$

---

$$\therefore q \vee r$$





# Thanks!

## Any questions?

You can find me at:  
[aegono@carsu.edu.ph](mailto:aegono@carsu.edu.ph)

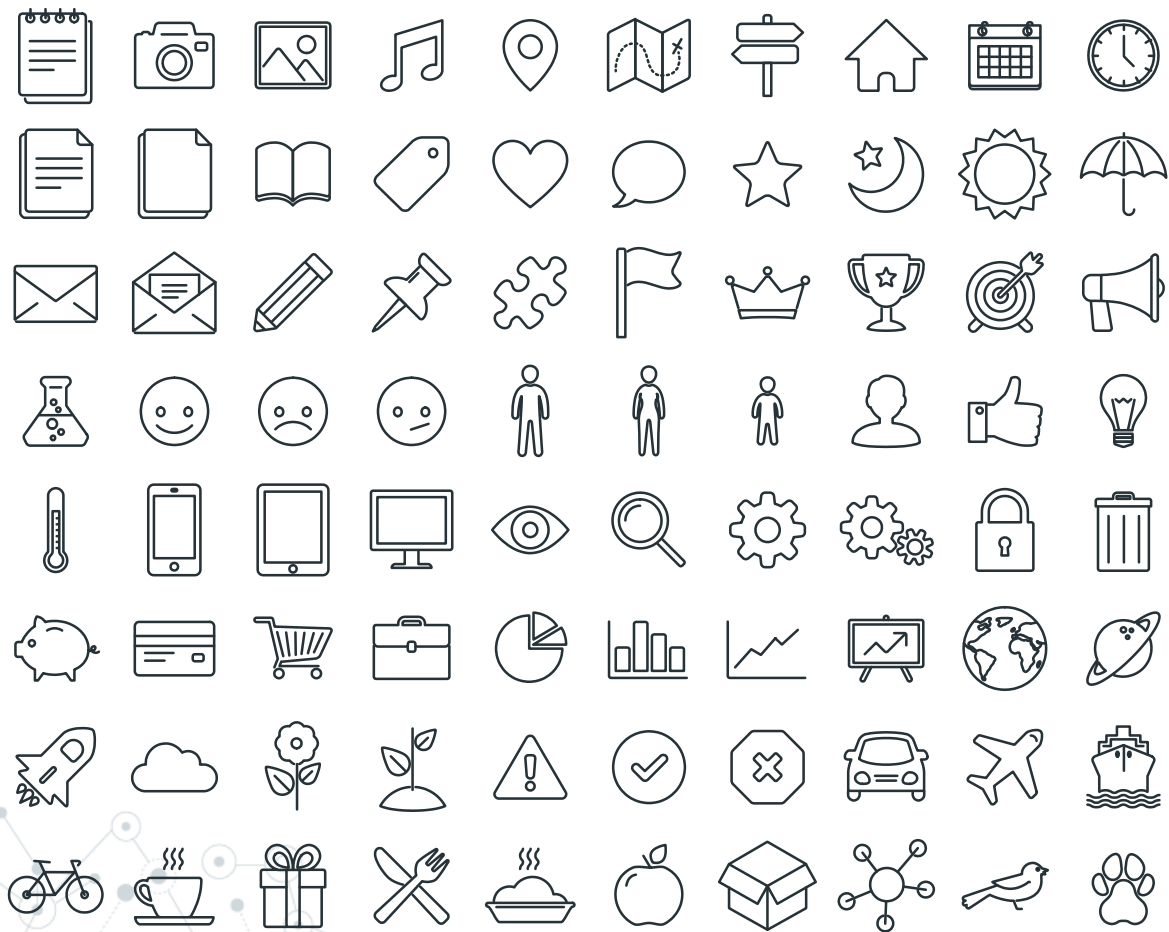
# References

Schaum's Outlines, Discrete Mathematics (Third Edition)

## Credits

Special thanks to all the people who made and released these awesome resources for free:

- ◎ Presentation template by [SlidesCarnival](#)
- ◎ Photographs by [Unsplash](#)



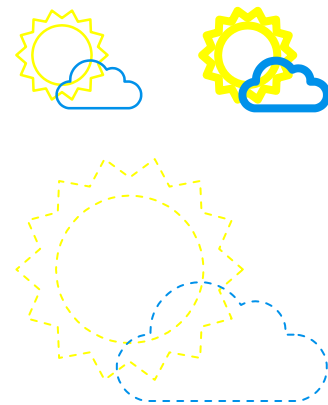
## SlidesCarnival icons are editable shapes.

This means that you can:

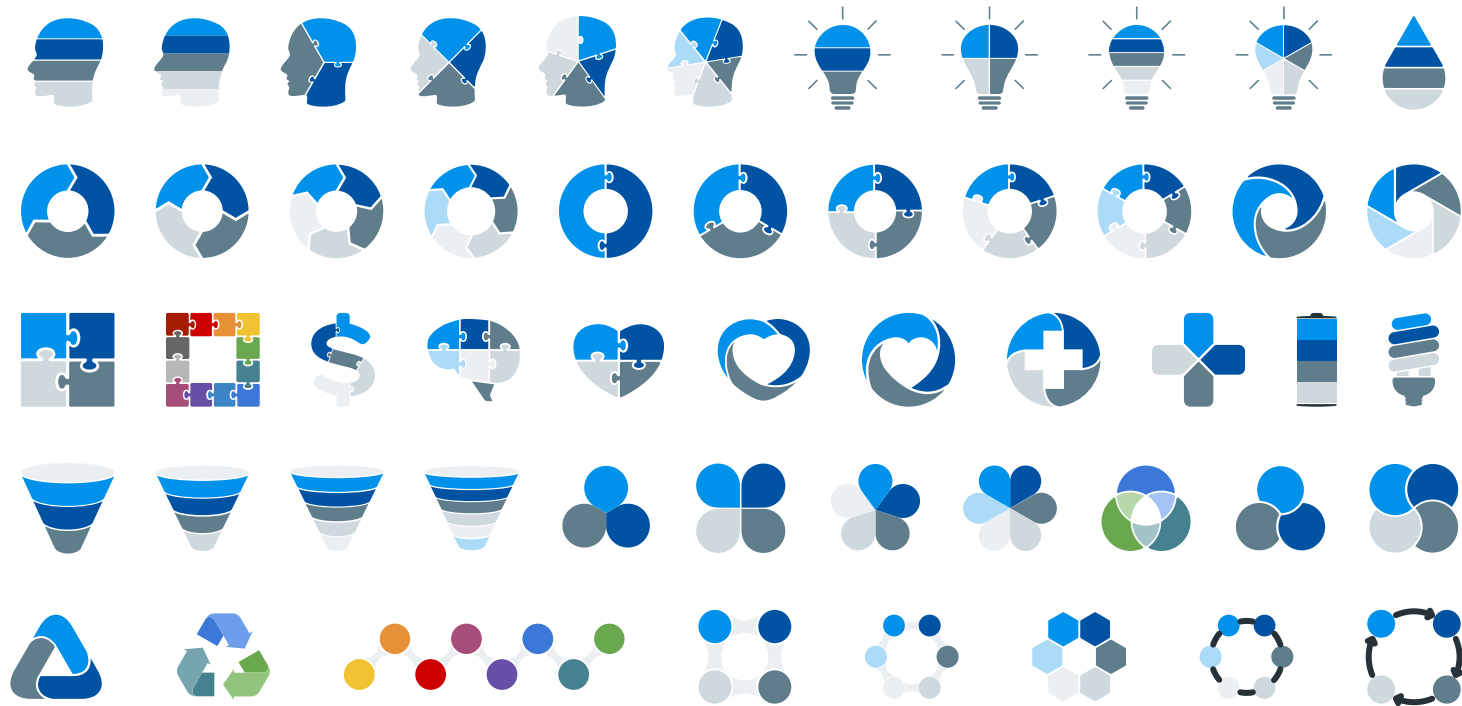
- Resize them without losing quality.
- Change line color, width and style.

Isn't that nice? :)

Examples:



# Diagrams and infographics



**You can also use any emoji as an icon!**

And of course it resizes without losing quality.

How? Follow Google instructions <https://twitter.com/googledocs/status/730087240156643328>

